

Numerical Production Contract for an Adaptive Chiral-Defect CFT

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This document is a preregistered numerical specification for the proposed PRX Quantum manuscript on a postselection-free adaptive class-A fermion circuit. It contains no numerical result and licenses no present-tense physical claim. Existing repository code and artifacts are implementation references only. The contract fixes the physical ensemble, system-size sequences, sample counts, fixed-duration logic, saved data, estimator order, controls, uncertainty analysis, acceptance tests, claim downgrades, and five-figure architecture. Every GPU circuit trajectory must be generated by the canonical `classA_U1FGTN_gpu.run_markov_circuit` entry point on an A100-class 40 GB runtime. The central acceptance standard is deliberately conjunctive: quantitative revalidation of the inherited BPJ bulk, a $c = 1$ and level-one $U(1)$ wall state, physical chiral response, an integer twist spectral flow, a wall-localized tangent sector controlling purification, and universal information recoverable from trajectory-resolved records must all survive thermodynamic, transverse, temporal, and protocol controls.

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I. SCOPE AND EVIDENTIARY RULES

All stochastic production campaigns below retain status PLANNED. The deterministic B0 review calibration is now completed and is recorded explicitly in its subsection. No saved plot, notebook output, or previous fit is admissible as production evidence. Repository artifacts may be cited only in a field labelled “legacy comparison,” and only when the observable and protocol have been checked at the path level. Such a comparison is never pooled with production data.

The physical object is the Born ensemble of adaptive trajectories. For a record $\xi = (m_1, \dots, m_T)$,

$$K_\xi = K_{m_T|\xi_{<T}} \cdots K_{m_1}, \quad p_\xi = \text{Tr}(K_\xi \rho_0 K_\xi^\dagger), \quad \rho_\xi = K_\xi \rho_0 K_\xi^\dagger / p_\xi. \quad (1)$$

TABLE I. Conjunctive manuscript gates. “Pass” means that every campaign-specific finite-size, transverse, temporal, and control test has also passed.

gate	required result	failure consequence
G1	fresh-ancilla $n_{\text{shell}} = 1$ realization reproduces the quantized BPJ bulk phase	remove the validated-topological-platform claim
G2	state data give $c_{\text{edge}} = 1$, $U(1)$ level $k = 1$, and the expected correlator	say “critical wall,” not “chiral defect CFT”
G3	total Born response is chiral and frozen-record flux has integer flow	do not claim operational chirality or a branch-wise integer wall index
G4	a wall-localized tangent gap closes and controls max-mix purification	retain tangent construction only as a finite-time diagnostic
G5	replica-calibrated record self-information and charge sectors reproduce universal defect data	remove the record-only universality claim

Perfect correction samples every m_t from its conditional Born probability and then applies the prescribed feedback. Forced postselection is a control. The mean channel is a second, inequivalent control. For every nonlinear observable,

$$\mathbb{E}_\xi[O(C_\xi)] \neq O(\mathbb{E}_\xi[C_\xi]) \quad (2)$$

in general. The estimator must therefore be evaluated per trajectory before equal-weight averaging. Since the trajectories are Born sampled, no second probability weight is applied.

The manuscript is ready for a PRX Quantum submission only if the five gates in Table I pass. A campaign may produce an interesting result while failing its predeclared gate; in that case the corresponding weaker wording, not a modified estimator, is used.

II. PRX QUANTUM THESIS AND INHERITED BPJ BASELINE

The numerical program is not designed to re-establish the premises already supplied by the Bhuiyan–Pan–Jian (BPJ) construction [1]. BPJ already provides the mEO/sTM setting, a postselection-free finite-range class-A adaptive instrument, Chern trajectory ensembles, programmable dynamical domain walls, finite-range parallelization, an ancilla–fSWAP implementation route, and quantum-simulation prospects on encoded-qubit or native-fermion hardware. It also reports coherent-noise robustness and distinct trajectory and averaged thresholds. Accordingly, this manuscript shall not advertise postselection freedom, ancilla feedback, Gaussian simulability, finite-range state preparation, or the mere existence of a domain-wall slow mode as its standalone novelty.

The proposed advance is the conjunction of results absent from that baseline. The perfect-correction instrument should program a *universal chiral defect* across typical retained Born trajectories: each trajectory may occupy a different definite system-charge sector, but the ensemble shares $c_{\text{edge}} = 1$ and level-one $U(1)$ wall data, its sector wandering is localized at the walls, its response and flux flow are oriented, and its slow purification is controlled by a singular conditional tangent sector. The strongest version further recovers the defect data from the classical record and shows precisely what survives or disappears in the unconditional channel. This is the unique intersection to which every headline panel must contribute.

The quantum-simulation proposition is therefore operational rather than computational. It is not a quantum-advantage claim, nor a claim that an adaptive platform can prepare a Chern state for the first time. It is a protocol for realizing and interrogating a programmable topological defect ensemble without rejecting records, with a hierarchy of record-only, number-measurement, covariance-tomography, and replica observables. A result that cannot be placed in this hierarchy or tied to one of Gates G1–G5 is supplemental or deferred even if it is numerically interesting.

III. CANONICAL PROTOCOL AND FIXED PRODUCTION CHOICES

Every GPU script or notebook that simulates the circuit shall configure and call `classA_U1FGTN_gpu.run_markov_circuit`. It shall not reproduce the cycle loop or call private local-update helpers. New dynamics functionality belongs in the canonical class. Every output manifest records this entry point explicitly.

The base physical protocol is

$$n_{\text{shell}} = 1, \quad \alpha_{\text{top}} = 1, \quad \alpha_{\text{triv}} = 30, \quad \text{perfect_correction}=\text{True}, \quad \text{postselect}=\text{False}. \quad (3)$$

The production implementation is the ordered serial instrument implemented by the canonical dynamics engine. A fresh random permutation of the active centers is drawn for each trajectory and cycle and stored in metadata; the four controller channels at a center are applied in the fixed declared engine order. Raster and reverse-raster comparisons are not part of the production campaign. Campaign V0 validates exact generation and replay of the declared random-serial schedule; historical raster data remain legacy comparisons only. The candidate quasi-one-dimensional sequence is

$$N_x = 20, \quad N_y \in \{24, 32, 40, 48, 64, 80\}, \quad (4)$$

but $N_x = 20$ is not accepted until Campaign W1 passes. Every stochastic parameter point uses a base batch of $S = 100$ independently initialized trajectories. There is no separate physical burn-in. Unless a campaign explicitly defines a shorter deterministic validation, each stochastic trajectory runs for a total of $2N_y$ complete cycles from its independent initial state. If a predeclared confidence interval is too broad to decide its gate, launch a second complete $S = 100$ batch for *every* member of the affected comparison family, with a new root seed. Never add samples only to an inconvenient point. The first and second batches remain identified in metadata and are combined only after a heterogeneity check.

All topology, tangent, twist, and fit inputs use complex128 or float64. Figure defaults are 3.375 inches per single-column panel, 300 dpi, CMU Sans Serif, and at least 8 pt text.

IV. QUANTUM-SIMULATION IMPLEMENTATION AND OPERATIONAL OBSERVABLES

The primary model is the fresh-ancilla, perfect-correction idealization of the BPJ adaptive circuit. For a controller mode χ_j with target occupation $s_j \in \{0, 1\}$, measure n_{χ_j} and retain the Born outcome m_j . If $m_j \neq s_j$, prepare a fresh local ancilla mode in occupation s_j , fSWAP it with χ_j , and trace out or reset the outgoing ancilla. Thus the system charge transfer is

$$Y_j = s_j - m_j, \quad (5)$$

and every outcome is a valid retained run. No success flag, trajectory rejection, or reweighting is permitted in the primary data. The older dynamical-bath implementation, where correction succeeds only when a reused ancilla happens to have the needed occupation, is an imperfect-feedback control rather than the definition of perfect correction.

For $n_{\text{shell}} = 1$, an instrument block may be compiled by a number-conserving Gaussian basis rotation on the 3×3 -cell patch that maps χ_j to a readout mode, followed by occupation readout, classical feedforward, target-occupation ancilla preparation, fSWAP, and the required inverse basis rotation. The hardware report shall give the two-mode-gate count and depth of this compilation as d_{inst} , the measurement and feedforward latency, the number of concurrently live ancillas, reset fidelity, and the colored-sweep bound in Eq. (E6). The same ancilla pool can be recycled between color/channel blocks. Fermionic locality does not automatically imply qubit locality in two dimensions: a qubit implementation must state its Jordan–Wigner, Bravyi–Kitaev, swap-network, or auxiliary-gauge encoding and include that mapping overhead. A native fermionic processor may use the fermionic compilation directly.

Because controllers within each conflict-graph color have disjoint support, their local instruments commute and admit simultaneous execution on a quantum simulator. A numerical evaluation of such a color block may nevertheless apply its instruments sequentially, which samples the identical joint Born law while avoiding a separate correlated multi-mode sampler. This optional finite-color compilation is a quantum-simulation resource advantage, not a requirement of the production numerics. Reordering noncommuting color blocks relative to the production random-serial word would define a different microscopic schedule and is not claimed to leave finite-size trajectories unchanged.

The operational hierarchy is:

1. the measurement/correction record, activity, transfer FCS, and total system-charge history require no state tomography;
2. boundary charge profiles, Eq. (27), and the real-time response require repeated occupation measurements and local source control;
3. the one-body covariance, local marker, and occupation spectrum require a correlation-matrix tomography protocol;
4. Rényi entropies require replicas, randomized measurements, or an equivalent interferometric protocol, while modular-Hamiltonian evolution may remain a trajectory-reconstruction diagnostic rather than a direct hardware primitive.

The simulation package shall attach shot-complexity and calibration estimates to each observable proposed as operational. Since Gaussian circuits are classically simulable, the manuscript does not claim computational quantum advantage. The quantum simulation value is controlled realization of a programmable topological defect and access to typical conditioned states and their associated trajectory records, without conditioning on an exponentially unlikely favorable record.

This implementation inherits rather than supersedes the BPJ hardware proposal. Its concrete new simplification is that a fresh ancilla is prepared in the target occupation s_j for each wrong branch, so the correction is deterministic once the Born outcome is known; no ancillary redistribution layer or finite bath-occupation success probability is part of the default model. Production resource tables must compare this fresh-ancilla cost with the reused-bath BPJ primitive instead of presenting the whole measurement-fSWAP idea as new.

The ensemble language must also remain precise. Starting from a definite total charge, each conditioned trajectory lies in a definite system-charge sector after every recorded gain/loss event, while the ensemble spans different sectors. The intended result is therefore a Born-sampled ensemble of boundary states with shared infrared data, not one deterministically prepared ket and not an individual charge-indefinite pure state. Campaign S1 must demonstrate that the trajectory-to-trajectory charge wandering is localized at the walls and separately extract the within-trajectory current-algebra coefficient.

V. REPRODUCIBILITY, SHARDING, AND OUTPUT SCHEMA

Each campaign has an immutable JSON manifest containing a schema version, git commit, canonical entry point, environment versions, device, dtype, complete geometry and protocol, observer definitions, root seed, global sample-index allocation, shard RNG seed, and analysis version. The present GPU engine uses one exact generator stream per immutable five-trajectory shard and checkpoints its pre-run and post-run states. It does not claim partition-invariant counter-based randomness: changing the shard partition defines a new versioned ensemble rather than an exact rearrangement of the same 100 random words.

A production directory contains

$$\text{manifest.json, shards/, merged/, analysis/, figures/, logs/}. \quad (6)$$

Sample indices are assigned to shards before execution. The merger rejects duplicate or missing indices and any manifest hash mismatch. Atomic permanent checkpoints are written after each observation block and contain the record cursor, RNG state, accumulated $-\log p$, compact observer accumulators, and, when present, tangent QR log sums and declared low-subspace products. A covariance can exist only in live memory or local scratch during the active stage; it is excluded from the permanent checkpoint schema. Replaying any immutable shard from its saved pre-run RNG state and frozen word must agree within declared tolerances. Arbitrary repartitioning of the current stateful RNG stream is not an exact-reproducibility operation.

Required raw products include per-trajectory scalars, not only ensemble averages; bit-packed replay records for every stochastic family; conditional log probabilities or lossless per-cycle sums; numerical residuals; censor/null masks; and a completion table. Covariances are evaluated on the fly. A parent covariance needed by H1-H3 is a transient shard-local input, not a permanent ensemble product: H1/H2 consume it during the live parent process, while H3 reconstructs it from the saved initial RNG state, ordered word, outcomes, and engine hash. Analysis writes figure-ready CSV/NPZ files and a fit JSON containing the input data hash, fit window, covariance model, bootstrap distribution, and sensitivity fits. Plotting scripts do not perform unrecorded fits.

VI. MASTER TRAJECTORY FAMILIES AND REUSE RULE

A new independent trajectory family is required only when at least one component of the following key changes:

$$\mathfrak{K} = (N_x, N_y, \text{controller construction and parameters, initial ensemble, serial schedule law, feedback/noise law, } n_{\text{shell}}). \quad (7)$$

Changing the observer, fit, source position, analysis window, or deterministic replay does not define a new Born family. All campaigns sharing $\mathfrak{K}$ must consume one common set of independently seeded trajectories rather than launch campaign-specific copies. One trajectory remains the independent sampling unit even when many cycle snapshots, periodic origins, subsystem positions, sources, or twists are derived from it.

For the primary pure-state wall and its matched wall-off control, run one master family per accepted size for exactly $2N_y$ complete cycles from an independently sampled product state. No physical cycles are discarded as burn-in. The same family supplies S1, T1, B2, the late-time states for H1, the unperturbed parents and time origins for H2, the

frozen records for H3, R1, and the Born comparison in M1. Response continuations in H2 and twist replays in H3 are derived jobs attached to a parent trajectory ID; they are not new independent samples.

For max-mix initialization, run one separate master family per wall/control geometry and size from $t = 0$ through exactly $2N_y$. It supplies P2 and the max-mix portion of S1. Uniform bulk, secondary-wall, changed- α , changed-noise, imperfect-feedback, postselected, changed- n_{shell} , and changed serial-word controls have different keys and therefore cannot be manufactured from a baseline trajectory. B0 and the B1 pilot are deterministic or small-pilot calculations; H1 modular evolution, H3 fixed-record flux evolution, and the exact mean maps in M1 are deterministic descendants of saved states or records.

Each pure-state master trajectory saves:

1. its sample seed, initialization hash, controller hash, complete ordered site/channel word, bit-packed outcomes, per-event conditional log probability, X_j , Y_j , and the system charge after every event or a losslessly reconstructible block representation;
2. compact observables computed at the declared transient-diagnostic, tangent-alignment, three late-time S1, H2-source, and final checkpoints; transient parent covariances are consumed shard-locally by H1/H2 and then discarded, while H3 regenerates them recordwise and the permanent archive retains the initial RNG state, ordered record, engine hash, checkpoint cursor, and replay hashes needed to regenerate them;
3. per trajectory, snapshot, and Rényi index, the y_0 -averaged full-strip entropy curve, selected y_0 -resolved validation curves, unnormalized wall-window contour sums, subsystem spectra, charge-susceptibility spectra $q_a = \nu_a(1 - \nu_a)$, matched entropy-variance residual curves, interval-charge cumulants, density and charge profiles, structure factors, the full fixed-slice array $C_{G,\xi}(x, r_y)$, its wall/bulk derived curves, topology and purity diagnostics, and the simultaneous $N_{\mathcal{D},\xi}$ needed for sector-resolved fits;
4. tangent QR log increments, null masks, aligned low-subspace projectors and densities, block- R factors at declared checkpoints, and the initial frame seed; and
5. observer residuals, censor masks, completion status, and enough accumulated score data or replay information to reconstruct the H2 conditional and score terms.

The max-mix master additionally saves every early-cycle scalar, logarithmically spaced late compact contours and occupation-side bridge spectra, the complete start-to-time record, and the reduced Choi/tangent objects required to test Eq. (43) recordwise at cycles 1, 2, 4, 8, 16, N_y , $3N_y/2$, $2N_y$ (with duplicates removed). The bridge observer receives C_t and the canonical reduced-Choi cross block in the same callback, then saves Gram-matrix, eigenvalue, and entropy residuals before either dense object is discarded. Scan-only families save the corresponding per-trajectory fitted scalars and compact profiles; they must not discard the sample axis or save only ensemble means. Neither family archives sample covariance matrices.

A. Minimum gate-first execution matrix

The minimum below assumes that B0 accepts $N_x^* = 20$, one base batch of $S = 100$ resolves every gate, the explicit interface is the primary wall, and the support-terminated construction is the secondary wall. It is a count of independently sampled trajectory families, not deterministic exact calculations, observer passes, response continuations, or fixed-record replays. The fixed core is

family block	$S = 100$ families trajectories	
P1 lean bulk baseline	20	2000
primary pure wall and matched control, six sizes	12	1200
primary max-mix wall and matched control, six sizes	12	1200
secondary pure wall and matched control, three sizes	6	600
W1 transverse-convergence additions	18	1800
core through G1–G4	68	6800
R1 extra wall/control families after the $L = 20$ reuse	5	500
core through G1–G5	73	7300
S2 nonbaseline α_{in} points	32	3200
fixed campaign including S2	105	10500

Here P1 uses pure topological and trivial states at all six square sizes, max-mix states only at $L = 20, 28$, and the $n_{\text{shell}} = 2$ check only for the pure topological and trivial states at $L = 20, 28$, giving $12 + 4 + 4 = 20$ families. The W1 count uses four protocols—two walls and their separate matched controls—at $N_x = 20, 24, 28$ and $N_y = 48, 80$. Of these 24 points, six are already master points, so only 18 are new. The S2 count is $9 \times 4 = 36$ points minus the four baseline $\alpha_{\text{in}} = 1$ families already present. R1 adds the primary wall and control at $N_y = 16, 20, 28$, except that the $N_x = N_y = 20$ matched all-trivial control reuses the pure $L = 20$ P1 trivial family by extending its saved stationary record. This reuse is allowed only after their controller, active-channel, schedule-law, and initialization hashes agree; otherwise the count increases by one.

All other fixed campaigns reuse those samples. B0 and the mean maps in M1 are deterministic. T1 is an observer of the live pure master record. H1 consumes live stationary states, H2 attaches response continuations to live parent origins, and H3 regenerates a parent from its pre-run RNG state and replays the saved record around the twist circle. B2 consumes the three common secondary-wall sizes. Raster/reverse-raster production families are omitted. The B1 train/test pilot and V0 exact checks are small validation jobs and are reported separately from the $S = 100$ production count.

The open perturbation grids add

$$N_{100} = 105 + 3n_{\text{imp}} + 17n_{\text{noise}}, \quad (8)$$

where n_{imp} is the number of nonperfect stochastic feedback points in M2 and n_{noise} is the number of nonzero noise values in M3. The deterministic postselected branch adds no stochastic family. Each added M3 noise value contributes five uniform square sizes and two wall geometries at six circumferences. Equation (8) excludes any escalation triggered by a failed gate. A second base batch adds one further $S = 100$ family for every member of the affected comparison block, never only for one selected point.

This minimum is valid only if every master run writes the complete save contract above. In particular the manifest and record must be sufficient to replay the exact ordered instrument; the three stationary live states must be reduced immediately to the declared state, modular, twist, and correlator products; the event stream must retain conditional log probabilities and signed charge transfers for B2 and R1; the tangent QR stream must span the full T1 window; and H2 must retain its declared parent snapshots, score accumulators, and source-time metadata. If one of these products is omitted, reconstructing it later may require a new independent family and invalidates the minimum count.

VII. VALIDATION LADDER

A. Campaign V0: engine and observer validation

a. Claim and status. Status PLANNED. The executable A100 gate checks exact one-cycle branch normalization, frozen-word replay, conditional log likelihoods, diagnostic-only observers, CPU-device versus GPU-device arithmetic for one common initial state and word, tangent-QR orthonormality, and the zero-covariance archive contract. It checks deterministic replay of each fixed shard; it does not assert bitwise invariance after repartitioning RNG streams.

b. Dependencies and run matrix. None. The executable bundle first uses a 1×2 dense system, two cycles and two records; it enumerates all $2^8 = 256$ Born words for one cycle. Larger 4×6 and 6×8 CPU-engine comparisons and centered finite-difference tangent tests remain repository regression tests rather than Drive production products.

c. Initialization, duration, and observation. No stochastic preparation interval is needed. Compare every elementary update and every cycle boundary.

d. Raw saves and estimator. Save branch probabilities, records, compact spectra, tangent QR actions, and RNG states, but no covariance matrices. Every CPU/GPU comparison passes an explicit common wall mask, `DW_loc`, active-index list, and controller-projector hash; it never relies on engine defaults, because the current CPU and GPU defaults choose different wall fractions. Enumerate all records on the smallest system and check $\sum_{\xi} p_{\xi} = 1$. The production gate checks finite QR factors and stabilized-frame orthonormality; the separate repository regression suite retains the stronger centered finite-difference test.

e. Controls, statistics, and gate. Require replay agreement, branch normalization, CPU-device–GPU-device agreement for a frozen word, observer neutrality, and tangent-QR residuals at float64 tolerances. Exact archive checksums provide restart recovery. Partition-invariant random streams and cycle-level GPU resume are not claimed by this build; the atomic unit is a complete five-trajectory shard. The H3 gate separately enforces 2π twist closure. Tolerances must be written in the manifest and justified from float64 roundoff. Any failure blocks all production.

f. Downgrade, figure, sharding, and legacy. There is no scientific downgrade and no main-text figure. No legacy comparison is needed.

B. Campaign V1: deferred serial-order sensitivity study

a. Claim and status. Status deferred. The production model is the random serial ordered instrument. Exact generation and replay of that schedule are validated in V0. A raster/reverse-raster physics comparison is not required for any manuscript gate and consumes no production GPU allocation.

b. Dependencies and run matrix. No production matrix is assigned. Existing small-system raster and reverse-raster artifacts may be retained as legacy diagnostics, but no $S = 100$ ordered-word family is launched. Any future schedule-sensitivity study requires an explicit amendment to this contract.

c. Initialization, duration, and observation. Not applicable to the production campaign.

d. Raw saves and estimator. V0 saves every cycle's ordered site list and seed, per-channel update counts, physical instrument-block count, and replay diagnostics. It verifies one-and-only-one visit to every active center and channel, periodic wrapping, exact replay, CPU-GPU agreement, and checkpoint restart for the production random-serial law.

e. Controls, statistics, and gate. There is no gate. The production claim is explicitly conditional on the declared random-serial microscopic schedule.

f. Downgrade, figure, sharding, and legacy. No new figure or shard family is produced. Existing raster results remain legacy comparisons and are not relabelled as random-serial data.

VIII. EXACT DOMAIN-WALL CALIBRATION

A. Campaign B0: deterministic analysis-pipeline benchmark

a. Claim and status. Status COMPLETED REVIEW CALIBRATION. The common analysis pipeline recovers the known chiral wall behavior of an exact class-A domain-wall Hamiltonian. This calibrates geometry and fit windows; it is not evidence about the adaptive circuit.

b. Dependencies and run matrix. Depends on V0. Use `classA.U1FGTN.domain_wall_hamiltonian` and `G.CI_domain_wall`. At $N_y = 48$, scan $N_x = 12, 16, 20, 24, 28, 32$. At the selected width scan the sequence in Eq. (4). Run both the exact coupled spatial-mass interface and `triv_region_local_mode`, a decoupled-exterior hard analogue. Neither is assumed identical to controller `dw.truncation`.

c. Initialization, duration, and observation. There is no stochastic preparation interval. Enforce exact half filling. Resolve zero-energy crossings with a small boundary twist and continuous overlap tracking; test both twist signs and the zero-twist limit. Never fill states solely by $E < -\epsilon$.

d. Raw saves and estimator. Save Hamiltonians or their reproducible construction metadata, low spectra and vectors, wall weights, entanglement spectra, the complete fixed- x correlator tensor for every x and $1 \leq r_y \leq \lfloor N_y/2 \rfloor$, modular responses, density responses, and twist-overlap matrices. Recompute the wall, exterior-neighbor, topological-bulk, trivial-bulk, wall-window, and full- x correlation curves and their fit tables at every benchmark size; no archived curve or fitted exponent is imported. Apply without alteration the estimators later used in S1 and the three chirality campaigns.

e. Controls, statistics, and gate. Extract edge velocity, transverse localization length, finite-width hybridization, $c_{\text{edge}} = 1$ entropy coefficients in the one- and two-wall geometries, an r^{-2} squared correlator, opposite modular/physical velocities, and unit twist flow. Select N_x only when the hybridization mass satisfies

$$m(w) \ll \frac{2\pi v}{N_y}. \quad (9)$$

Failure means the analysis pipeline or geometry must be repaired before stochastic runs.

f. Downgrade, figure, sharding, and legacy. The calibration becomes Fig. S1, not a primary result. Deterministic sizes may be sharded independently. Path-audited visual references are collected under `exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/`. They are not production inputs.

g. Current audited campaign. The authoritative version-2 calculation is campaign `experiment_review/b0_exact_domain_wall/results/20260816_191957/`; all entries in its machine-readable acceptance ledger pass. It preserves but supersedes version-1 campaign `20260816_184251` for interpretation. The first run used a normalized packet at the center of the retained half-cylinder and fitted an absolute center of mass. That estimator mixed oppositely directed modular components and returned small same-sign slopes. It also selected $N_x = 12$ from the hybridization mass alone, even though the coupled local correlator and corrected modular response remain underconverged at $N_x = 12, 16$. The discrepancy is classified as an estimator mismatch together with a finite-width convergence effect, not as a failure of modular charge spreading.

Version 2 retains the mass gate but requires every single-geometry calibration to pass at both the accepted width and the next larger scanned width. This selects $N_x^* = 20$ because $N_x = 20$ and 24 pass whereas 12 and 16 fail the

TABLE II. Audited version-2 B0 results at $N_x = 20$, $N_y = 48$. The complete campaign also contains the accepted-width sequence $N_y = 24, 32, 40, 48, 64, 80$, all raw packet fields, and the 33/65/129-point twist controls.

construction	representative values	audited implication
Coupled spatial-mass interface	$c_1 = 1.0001$, $\beta = 2.07447$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.93045, +4.53563)$	The three-column modular source retains at least 0.3245 charge in the primary fit window, all sensitivity signs agree, and the norm drift is 3.3×10^{-15} . The wall twist flows are $(-1, +1)$.
Hard-exterior analogue	$c_1 = 1.0001$, $\beta = 2.07437$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.59985, +4.40578)$	The minimum modular retention is 0.3349, all sensitivity signs agree, and the norm drift is 4.0×10^{-15} . The wall twist flows are again $(-1, +1)$.
Legacy-protocol reproduction at 20×40	Aggregate initial charges 8 and 24; $0 \leq t \leq 40$ with $\Delta t = 0.05$	The complete aggregate $N_{xy}(t)$ fields are saved with norm drift 1.4×10^{-14} , while the separate-source data make handedness identifiable.

TABLE III. Post-campaign transverse-width plateau audit from $N_x = 20$ to 28 at $N_y = 48$. Entries are $100|O(28)-O(20)|/|O(20)|$. “Numerical zero” denotes changes below $10^{-6}\%$. Modular slopes are quoted only as diagnostics in modular-time units.

observable	coupled interface	hard exterior
spectral $ v $	numerical zero	numerical zero
transverse localization length $\xi_\perp$	0.147%	numerical zero
von Neumann coefficient c_1	0.00044%	0.00044%
Rényi coefficient c_2	0.000085%	0.000049%
Rényi coefficient c_3	0.00324%	0.000010%
wall correlator exponent β	0.00910%	numerical zero
physical-response $ v $	numerical zero	numerical zero
modular-slope magnitude, lower wall	0.684%	0.536%
modular-slope magnitude, upper wall	0.0675%	0.0470%

coupled local checks. For modular flow it evolves the $y = 0$ and $y = N_y/2 - 1$ sources separately at each physical wall and fits

$$D_w(t) = \frac{1}{2} [\bar{y}_{w,0}(t) + \bar{y}_{w,L-1}(t) - (L-1)], \quad L = N_y/2, \quad (10)$$

on $0.1 \leq t \leq 2.0$. One- and three-column sources and three alternate time windows give the same sign. Table II records the accepted-width values; modular slopes use modular-time units and are not compared in magnitude with physical velocities.

h. Transverse-width plateau audit. The version-2 selector above did not preregister a successive-width percentage-change threshold: it required absolute single-geometry gates to pass at $N_x = 20$ and at the next width $N_x = 24$. Thus its recorded choice $N_x^* = 20$ must not be described as the output of a percentage-plateau test. To expose the distinction, Table III reports the direct fractional change from $N_x = 20$ to 28 at fixed $N_y = 48$, normalized to the $N_x = 20$ value. Every listed continuous observable changes by less than one percent; the largest change is the coupled lower-wall modular slope, 0.684%. The signed twist flows remain exactly $(-1, +1)$ in both constructions.

A post hoc one-percent criterion applied only to the direct $20 \rightarrow 28$ comparison would therefore accept $N_x = 20$. A stricter criterion requiring each adjacent increment to be below one percent, and treating the nonuniversal modular-slope magnitude as a convergence observable, would not: the lower-wall modular magnitude changes by 1.27% and 1.15% from $N_x = 20$ to 24 in the coupled and hard-exterior constructions, respectively. The corresponding $24 \rightarrow 28$ changes are 0.58% and 0.61%, and all listed $28 \rightarrow 32$ changes are below 0.063%. Under that stricter definition the smallest supported width would be $N_x = 24$; under a two-percent adjacent-width threshold, or when modular flow is tested only for its preregistered stable sign, $N_x = 20$ remains supported. Relative changes of the hybridization ratio and contour factor-of-two residual are omitted from the table because both are already near numerical-zero floors; their relative percentages are not stable measures of physical convergence. The production choice must therefore state the percentage threshold and observable set before invoking marginal gains.

i. Legacy comparison. Assessment: promising and already a substantial single-geometry calibration, but not a B0 pass. A repository-wide audit finds quantitative exact-state entropy and correlator fits for the coupled interface, corresponding fit tables for `triv_region_local_mode`, and raw exact modular-evolution arrays. The strongest path-audited results are summarized in Table IV. Here m_{full} is the two-wall full-width von Neumann log-chord slope, m_{wall} is the unnormalized single-wall contour-window slope, and β is defined by $C_G \sim \ell_y^{-\beta}$.

TABLE IV. Legacy B0 evidence. These deterministic results motivate the production fit windows but are not pooled with the preregistered campaign.

artifact and coverage	strongest quantitative indication	implication and limitation
Coupled spatial-mass interface in the exact-DW material under <code>notebooks/</code> and <code>exact_DW_benchmark_notes/</code> , 20×40 , two walls at $x = 6, 14$	$m_{\text{full}} = 0.343924$ ($3m_{\text{full}} = 1.03177$, $R^2 = 0.999080$); $m_{\text{wall}} = 0.168600$ – 0.168642 ($R^2 \simeq 0.999875$); $\beta_{\text{wall}} = 2.024641$ ($R^2 = 0.999835$) on $r_y = 3, \dots, 15$. The saved spectrum shows two low-energy crossing branches, while the marker and modular current maps are wall localized.	Strong evidence that the existing geometry and estimators recover the expected $c = 1$ and r^{-2} structure at one size. It is a two-wall calculation only and has no finite-width or longitudinal-size extrapolation.
Exact fit tables under <code>notebooks/flattened_hamiltonian_analysis/data/</code> , 20×40	For <code>triv_region_local_mode</code> , $m_{\text{full}} = 0.343923$, $m_{\text{wall}} = 0.168600$ – 0.168642 , and $\beta_{\text{wall}} = 2.024671$ with $R^2 = 0.999835$. The coupled and hard-exterior values differ by less than 2×10^{-6} in the entropy slope and 3.1×10^{-5} in β .	Very promising coupled-versus-hard agreement, but only for the same 20×40 geometry. The table preserves fits rather than Hamiltonians, eigenvectors, or complete spectral data.
Current-normalization correlation archive in <code>colab_small_system_testing/analysis_outputs/pure_state_square_correlations_cpu/</code> , 16×30 and 16×40	Complete scalar $C_G(x, r_y)$ curves are saved for every x and $0 \leq r_y \leq N_y/2$. The wall exponents are respectively $\beta = 2.042461$ and 2.040144 , with $R^2 = 0.999742$ and 0.999753 on $r_y = 3, \dots, 10$; the full- x averages give $\beta = 2.11628$ and 2.10899 .	Confirms modest N_y stability using the normalization of Eq. (24) and supplies reusable raw curves and fit rows. It still lacks the B0 N_x scan and the preregistered $r_y = 3, \dots, 15$ model comparison.
Exact modular data under <code>notebooks/modular_charge_spreading/exact_dw_gs/</code> and static-map material under <code>exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/</code> , 20×40	The static J_y map has opposite signs at the two interfaces. Two saved exact evolutions contain the full $N_{xy}(t)$ array for $0 \leq t \leq 40$ with total-charge drift below 4.4×10^{-7} .	Provides raw material for a modular-drift calibration, but no wall velocity was fitted; there is no corresponding physical density-response velocity or orientation-reversal control.

The principal defect is the zero-mode prescription. The archived periodic spectrum has two modes at numerical zero, whereas `G_CI_domain_wall` occupies only $E < -\text{tol}$. Reconstructing the archived 20×40 , $x = 6, \dots, 14$ geometry with that audited rule selects 799 modes for the coupled interface rather than the required 800. For the hard-exterior Hamiltonian it selects 1239 modes because the decoupled exterior orbitals lie at energy -1 . Thus the excellent legacy slopes do not yet constitute the exact-half-filled, twist-tracked benchmark demanded above. The artifacts also predate the present domain-wall convention: they use walls at $x = 6, 14$, and some notebooks use the historical reversed `alpha_1/alpha_2` naming, whereas the current source uses a wider slab and explicit topological/trivial aliases.

At the time of the legacy audit, no completed artifact was found for the required $N_y = 48$ scan over $N_x = 12, 16, 20, 24, 28, 32$, the accepted-width scan over Eq. (4), a one-wall entropy geometry, edge velocity, transverse localization length, hybridization mass, physical density response, twist-overlap tracking, or unit twist flow. An exact entropy size-sweep driver exists under `scripts/`, but its declared output is absent and it saves no numerical fit table, so it is not counted as a completed legacy run.

B. Pilot B1: signed controller-frame incompatibility diagnostic

a. Claim and status. Status PLANNED exploratory pilot. This calculation asks only whether a static fixed-charge incompatibility diagnostic has enough quantitative predictive value to merit later expansion. It is not a manuscript gate, a model of the ordered trajectory dynamics, or evidence for topology, a wall CFT, or slow conditioned dynamics.

b. Formal object. For controller targets $s_j \in \{0, 1\}$ and normalized rank-one projectors P_j , define

$$H_{\text{frame}} = \sum_j (1 - 2s_j) P_j. \quad (11)$$

The static target cost over rank- N Slater projectors is

$$\mathcal{F}[C] = \sum_j s_j + \text{Tr}(H_{\text{frame}} C), \quad C^2 = C, \quad \text{Tr } C = N. \quad (12)$$

Ky Fan minimization therefore gives

$$C_\star = \sum_{a=1}^N |u_a\rangle\langle u_a|, \quad F_\star = \sum_j s_j + \sum_{a=1}^N \epsilon_a^\dagger(H_{\text{frame}}). \quad (13)$$

This evaluates all constraints on one common covariance. Perfect correction instead changes both the covariance and, on wrong branches, the system charge after each ordered event. Consequently $F_\star(N)$ is neither a lower bound nor a dynamical variational principle for the realized activity. The object called the flattened Hamiltonian in the legacy notebooks is therefore used here only as a signed controller-frame diagnostic; its spectrum is not generally ± 1 , and it is neither the conditioned tangent operator nor the mean channel.

c. Dependencies and run matrix. Depends on V0. Use $N_x = 20$, $N_y = 24, 40$, and $n_{\text{shell}} = 1$ for the two wall constructions and their geometry-matched wall-off controls. Evaluate nominal half filling and the charge sectors encountered in a four-trajectory stationary pilot. Do not launch a width, support, or parameter sweep at this stage.

d. Raw saves and estimator. Save the frame hash, spectrum of H_{frame} , $F_\star(N)$, selection gaps, and the charge-resolved transverse residual profiles. Verify Eqs. (12) and (13) by direct residual summation. Compare the residual profiles, without shifting or refitting their spatial origins, with the wrong-outcome activity measured on a disjoint four-trajectory test set. Do not run the CFT, topology, twist, or chirality pipeline on $C_{\star,N}$.

e. Controls, statistics, and gate. Nonzero incompatibility is generic for nonorthogonal or overcomplete constraints and is not called topological frustration. If a selection gap closes, report the allowed degenerate-subspace variation rather than choosing a favorable minimizer. The pilot is expanded only if its fixed, charge-resolved residual predicts more than generic wall colocalization: require an out-of-sample profile overlap and wall-to-bulk contrast that both improve substantially over the matched wall-off profile.

f. Downgrade, figure, sharding, and legacy. There is no main-text figure and no failure consequence. Unless the pilot passes its expansion criterion, stop the static analysis and mention at most the defining algebra in the Supplement. Available legacy inputs are under `notebooks/flattened_hamiltonian_analysis/` and `topological_frustration_diagnostics/docs/`. They guide sizes and estimator debugging only. Legacy postselected Frobenius convergence does not validate the Born-sampled perfect-correction circuit.

C. Campaign B2: feedback activity and slow dynamics

a. Claim and status. Status PLANNED. This campaign tests the dynamical association

$$\text{Born feedback activity} \longleftrightarrow \text{wall-localized slow tangent and purification sectors}, \quad (14)$$

rather than treating several wall-centered plots as independent observations. Pilot B1 is an optional comparison and is not required for this campaign.

b. Exact branch relation. Immediately before channel j , the covariance is C_{j-} and the wrong-outcome indicator and signed system transfer are

$$X_j = |m_j - s_j|, \quad Y_j = s_j - m_j. \quad (15)$$

For the perfect-correction primitive,

$$\mathbb{E}[X_j | C_{j-}] = s_j + (1 - 2s_j) \text{Tr}(P_j C_{j-}), \quad |Y_j| = X_j \quad (16)$$

event by event. This identity, evaluated on the actual pre-event covariance, is the primary microscopic statement. No simultaneous-frame cost or static variational bound is assumed for the ordered sweep.

c. Dependencies and run matrix. Depends on T1 and P2. Reuse the production serial schedule and controller hashes at the common secondary-wall sizes $N_y = 32, 48, 64$, $S = 100$, for both wall constructions and their separate geometry- and update-count-matched wall-off controllers. These are the S1/W1 master families, so B2 launches no independent production trajectories. P1 supplies the uniform bulk reference. Changed n_{shell} , support, or slab width is an escalation only if the primary wall-versus-control association is unresolved; such an escalation receives its own preregistered grid and is not part of the minimum count.

d. Raw saves and estimator. Save every X_j , Y_j , controller label, position, pre-branch probability, schedule position, and observation-block index. Measure stationary signed and absolute activity profiles, wall/interior rates, inter-event correlations, and the joint fluctuations of activity, total system charge, tangent gap, and residual max-mix entropy. Compare the activity profile directly with the globally normalized low-tangent-subspace density and purification residual using profile overlap, Spearman rank correlation, and independently fitted localization lengths. Condition

the ordered within-cycle record on its instantaneous N_{j-} where useful. A B1 residual may be shown as an explicitly exploratory overlay only if its independent pilot criterion passed. Correlations across different times from one trajectory do not increase the trajectory sample count.

The empirical activity function

$$\theta_R(s, T) = T^{-1} \log \mathbb{E}_\xi [e^{-s \sum_{j \in R, T} X_j}] \quad (17)$$

may be reported with log-sum-exp stabilization and reweighting effective sample size. It is a finite-time annealed record statistic, not the replica-calibrated defect self-information in R1, not a trajectory-resolved CFT partition function, and not a tilted-channel eigenvalue unless the tilted operator is explicitly constructed and its long-time limit is converged.

e. Controls, statistics, and gate. Validate Eq. (16) branch by branch and the mean profile against the M1 deterministic covariance prediction. Require a stationary interface excess over the matched active-channel control. To support a dynamical association, the same parameter sweep must link increased activity to the closing tangent sector and delayed purification with a preregistered monotone or scaling relation. Wall colocalization alone does not pass, and causation is not inferred from correlation.

f. Downgrade, figure, sharding, and legacy. If only Eq. (16) and a wall excess pass, report boundary-localized controller work without claiming that activity causes the CFT or tangent closing. The result is supplemental unless it quantitatively explains a Fig. 5 trend. Legacy parameter guidance is available in `topological_frustration_diagnostics/results/campaigns/20260813\_121340\_Nx20\_perfect\_Ny20\_22\_24/`; its $N_x = 20$, $N_y = 20, 22, 24$ perfect-correction activity and spectral runs use only 25 trajectories, raster order, and short size-dependent times, so they are not production evidence. The corresponding formal reference is under `topological_frustration_diagnostics/docs/`. Local click words and regularized-Choi endpoint spectra are not part of B2.

g. Current legacy result. Assessment: a quantitatively promising precursor, but not a B2 pass. The completed CPU campaign at the path above used the canonical `classA_U1FGTN.run_markov_circuit` entry point for a hard-truncated wall and a uniform reference at $N_x = 20$, $N_y = 20, 22, 24$, with 25 trajectories per geometry, raster ordering, a burn-in of N_y cycles, and a total duration of $2N_y$ cycles. The saved records verify $|Y_j| = X_j$ in every run. Across the three sizes, the normalized all-channel wrong-outcome rate is 0.01691–0.01710 for the wall geometry and 0.00353–0.00382 for the uniform reference. Within the wall geometry the interface rate is 0.04698–0.04721, compared with 0.00355–0.00374 in the interior. The mean final Lyapunov gap is simultaneously much smaller for the wall, 0.1014–0.1125, than for the uniform reference, 0.8936–0.8964, and the associated wall slow mode has 0.9922–0.9958 of its weight in the interface region. Table V records what this calculation establishes and what remains outside its scope.

These results show boundary-localized controller work and a boundary-localized slow one-body sector in the same finite-size range. They do not yet establish the B2 association in Eq. (14): activity and spectra were collected in separate trajectory suites, the uniform reference is not a geometry- and update-count-matched active controller, and no purification residual or joint-fluctuation observable was saved. Thus their coexistence is suggestive rather than a measured trajectory-level or scaling relation.

IX. TOPOLOGY AND WALL CONSTRUCTION

A. Campaign P1: quantitative revalidation of the BPJ bulk baseline

a. Claim and status. Status PLANNED. The fresh-ancilla $n_{\text{shell}} = 1$ realization quantitatively reproduces the postselection-free Chern-one bulk phase inherited from BPJ. This licenses the platform used by the wall campaigns; it is not the new universal result.

b. Dependencies and run matrix. Depends on V0. Use uniform periodic $L \times L$ systems with $L = 12, 16, 20, 24, 28, 32$, topological and trivial controller parameters, pure product-state initialization, and $S = 100$ per point. At $L = 20, 28$ also run max-mix initialization and repeat the pure-state topological and trivial points at $n_{\text{shell}} = 2$. This is the 20-family P1 block in the minimum execution matrix. Additional max-mix sizes are launched only if these two sizes or the wall purification campaign expose a bulk finite-size ambiguity.

c. Initialization, duration, and observation. Pure-state and max-mix runs begin from their independently sampled declared initial states at $t = 0$ and run for $2L$ cycles. There is no discarded physical burn-in. Time-resolved marker and purity diagnostics determine whether the final-cycle estimator is usable.

d. Raw saves and estimator. Save purity spectra, local Chern markers, real-space Chern estimators, Bott data, density, and covariance checkpoints. For mixed covariances define $P_\xi = \Theta(C_\xi - \frac{1}{2})$ only if

$$\Delta_{\text{pur}, \xi} = \min_j |\nu_{j, \xi} - \frac{1}{2}| \quad (18)$$

exceeds the declared numerical and finite-size threshold. Compute each invariant per trajectory and then average its reported scalar/distribution.

TABLE V. Completed legacy work relevant to B2. Rates are normalized per active channel. The wall and uniform entries are finite-size parameter pilots and are not pooled with the preregistered production campaign.

artifact and coverage	strongest quantitative result	B2 implication and limitation
Perfect-correction activity records, $N_x = 20$, $N_y = 20, 22, 24$, $S = 25$	The branch identity $ Y_j = X_j$ passes in all six geometry-size runs. Wall and uniform all-channel rates are respectively 0.01691–0.01710 and 0.00353–0.00382; the wall interface rates lie in $[0.04698, 0.04721]$, while interior rates lie in $[0.00355, 0.00374]$.	Establishes a stationary wall-localized excess for this hard-truncated raster pilot. The uniform reference does not match the wall geometry or active update count, and the records use neither the production random-serial schedule nor $S = 100$.
Lyapunov tangent records at the same three sizes and sample count	The wall final-gap means are 0.1125, 0.1083, and 0.1014, versus 0.8936, 0.8964, and 0.8960 for the corresponding uniform runs. Wall interface weights are 0.9937, 0.9958, and 0.9922.	Establishes a strongly wall-localized slow one-body sector. It was measured in a separate trajectory suite, so no activity-gap covariance, profile overlap, rank correlation, or common-sweep scaling relation has been tested.
Auxiliary regularized-Choi particle-transfer records	The wall finite-sector gaps are 0.04165, 0.04424, and 0.03773, with interface weights 0.8646, 0.8975, and 0.8587; the uniform endpoints saturate before the final cycle.	Supplies an independent slow-sector diagnostic for method development. Endpoint spectra are explicitly outside B2 and do not substitute for the required max-mix purification residual.
Unrun B2 production and cross-analysis	No accepted $N_y = 32, 48, 64$, $S = 100$ master family or unified activity-tangent-purification record is present.	Still required are both wall constructions, separate geometry- and update-count-matched active controls, the random-serial schedule, purification residuals, and preregistered profile-overlap, Spearman, localization-length, joint-fluctuation, and monotone/scaling tests.

TABLE VI. Legacy uniform-topological real-space Chern data relevant to P1. The first reported value is the first saved post-cycle estimator, not a $t = 0$ invariant. Means, sample standard deviations, and ranges are taken directly over the 25 trajectories in each saved array.

N_y	(S, C)	$\langle \mathcal{C}_R(1) \rangle$	$\langle \mathcal{C}_R(C) \rangle \pm \text{SD}$	final trajectory range
30	(25, 30)	0.762876	0.999995 ± 0.000012	[0.999959, 1.000014]
40	(25, 40)	0.761324	0.999687 ± 0.001488	[0.992548, 1.000007]
50	(25, 50)	0.784492	0.999927 ± 0.000299	[0.998506, 1.000000]

e. Controls, statistics, and gate. Bootstrap trajectories. Compare at least two finite-size correction forms. Require the topological bulk marker to extrapolate to one, the trivial marker to zero, an open purity gap, and Bott index $+1$ in at least 95 of 100 accepted large-size trajectories. Require stability under the $n_{\text{shell}} = 2$ check.

f. Downgrade, figure, sharding, and legacy. If $n_{\text{shell}} = 1$ approaches a noninteger plateau, report that plateau or change the primary cutoff; do not claim convergence to one. Results map to Fig. 1(c,d). Shard by trajectory. Path-audited visual references are under `colab_lyapunov/figs/`; they are reference only.

g. Current legacy result. **Assessment: compelling single-width evidence for the intended Chern-one bulk, but not a P1 pass.** The strongest path-audited precursor is `colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. It used the canonical `classA_U1FGTN_gpu.run_markov_circuit` entry point with a uniform $\alpha_1 = \alpha_2 = 1$ controller, $n_{\text{shell}} = 1$, default product-state initialization, and perfect correction. At $N_x = 20$ it ran 25 trajectories for $N_y = 30, 40, 50$, with depth $C = N_y$. The saved Chern arrays contain a radius-eight tripartition estimator $\mathcal{C}_R(t)$ for every trajectory and saved cycle. As summarized in Table VI, its final mean is within 3.2×10^{-4} of one at all three sizes, and even the smallest final trajectory value is 0.992548.

The data therefore show rapid and numerically stable convergence of this uniform topological controller to a Chern-one tripartition estimator over the tested rectangular geometries. They do not establish the full P1 claim. The archive has no uniform trivial controller, no square sequence $L = 12, 16, 20, 24, 28, 32$, no $S = 100$ ensemble, no $n_{\text{shell}} = 2$ comparison, and no max-mix purity-spectrum run. It also contains neither the required local-marker profile nor a Bott-index implementation or output. Consequently no topological-versus-trivial extrapolation, open-purity-gap test, bootstrapped finite-size comparison, or 95-of-100 Bott acceptance test can be performed from this legacy campaign.

B. Campaign W1: wall realization and transverse-convergence gate

a. Claim and status. Status PLANNED. An explicit topological-trivial controller interface and a support-terminated controller realize the same long-distance wall theory after transverse convergence.

b. Dependencies and run matrix. Depends on V0, B0, and P1; B1 is a nonblocking mechanism analysis. At $N_y = 48, 80$, use $N_x = 20, 24, 28$, $S = 100$, and compare: an explicit topological-trivial controller interface with both sides active and no support mask; a support-terminated `dw.truncation` controller with the exterior prepared and excluded according to the active-slab convention; a *separate* geometry- and update-count-matched trivial controller for each construction. The exact coupled spatial-mass Hamiltonian and P1 uniform topological runs are common benchmarks, not extra stochastic W1 protocols. Six of these 24 points coincide with accepted-width master families, leaving 18 new families. If $N_x = 20$ fails, extend the same grid upward in increments of four rather than preregistering unused widths.

c. Initialization, duration, and observation. Use independent product-state starts and evolve each trajectory for a total of $2N_y$ cycles, with checkpoints every $N_y/4$. No separate physical burn-in or subsequent observation extension is used.

d. Raw saves and estimator. Save bulk-marker profiles, one- and two-wall entropy curves, charge FCS, correlators, tangent wall weights, and opposite-wall overlap. Compare asymptotic coefficients, not pointwise covariances.

e. Controls, statistics, and gate. The candidate $N_x = 20$ passes only if all headline coefficients agree with the next two widths within joint 95% intervals at both tested N_y , the fitted transverse localization length is well below wall separation, and opposite-wall low-mode weight is below 5%. The exact benchmark must also satisfy the hybridization inequality in Appendix E. Passing at $N_y = 48$ alone does not license a fixed- N_x thermodynamic claim because the required separation can grow as $\xi_\perp \log N_y$.

f. Downgrade, figure, sharding, and legacy. If $N_x = 20$ fails, all later campaigns use the smallest passing width. If no fixed width passes, use a joint N_x, N_y scaling. If the explicit-interface and support-terminated walls disagree, present them as distinct programmable walls rather than one universality class. Results map to Fig. 1(b,e) and Fig. S2.

g. Current legacy result. **Assessment: the support-terminated wall and exact transverse benchmark are well supported, and a correct-feedback untruncated-interface purification comparison is now identified, but the full W1 construction-to-construction gate has not been run.** The completed B0 exact campaign supplies the common benchmark required by W1. At $N_y = 48$ its coupled spatial-mass and hard-exterior constructions both pass the consecutive-width gate and select $N_x = 20$; the fitted localization lengths are 0.29557 and 0.222754, respectively. The corresponding hybridization ratios at $N_x = 20$ are 1.63×10^{-8} and 1.51×10^{-8} , far below the preregistered 0.1 threshold. This establishes transverse convergence for the exact models, not for either stochastic controller.

The strongest correct-feedback stochastic precursor is the support-terminated $N_x = 20$ campaign in `colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. For $N_y = 30, 40, 50$, 25 perfect-correction trajectories give final mean tangent gaps 0.02698, 0.02431, and 0.01968 and final mean tripartition Chern estimates 0.9781, 0.9543, and 0.9535. The minimum-gap vector places 0.9326, 0.9349, and 0.9435 of its norm directly on the two wall columns; including one neighboring column on each side raises these values to 0.9611, 0.9672, and 0.9708. Because hard support truncation forces exactly zero exterior weight, these numbers demonstrate wall localization but do not supply the fitted leakage or localization-length comparison required by W1.

Independent support-terminated entropy runs at $N_x = 20$, $S = 100$, and fixed depth $C = 50$ give full-transverse log-chord slopes 0.35572, 0.35092, and 0.34920 for $N_y = 30, 40, 50$, approaching the two-wall target $1/3$. A larger $N_y = 80$ pilot gives 0.35316, but uses only ten trajectories and depth $C = 40$. Table VII separates these useful one-construction results from the missing W1 comparison.

The Colab-wide audit identifies an additional correct-feedback precursor in `colab_charge_fluctuations/cpu_data/purification_charge_sharpening_alpha_sweep/`. It keeps `DW=True` and $\alpha_{\text{out}} = 30$ while setting `dw.truncation=False`, so the spatial interface remains and the controller projectors overlap it. This is not a homogeneous `DW=False` control. At $N_x = 16$, $N_y = 16, 24, 32$, and $T = 2N_y$, the completed perfect-correction arm contains ten trajectories at each of nine α_{in} values, for 27 configurations and 270 trajectories. At $N_y = 32$, its final entropy is 0.10488, 0.26286, and 6.06×10^{-5} at $\alpha_{\text{in}} = 1, 1.5, 1.9$, compared with 0.18509, 0.45161, and 0.06416 in the support-terminated arm. The corresponding untruncated intrinsic charge variances are 0.03279, 0.07807, and 4.98×10^{-6} , versus 0.04626, 0.13334, and 0.02241. Both constructions reach the numerical purity floor near $\alpha_{\text{in}} = 2$, but the untruncated interface generally purifies faster below the transition. Because it saves full-layer purification and charge observables rather than Chern, log-chord, tangent, or leakage data, it upgrades the W1 construction inventory without satisfying the W1 equivalence gate.

Consequently W1 remains planned. The newly identified untruncated campaign supplies the second stochastic construction for purification and charge sharpening at smaller sizes, but no archive provides both constructions and their separate geometry- and update-count-matched trivial controls at $N_y = 48, 80$, $N_x = 20, 24, 28$, and $S = 100$. The

TABLE VII. Completed and implemented legacy work relevant to W1. Stochastic entries use the canonical perfect-correction dynamics unless explicitly marked otherwise. Exact B0 results are common benchmarks and are not counted as stochastic W1 protocols.

artifact and coverage	strongest quantitative result	W1 implication and limitation
Exact B0 coupled and hard-exterior walls, $N_y = 48$, $N_x = 12\text{--}32$	Both constructions pass at $N_x = 20$ and the next larger width. At $N_x = 20$, $\xi_\perp = 0.29557$ and 0.222754 and the hybridization ratios are 1.63×10^{-8} and 1.51×10^{-8} .	Passes W1's exact-model transverse benchmark and supports $N_x = 20$ as a candidate. It does not establish convergence or equivalence of the stochastic controllers.
Support-terminated tangent and Chern data, $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$	Final mean gaps are 0.02698, 0.02431, and 0.01968; final mean Chern estimates are 0.9781, 0.9543, and 0.9535. Direct wall-column weights are 0.9326, 0.9349, and 0.9435.	Establishes a topological strip with a sharply wall-localized slow sector for one construction. There is no N_x scan, stochastic localization fit, or valid opposite-wall leakage gate.
Support-terminated entropy data, $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, $C = 50$	Full-transverse log-chord slopes are 0.35572, 0.35092, and 0.34920, versus the two-wall target $1/3$; all three fits have $R^2 > 0.99997$.	Supplies a strong asymptotic-coefficient precursor for one wall realization, but not the independently initialized, fixed- $2N_y$ comparison at $N_y = 48, 80$ across $N_x = 20, 24, 28$.
Support-terminated large-circumference pilot, $N_x = 20$, $N_y = 80$, $S = 10$, $C = 40$	The log-chord slope is 0.35316 with $R^2 = 0.999992$.	Reaches one W1 circumference, but the sample count, depth, observation window, width coverage, and missing comparison arm exclude it from the W1 gate.
Correct-feedback untruncated overlapping interface, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	With $\alpha_{\text{out}} = 30$ and nine α_{in} values, all 270 trajectories are complete. At $N_y = 32$, entropy drops from 0.26286 at $\alpha_{\text{in}} = 1.5$ to 6.06×10^{-5} at 1.9 and reaches the numerical floor at 2.	Establishes a genuine overlapping-interface perfect-correction precursor. It has no N_x scan, $N_y = 48, 80$ arm, Chern or entropy coefficient, tangent localization length, or opposite-wall overlap, so it does not pass W1.

required independent starts, total $2N_y$ trajectories, windows, joint 95% coefficient comparisons, stochastic localization-length fits, and opposite-wall low-mode weight below 5% therefore remain untested. The legacy record supports the support-terminated wall as a strong production candidate, but does not license a common stochastic universality class or a fixed- N_x W1 pass.

X. UNIVERSAL PREPARED-STATE DATA

A. Campaign S1: state CFT and charge sector

a. Claim and status. Status PLANNED. Stationary Born trajectories carry wall-localized complex-fermion state data with $c_{\text{edge}} = c_L + c_R = 1$ and a level-one $U(1)$ current algebra. This campaign does not establish handedness; chirality is licensed only after H1–H3.

b. Dependencies and run matrix. Depends on W1. Use the accepted N_x , Eq. (4), and $S = 100$. The primary wall and its matched control use both an onsite pure product state and max-mix initialization, the latter entering only after purification. At $N_y = 32, 48, 64$ also run the secondary wall construction and its separate matched control from pure product states. These are respectively the 12, 12, and 6 master-family blocks in the minimum matrix.

c. Initialization, duration, and observation. Pure and max-mix runs begin independently at $t = 0$ and last a total of $2N_y$ cycles. There is no separate physical burn-in. Save the three preregistered late-time snapshots at $t = N_y, 3N_y/2, 2N_y$ (rounded to whole cycles when necessary). Max-mix snapshots enter this analysis only if the P2 purity and stability diagnostics pass.

d. Raw saves and estimator. For

$$X_L(\ell) = \log \left[\frac{L}{\pi} \sin \frac{\pi \ell}{L} \right], \quad (19)$$

fit the two-wall full-width interval to

$$\mathbb{E}[S_q(\ell)] = \frac{c_{\text{edge}}}{6} \left(1 + \frac{1}{q} \right) X_L(\ell) + b_q + \cdots, \quad q = 1, 2, 3, \quad (20)$$

using the full trajectory-bootstrap covariance across ℓ . Because one realization prepares one conditioned state, define $c_{\xi,q}$ trajectory by trajectory using the same preregistered fit window, design matrix, and linear estimator for every

sample. Two co-primary summaries are reported: the coefficient c_q^{Born} fitted from the Born-averaged conditional entropy curve $\mathbb{E}_\xi[S_{\xi,q}(\ell)]$, and the full Born distribution of $c_{\xi,q}$ with its mean and whole-trajectory bootstrap interval. With a shared linear estimator, $c_q^{\text{Born}} = \mathbb{E}_\xi[c_{\xi,q}]$ algebraically, while displaying both the mean curve and the individual distribution separately tests ensemble scaling and trajectory typicality. Resolve the latter by the definite trajectory charge $N_{\mathcal{D},\xi}$ and report sector-conditioned means $c_q(N)$ wherever the sector count is sufficient. A broad $P(N)$ together with mutually compatible $c_q(N)$ is the direct test of a charge-fluctuating ensemble of critical states with shared infrared data. Do not average results from trajectory-dependent fit windows or model selections, and never extract a central charge from the entropy of the averaged covariance. Average multiple stationary snapshots within a trajectory before fitting or bootstrapping so they do not become independent samples.

e. Legacy-compatible entropy processing and production upgrade. The established repository estimator uses full-transverse strips

$$A(y_0, A_y) = [0, N_x) \times \{y_0, y_0 + 1, \dots, y_0 + A_y - 1\} \quad (21)$$

with periodic y , computes the entropy from the eigenvalues of the restricted one-body correlation matrix, and averages over all strip origins within each trajectory,

$$\bar{S}_{q,\xi}(A_y, t) = \frac{1}{N_y} \sum_{y_0=0}^{N_y-1} S_{q,\xi}[A(y_0, A_y), t]. \quad (22)$$

The legacy full-strip size campaign then averaged Eq. (22) over samples before an ordinary least-squares fit with an intercept against $X_{N_y}(A_y)$ on $A_y = 8, \dots, \lfloor N_y/2 \rfloor$ at a fixed final cycle. Its target von Neumann slope was $m_1 = 1/3$. Earlier wall-window contour scripts similarly summed the unnormalized contour over two-cell windows centered on each wall, averaged those curves over samples, and fitted the two complementary A_y branches separately.

Production retains this geometry, origin average, legacy fit window, intercept, and full-width target as the primary compatibility analysis, but preserves the sample axis before fitting. This follows the newer streaming repository path whose saved entropy array has axes (`sample`, `cycle`, `Ay`) after the y_0 average. For each q and trajectory, first average Eq. (22) over the declared stationary snapshots, then perform the common linear fit

$$\bar{S}_{q,\xi}(A_y) = m_{q,\xi} X_{N_y}(A_y) + b_{q,\xi}, \quad c_{\xi,q} = \frac{6m_{q,\xi}}{1 + 1/q}. \quad (23)$$

Thus $c_{\xi,1} = 3m_{1,\xi}$. Save $m_{q,\xi}$, $b_{q,\xi}$, $R_{q,\xi}^2$, residual vectors, and the number of fit points, along with the full per-trajectory entropy curves. The regression standard error across A_y used in legacy tables diagnoses deviation from the fitted shape; it is not the sampling uncertainty of the Born ensemble. All quoted uncertainties on c_q^{Born} , sector-conditioned $c_q(N)$, or distributional statistics come from resampling whole trajectories. Periodic origins and stationary snapshots remain correlated averages internal to one trajectory.

Repeat the fits after moving the lower endpoint from $A_y = 8$ by the B0-declared sensitivity increments, deleting the largest included A_y , and including the preregistered finite-size correction model. Check $S_{q,\xi}(A_y) = S_{q,\xi}(N_y - A_y)$ within tolerance for pure snapshots; fit the two branches separately only as a diagnostic. The primary wall-window result uses per-trajectory unnormalized contour sums before Born averaging, upgrading the older sample-averaged contour procedure without changing its spatial windows. The audited implementation references are under `colab_large_entanglement_scaling_N20/src/` and `colab_charge_fluctuations/src/`; their legacy outputs remain comparison data rather than production inputs.

f. Production boundary-bulk correlator campaign. All correlator observables and fits are recalculated from the newly generated B0 covariances and S1 trajectory snapshots. Prior repository results motivate the observable and expose likely fit-window sensitivity, but none of their curves, averages, or fitted exponents enter the production estimates. This recalculation requires no additional Born families because it is performed on every covariance snapshot already generated for S1. Use the fixed-slice squared Green function

$$C_{G,\xi}(x, r_y) = \frac{1}{2N_y} \sum_{y,\mu,\nu} |C_{\xi;(x,y,\mu),(x,y+r_y,\nu)}|^2 \quad (24)$$

for every transverse slice x and every independent separation $1 \leq r_y \leq \lfloor N_y/2 \rfloor$. The exact B0 calculation and the stochastic S1 calculation use identical normalization, slice definitions, chord coordinate, candidate models, and fit windows.

For every trajectory retain the full $C_{G,\xi}(x, r_y)$ array and derive, without discarding the sample axis, the two individual wall curves, exterior-neighbor curves, distance-from-wall profiles, declared topological- and trivial-bulk

averages, fixed wall-window sums, and the full- x average. Save all per-trajectory curves as well as the Born means, covariance across r_y , fit parameters, residuals, R^2 , point counts, and model-comparison scores. Source-position and stationary-snapshot averages remain correlated averages internal to a trajectory.

B0 fixes the primary distance window before the stochastic data are examined, using $r_y \in [3, 15]$ where the circumference permits. Fit the wall data to chord-power, exponential, chord-power-plus-floor, and exponential-plus-floor models; apply the same models to matched bulk data rather than assuming exponential decay. Endpoint and window sweeps are sensitivity diagnostics and may not select the reported exponent. Report the Born-averaged fit and the complete trajectory-level exponent distribution, including its dependence on the trajectory charge sector, exactly as for the central-charge analysis. The primary conclusion is spatial: stable algebraic behavior should be wall localized and absent from the bulk. Quote a precision exponent near two only if it is stable under the preregistered window and model checks; otherwise retain the freshly calculated boundary–bulk contrast as qualitative support and let entropy and charge fluctuations carry the universal claim. Bootstrap whole trajectories.

The audited exact references are under `exact_DW_benchmark_notes/`. Earlier stochastic correlator diagnostics are under `colab_charge_fluctuations/analysis_outputs/`. They document what was tried before and preregister failure modes; every plotted curve and quoted number in the new manuscript is regenerated under the production protocol.

Test whether the contour integrated over a declared single-wall window carries half that coefficient. This window sum is a diagnostic decomposition of the full-strip entropy, not the entropy of an independent wall subsystem. Save every trajectory’s subsystem spectra, entropy curves, unnormalized spatial contour $s_{A,\xi}(x, y)$, charge-generating function, wall-projected connected density structure factor, and globally normalized real-space correlator. Form $s_A(x, y) = S^{-1} \sum_{\xi} s_{A,\xi}(x, y)$ without rescaling individual contours, and verify $\sum_{x,y} \overline{s_A} = \mathbb{E}[S_A]$. Average S_q , quantum cumulants, structure factors, and squared correlations only after their per-trajectory evaluation. The interval variance and its small- k structure factor are primary current diagnostics; the real-space squared Green function is an appendix cross-check. Appendix C fixes the FCS conventions.

Separate quantum charge fluctuations from classical record-to-record profile fluctuations. For every interval A save

$$F_{\xi}^q(A) = \langle N_A^2 \rangle_{\xi} - \langle N_A \rangle_{\xi}^2 = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})], \quad (25)$$

$$F^{\text{cl}}(A) = \text{Var}_{\xi}(\langle N_A \rangle_{\xi}), \quad (26)$$

and verify the law of total variance

$$\text{Var}(N_A) = \mathbb{E}_{\xi}[F_{\xi}^q(A)] + F^{\text{cl}}(A). \quad (27)$$

Only the logarithmic coefficient of $\mathbb{E}_{\xi}[F_{\xi}^q(A)]$ is used to infer the $U(1)$ current level. The classical term is not purely sector wandering. Let $\mathcal{D}$ denote the declared dynamical subsystem—the active slab for support termination and the full system for the explicit interface—and let $N_{\mathcal{D}}$ be its charge. For a definite-charge input, each pure conditioned trajectory has a definite $N_{\mathcal{D}}$ and

$$F^{\text{cl}}(A) = \mathbb{E}_{N_{\mathcal{D}}}[\text{Var}_{\xi|N_{\mathcal{D}}}(\langle N_A \rangle_{\xi})] + \text{Var}_{N_{\mathcal{D}}}(\mathbb{E}_{\xi|N_{\mathcal{D}}}[\langle N_A \rangle_{\xi}]). \quad (28)$$

The first term is fixed-sector shape variation and the second is the contribution from sector wandering. Max-mix trajectories enter this decomposition only after both the global purity deficit and the within-trajectory quantum variance of $N_{\mathcal{D}}$ fall below declared tolerances.

To test where the total-sector fluctuations couple spatially, define

$$q_{\xi x} = \sum_{(y,\mu):(x,y,\mu) \in \mathcal{D}} \langle n_{xy\mu} \rangle_{\xi}, \quad N_{\mathcal{D},\xi} = \sum_x q_{\xi x}. \quad (29)$$

Save the complete transverse covariance matrix and its signed additive row sum,

$$K_{xx'}^{\text{cl}} = \text{Cov}_{\xi}(q_{\xi x}, q_{\xi x'}), \quad \kappa_{\text{cl}}(x) = \sum_{x'} K_{xx'}^{\text{cl}} = \text{Cov}_{\xi}(q_{\xi x}, N_{\mathcal{D},\xi}), \quad \sum_x \kappa_{\text{cl}}(x) = \text{Var}_{\xi}(N_{\mathcal{D},\xi}). \quad (30)$$

Also save the nonnegative onsite-heterogeneity profile $v_{\text{cl}}(x) = \sum_{y,\mu} \text{Var}_{\xi}(\langle n_{xy\mu} \rangle_{\xi})$. Report signed κ_{cl} without taking its absolute value, its signed partial sums in fixed wall windows, and its absolute leakage outside those windows. Full-system charge bookkeeping is saved separately. The profile v_{cl} omits inter-site covariances, so it neither sums to $\text{Var}(N_{\mathcal{D}})$ nor defines a variance weight; use it only as a descriptive localization control. Match every system-charge change to the signed gain/loss record. These quantities test whether sector wandering is coupled to the walls; they are not themselves universal CFT coefficients.

g. Matched entropy–variance spectral test. For every trajectory, stationary snapshot, origin, and interval used in the entropy fit, compute the quantum internal variance from the same restricted covariance and perform the same internal snapshot and origin averages. With

$$\bar{F}_\xi^q(A_y) = f_\xi X_{N_y}(A_y) + b_{F,\xi} + \dots, \quad (31)$$

define the paired trajectory statistic and its Born mean by

$$\delta_{SF,\xi} = m_{1,\xi} - \frac{\pi^2}{3} f_\xi, \quad \delta_{SF}^{\text{Born}} = \mathbb{E}_\xi[\delta_{SF,\xi}] = a_S - \frac{\pi^2}{3} a_F. \quad (32)$$

The primary analysis fits the paired residual curve

$$R_{SF}(A_y) = \mathbb{E}_\xi \left[\bar{S}_{1,\xi}(A_y) - \frac{\pi^2}{3} \bar{F}_\xi^q(A_y) \right] = \delta_{SF}^{\text{Born}} X_{N_y}(A_y) + b_{SF} + \dots \quad (33)$$

with the full paired covariance across A_y and a whole-trajectory bootstrap. Report a_S/a_F only as a secondary statistic because a ratio is unstable when its denominator is uncertain. For Rényi index q , also test the free-fermion prediction

$$a_{S_q} - \frac{\pi^2}{6} \left(1 + \frac{1}{q} \right) a_F = 0. \quad (34)$$

Appendix D derives these relations and their limits. This is an observer analysis of the existing S1 master covariances and launches no new trajectory family.

h. Controls, statistics, and gate. Use a trajectory bootstrap with the full ℓ covariance matrix. Predeclare fit windows from B0 and repeat fits after deleting each endpoint window and the smallest size. Require $c_{\text{edge}} = 1$ for all three Rényi indices, the factor-of-two relation between one and two walls, and level $k = 1$ from both interval variance and the small- k structure factor. Using the same fit windows and finite-size model, require the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ to be compatible with zero at 95% confidence and stable under the declared endpoint and smallest-size deletions. Apply the same paired test to the exact-Hamiltonian benchmark and require no false logarithmic signal in the matched trivial control. The wall squared-Green-function exponent must be compatible with two on the benchmark-selected window and remain stable under the declared sensitivity checks; no matched bulk slice may support a comparably stable algebraic fit. If the exponent remains window sensitive, downgrade it to a boundary-localization diagnostic rather than tuning the window. Require a log-chord model to outperform an area-law constant in the wall geometry and not in the trivial control.

i. Current legacy result. A repository-wide audit finds a clean deterministic calibration of the contour-decomposed tripartite-information estimator, but no production-compatible stochastic S1 result. In the exact-DW benchmark material under `notebooks/`, the subsystem is first defined across the full transverse width and only the final entanglement-contour sum is restricted in x . For the exact 20×40 domain-wall ground state, $L_A = L_C = 3$, and fixed $y_0 = 0$, the two three-column wall windows each give a slope $0.166967668 \pm 0.000027005$ in $I_2 - I_3$ versus $\log[1/(1-x)]$, with $R^2 = 0.99999945$. The full-width sum gives $0.334232464 \pm 0.000095377$, with $R^2 = 0.99999829$. These values validate the expected $1/6$ single-wall coefficient, $1/3$ two-wall coefficient, and factor of two within the deterministic analysis pipeline. Equivalently, the direct TMI central-charge estimates are

$$c_{\text{TMI}}^{\text{wall}} = 6m = 1.0018060(1620), \quad c_{\text{TMI}}^{\text{full}} = 3m = 1.0026974(2861), \quad (35)$$

making this one of the cleanest archived $c = 1$ calibrations.

The strongest admissible stochastic input is the full-covariance history under `sample_average_testing/`. It contains $S = 100$ trajectories on 12×31 through $C = 100$ and evaluates the entropy combinations trajectory first, followed by the y_0 and Born averages. At $S = 100$ and $L_A = L_C = 4$, the three-column wall sum has slope 0.2487 ± 0.0014 and the full-width sum has slope 0.5112 ± 0.0014 , both with R^2 rounded to 1.000; the separate $L_A = L_C = 3$ full-width analysis gives 0.5251 ± 0.0016 with $R^2 = 0.9999$. The approximate factor of two is visible, but both coefficients lie roughly 50% above the exact targets. In direct central-charge units these are 1.4922(84), 1.5336(42), and 1.5753(48), respectively. Moreover this run uses $n_{\text{shell}} = \text{None}$, nonperfect stochastic feedback, the legacy `dw_symmetric` schedule, one circumference, and weak source provenance, so it is a reusable estimator pilot rather than evidence for the S1 production model.

A second, previously omitted Colab archive provides a useful physical-state cross-check. The canonical GPU covariance snapshots and downstream CPU analyses under `colab_small_system_testing/` cover $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, $S = 10$, and $C = 50$ with perfect correction. Fitting the final full-width entropy over $8 \leq A_y \leq N_y/2$ gives

	$n_{\text{shell}} = 1$	$n_{\text{shell}} = 2$
$N_y = 30$	$c_{\text{eff}} = 1.04583(13)$	$1.05513(16)$
$N_y = 40$	$c_{\text{eff}} = 1.03244(22)$	$1.08650(21)$

(36)

with $R^2 \geq 0.99995$; the parentheses here are regression standard errors across the A_y fit points, not trajectory bootstrap errors. On the fixed correlator window $2 \leq r_y \leq N_y/4$, the two-wall-window squared-correlator exponents are 2.154(36), 2.126(33), 2.191(36), and 2.092(34) in the same order, while the matched nonwall slices are much steeper, 4.11–6.37. These numbers independently support a wall-localized algebraic sector and show that it persists at both shell depths. They are not a precision exponent determination: the dedicated window audit under `colab_charge_fluctuations/analysis_outputs/correlator_scaling_diagnostics/` finds order-one drift when the short- and long-distance endpoints are varied. This is a fit-window/finite-size limitation, not a contradiction with the stable selected-window fits.

The same raw cycle-50 covariances also contain a previously unreported charge-sector observable. Reconstructing $C = (G + 1)/2$ and evaluating the full- x interval quantum variance trajectory by trajectory,

$$F_{A,\xi}^q = \text{Tr}[C_{A,\xi}(\mathbf{1}_A - C_{A,\xi})], \quad (37)$$

gives the two-wall log-chord fit

$$F_A^q = \frac{k_{\text{eff}}}{\pi^2} \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b. \quad (38)$$

Over $8 \leq A_y \leq N_y/2$, the fitted levels are

$$\begin{array}{c|cc} & n_{\text{shell}} = 1 & n_{\text{shell}} = 2 \\ \hline N_y = 30 & 1.0409(166) & 1.0468(106) \\ N_y = 40 & 1.0328(346) & 1.0853(449), \end{array} \quad (39)$$

where the parentheses are standard errors over ten trajectory-resolved coefficients. The ensemble-mean curves have $R^2 > 0.99996$, and the four values track the matched entropy coefficients above. This upgrades the archive from an entropy-only $c \simeq 1$ precursor to a matched $c \simeq k \simeq 1$ precursor on the same conditioned ensembles. It remains finite-size evidence rather than a thermodynamic current-algebra measurement.

Table VIII records the admissible results and the source–output mismatch that is otherwise easy to mistake for stronger evidence. The saved $S = 250$ plots in the legacy analysis material under `notebooks/` do not match the current source configuration and are not treated as reproducible results.

TABLE VIII. Legacy tripartite-information evidence relevant to S1. Slopes refer to the repository’s contour-decomposed $I_2 - I_3$ fit against $\log[1/(1-x)]$; a wall-window sum is a decomposition of a full-width subsystem entropy, not an independent wall-subsystem entropy. The excluded source–output mismatch does not enter the scientific summary.

Artifact and protocol	Result	S1 interpretation
Exact DW ground state, 20×40 , $L_A = L_C = 3$	Each wall: $c_{\text{TMI}} = 1.0018060(1620)$; full width: $c_{\text{TMI}} = 1.0026974(2861)$, with $R^2 > 0.999998$.	Strong deterministic estimator calibration; passes the one-wall/two-wall coefficient and factor-of-two checks, but contains no Born dynamics.
Canonical pure-state covariance snapshots, $16 \times (30, 40)$, $n_{\text{shell}} = 1, 2$, $S = 10$, $C = 50$	Late-window entropy gives $c_{\text{eff}} = 1.032$ – 1.086 ; matched interval fluctuations give $k_{\text{eff}} = 1.033$ – 1.085 ; fixed-window wall exponents are 2.092–2.191, versus nonwall values 4.11–6.37.	Strong perfect-correction matched entropy–charge cross-check. Only two sizes and ten trajectories are present, and the correlator exponent is window sensitive.
Full covariance history, 12×31 , $C = 100$, $S = 100$, $n_{\text{shell}} = \text{None}$, nonperfect feedback	For $L_A = L_C = 4$, one wall gives $c_{\text{TMI}} = 1.4922(84)$ and full width gives 1.5336(42); the $L_A = L_C = 3$ full-width value is 1.5753(48).	Best reusable dynamical TMI input, but wrong protocol, one small geometry, and coefficients well above the exact values.
Saved dynamic TMI output under <code>notebooks/</code>	Plot output claims $S = 250$, while the current source/configuration does not reproduce that sample contract.	Excluded as a stale source–output mismatch.

Thus the exact estimator and its factor-of-two decomposition are established, one longer stochastic history can be reanalyzed as a protocol pilot, and a separate canonical perfect-correction archive already supplies matched von Neumann entropy and within-trajectory quantum charge variance. No admissible adaptive dataset combines the production channel word and random-serial schedule, the accepted circumference sweep and sample count, stationary snapshots, $q = 1, 2, 3$ entropies, and the full matched observable bundle. S1 therefore remains planned.

j. Downgrade, figure, sharding, and legacy. If only entropy is compatible with one, say “consistent with a $c = 1$ wall” and do not claim a charge-resolved CFT. If c and k are individually stable but Eq. (32) fails, report them separately and do not identify the wall as a minimal charged $U(1)_1$ sector. This campaign supplies all of Fig. 2. Shard by trajectory; source-position averages are internal correlated averages. Path-audited visual references are under `colab_large_entanglement_scaling_N20/docs/figures/`. The contour is a localization and sum-rule check, not a standalone novelty claim; BPJ already used entanglement contours to identify slow domain-wall structure. The exact and stochastic TMI pilots in Table VIII do not close the production gate. The newly reduced $N_x = 16$ pure-state covariance archive does contain both full-strip entropy and matched within-trajectory interval variance, but only at $N_y = 30, 40$ with ten trajectories and without the production schedule, Rényi bundle, one-wall decomposition, or thermodynamic extrapolation. The saved max-mix total variance remains a purification observable rather than the S1 current-algebra variance.

B. Campaign S2: topological-to-trivial sweep

a. Claim and status. Status PLANNED. The logarithmic wall sector and near-marginal tangent sector occur when the two sides have different bulk topology.

b. Dependencies and run matrix. Calibrate α_c from the exact uniform Hamiltonian. Fix $\alpha_{\text{out}} = 30$ and take

$$\alpha_{\text{in}} = \alpha_c + \{-1, -0.5, -0.25, -0.125, 0.125, 0.25, 0.5, 1\}, \quad (40)$$

plus $\alpha_{\text{in}} = 1$, at $N_y = 24, 32, 48, 64$, the accepted N_x , and $S = 100$.

c. Initialization, saves, estimator, and controls. Use independent pure-state starts and a total duration of $2N_y$. Save the bulk marker and purity gap, entropy and charge coefficients, correlator, and tangent gap. Do not fit the bulk-critical point itself to a pure wall CFT. Compare deep topological, crossover, and deep trivial regimes.

d. Statistics and acceptance gate. Bootstrap the joint location of the topological-marker change, disappearance of the logarithmic coefficient, and opening of the tangent gap. The gate requires a common finite-size crossover interval, not identical finite-size pseudocritical points.

e. Current legacy result. The exact uniform Bloch Hamiltonian places the positive-mass transition at $\alpha_c = 2$. Existing legacy sweeps consistently resolve a dynamical crossover there, but none measures the complete S2 observable bundle. The closest production-path input is the canonical GPU particle–Choi transfer campaign under `colab_no_feedback_alpha_sweep_transfer/`. At $N_x = 20$, $N_y = 20, 30, 40$, $S = 10$, $C = 2N_y$, $\alpha_{\text{out}} = 30$, and $\alpha_{\text{in}} = 1, 1.5, 1.8, 1.9, 2, 2.1, 2.2, 2.5, 3$, the no-feedback branch is complete at all three sizes. Its mean smallest finite transfer exponent is respectively 0.01215(229), 0.00808(182), and 0.00249(90) at $\alpha_{\text{in}} = 1.9$, but rises to 0.04366(577), 0.03440(527), and 0.03683(450) at 2.1. At 2.2 the corresponding values are 0.12059(1268), 0.09505(841), and 0.08394(908), with only eight of ten finite samples at $N_y = 40$; by 2.5 no finite transfer mode remains. Thus a finite particle–Choi sector softens with circumference on the topological side and is lost across the crossover. A missing finite mode is a censored transfer observable, not an infinite measured gap.

The perfect-correction branch gives the same qualitative boundary more sharply. At $\alpha_{\text{in}} = 1.5$ its gap decreases from 0.03065(805) to 0.02244(316) to 0.01847(199) across $N_y = 20, 30, 40$, whereas at 1.9 it is already 0.10730(938), 0.11146(838), and 0.09887(581); the last value has eight finite samples. The $N_y = 40$ perfect-correction sweep stops at 1.9, and at smaller sizes the finite sector is almost entirely absent at and above 2. These are singular exponents of the particle–Choi transfer construction, not the physical covariance tangent exponents required by S2.

The executed finite-mode analysis in the same Colab folder adds spatial information that was absent from the original scalar summary. At $N_y = 40$ and $\alpha_{\text{in}} = 1$, the slowest finite particle–Choi mode has mean weight 0.45187 within one column of the reduced-slab edges without feedback, versus 0.87933 under perfect correction. Its mean edge distance correspondingly falls from 1.9944 to 0.5745 cells, and its site participation ratio falls from 156.24(728) to 56.72(278). Approaching the transition on the perfect-correction branch, the edge weight decreases to 0.65428 at $\alpha = 1.5$ and 0.46673 at 1.9, while the mean edge distance grows to 1.2694 and 1.9432. There are no parse failures in the 7,151-mode analysis. This is a strong localization diagnostic for the conditioned transfer construction, but the reduced Choi-slab basis and finite-sector censoring prevent its reinterpretation as a physical tangent eigenmode.

A complementary canonical CPU max-mix campaign under `colab_charge_fluctuations/cpu_data/` completes all 63 combinations of $N_x = 16$, $N_y = 16, 24, 32$, 21 masses from 1 to 3 in steps of 0.1, and $S = 10$. At $N_y = 32$ the final slab entropy and total charge variance are 0.1954 and 0.04793 at $\alpha_{\text{in}} = 1$, and 0.1244 and 0.03447 at 1.8, but fall to approximately 2.04×10^{-8} and 7.9×10^{-12} at 2. The saved projector diagnostics flag zero minimum Bloch norm at the critical point itself, so that point is not interpreted as a wall-CFT fit. Moreover these are purification observables from a maximally mixed initial state, not stationary pure-state wall entropy and current coefficients.

Table IX includes the exact and CPU particle–Choi cross-checks. The exact flattened-Hamiltonian sweep fixes $\alpha_{\text{in}} = 1$ and varies α_{out} , so it calibrates the hard-wall interpretation rather than executing the S2 scan. The CPU

Choi campaign completes 27 runs at $N_x = 16$, $N_y = 16, 24, 32$, and the nine masses above. At $N_y = 32$ its mean finite exponent grows from 0.04936(560) at 1.8 to 0.08948(524) at 1.9; at 2 no sample has a finite mode, while the apparent values near 0.21 at 2.1 and 2.2 each come from only two of ten samples and are therefore censored tails.

TABLE IX. Legacy evidence relevant to the S2 topological-to-trivial sweep. Parentheses give the standard error in the last quoted digits. Particle–Choi gaps are reported only over finite samples; loss of the finite sector is recorded separately and is not converted into a numerical gap.

Artifact and protocol	Result	S2 interpretation
Exact flattened Hamiltonian, $\alpha_{\text{in}} = 1$, exterior-mass sweep	The physical-wall estimate crosses from $c_{\text{est}} = 2.006$ at $\alpha_{\text{out}} = 1$ to 1.079 at 2 and 1.004 at 3.	Exact calibration of $\alpha_c = 2$ and of the hard-wall limit; the sweep direction differs from S2.
Canonical GPU particle–Choi transfer, $20 \times (20, 30, 40)$, $C = 2N_y$, $S = 10$	No feedback: the gap changes from 0.01215, 0.00808, 0.00249 at $\alpha = 1.9$ to 0.04366, 0.03440, 0.03683 at 2.1. Perfect correction shows a sharper rise by 1.9 and extensive censoring at and above 2.	Strongest direct dynamical transition precursor; canonical source and useful size trend, but a particle–Choi proxy rather than the S2 tangent/entropy/marker bundle.
Finite particle–Choi mode profiles from the same GPU sweep	At $N_y = 40$, $\alpha = 1$, perfect correction raises one-column edge weight from 0.45187 to 0.87933 and reduces mean edge distance from 1.9944 to 0.5745 cells; localization weakens again by $\alpha = 1.9$.	Spatially resolves the feedback-dependent conditioned transfer mode. It remains a reduced-basis proxy and is not a physical covariance-tangent localization result.
Canonical CPU max-mix purification, $16 \times (16, 24, 32)$, 21 masses, $C = 2N_y$, $S = 10$	All 63 runs complete. At $N_y = 32$, entropy/variance change from 0.1244/0.03447 at 1.8 to $2.04 \times 10^{-8}/7.9 \times 10^{-12}$ at 2.	Direct fine mass scan, but it measures transient max-mix purification; the critical projector is singular and no stationary wall coefficients were saved.
Canonical CPU particle–Choi transfer, $16 \times (16, 24, 32)$, nine masses, $S = 10$	All 27 runs complete. At $N_y = 32$ the finite gap rises from 0.04936(560) at 1.8 to 0.08948(524) at 1.9; the finite sector is absent or severely censored thereafter.	Independent CPU confirmation of the crossover, but still a particle–Choi observable and not a thermodynamic tangent-gap determination.

f. Downgrade, figure, sharding, and legacy. Preemptive wall crossover is allowed and must be reported. This is Fig. 2(e) if compact; otherwise Fig. S3. Shard by trajectory. The legacy results in Table IX locate a common crossover near $\alpha = 2$, but no saved dataset jointly provides the bulk marker and purity gap, stationary wall entropy and charge coefficients, correlator, and physical tangent gap at $S = 100$ and $N_y = 24, 32, 48, 64$. S2 therefore remains planned.

XI. TANGENT DYNAMICS AND PURIFICATION

A. Campaign T1: quenched tangent spectrum and localization

a. Claim and status. Status PLANNED. The differential of the normalized realized covariance map has a wall-localized near-marginal sector whose gap closes with N_y although exact projectors make a conventional square transfer product singular.

b. Dependencies and run matrix. Depends on W1. Use the accepted N_x , Eq. (4), $S = 100$, wall and matched trivial controls, and enough tangent vectors to retain at least 12 physical modes nearest zero. At $N_y = 32, 48$, double the tangent-frame rank and repeat at least four independently randomized initial tangent frames on the same replayed physical records.

c. Initialization, duration, and observation. Initialize the physical trajectory and a common physical tangent subspace together at $t = 0$. Evolve for a total of $2N_y$ cycles. Use the first N_y cycles only to align the tangent frame; discard their QR log sums but do not call or treat these physical cycles as burn-in. Without resetting the physical trajectory or aligned frame, accumulate the reported exponents over the final N_y cycles, with cumulative checkpoints at $N_y/4, N_y/2, 3N_y/4, N_y$ within that accumulation window. Fixed-size finite-time drift across these checkpoints is reported as a systematic diagnostic. Production is not automatically extended beyond $2N_y$; failure to stabilize is a failed temporal gate and requires a separately approved amendment.

d. Raw saves and estimator. Save per-cycle QR log increments, finite/null masks, gauge directions, aligned low vectors, orthogonality residuals, and stable cocycle snapshots. A snapshot consists of the physical covariance at its left endpoint, the ordered realized branch/channel word for the window, the aligned input and output QR frames, block- R factors with their diagonal log scales, and a replay hash. On validation sizes also save the explicit window product and

its SVD; at production sizes reconstruct it from the stabilized factors rather than multiplying a long ill-conditioned dense word. For max-mix bridge snapshots save the reduced endpoint Choi frames A_K, B_K and Gram matrix needed to verify $\tilde{J}_K^\dagger \tilde{J}_K = 4C_K(\mathbf{1}_K - C_K)$. For trajectory s , define the one-leg half-Jacobian growth rates

$$\ell_{j,s}(T) = \frac{1}{T} \sum_{\tau=1}^T \log |(R_{\tau,s})_{jj}|, \quad \Delta_{1\text{leg}}(T) = \min_{j \in \text{physical}} |\bar{\ell}_j(T)|. \quad (41)$$

The primary quenched spectrum averages log stretches at continuously tracked Oseledec index before taking the gap. If decay-rate notation is desired, use $\gamma_j = -\ell_j$ and state the sign convention. The full covariance-tangent map is a congruence, so its exponents are pair sums $\ell_a + \ell_b$ (or decay rates $\gamma_a + \gamma_b$), not the one-leg spectrum itself. The mean and median of per-trajectory minimum gaps are reported as sensitivity diagnostics. Exact null and normalization directions are removed before defining the physical reference exponent. Appendix B fixes the cycle-time normalization, the equivalent finite-time log-strain operator, the stabilized reconstruction of singular modes, and the distinction between a quenched spectrum, an averaged log-strain operator, and an annealed second-moment object.

For a near-degenerate low set $\mathcal{L}_s$ of rank r_s , localize its projector rather than a basis-dependent QR vector. The primary density is

$$\rho_s(x, y) = \frac{1}{r_s} \sum_{a \in \mathcal{L}_s, \mu} |v_{s,a,\mu}(x, y)|^2, \quad \sum_{x,y} \rho_s(x, y) = 1. \quad (42)$$

Never normalize the orbital components independently. Resolve both walls by diagonalizing a projected transverse-position operator within $\mathcal{L}_s$ or by reporting left/right-window weights; do not select whichever individual QR vector looks most localized.

e. Controls, statistics, and gate. Compare aN_y^{-z} , a free-intercept power law, and a gapped constant plus corrections, using at least five sizes and bootstrap fits. Require the wall gap to have zero intercept, preferably $z = 1$, while the matched trivial control retains a positive intercept. Require the fixed- N_y temporal extrapolation above, initial-frame and rank convergence, near-degenerate-subspace interface weight above 70%, representation of both walls, and transverse localization satisfying W1.

f. Current legacy result. **Assessment: finite-size softening and wall localization are established as precursors, but a thermodynamic tangent-gap closing is not.** The strongest fully archived CPU result is the canonical perfect-correction campaign in `topological_frustration_diagnostics/results/campaigns/20260813_121340_Nx20_perfect_Ny20_22_24/`. At $N_x = 20$, $N_y = 20, 22, 24$, and $S = 25$, the mean final wall gaps are 0.11249(470), 0.10830(519), and 0.10142(464), whereas the uniform-control gaps are 0.89365(1721), 0.89638(1259), and 0.89600(916). The selected wall slow vector carries 0.99365(222), 0.99584(119), and 0.99222(333) of its norm in the interface region. This is a sharp geometry-resolved separation between a slow wall sector and an order-one bulk control. It is nevertheless a three-size raster pilot with $2N_y$ total cycles; its cumulative spectrum does not implement T1's declared N_y tangent-frame alignment followed by a separate N_y accumulation. Moreover, the selected individual QR vector lies almost entirely on one wall, so it cannot replace the basis-independent low-subspace and two-wall test required by the current estimator.

The broadest valid GPU precursor is the canonical campaign under `colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. For the support-terminated wall at $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$, and $T = N_y$, its final mean one-body gaps are 0.02698(313), 0.02431(287), and 0.01968(225). The corresponding uniform controls remain at 0.72169(1097), 0.74833(863), and 0.76760(874). Directly on the two wall columns, the saved minimum-gap vector has weights 0.9326, 0.9349, and 0.9435; including the adjacent columns gives 0.9611, 0.9672, and 0.9708. These GPU data extend the softening trend to larger circumferences and independently confirm strong wall localization. They still use only $T = N_y$, the default initial state, one saved basis-dependent minimum-gap vector, and no alignment, rank, initial-frame, or temporal convergence repeats.

The canonical CPU CFT-extraction archive supplies the decisive finite-time warning. With ten trajectories per size and no burn-in, the $T/L = 2$ tangent gaps at $L = 20, 30, 40$ are 0.04511, 0.04002, and 0.02839; at $T/L = 5$ they are 0.05914, 0.04180, and 0.03651. The latter final-quarter tangent drifts remain 18.2%, 15.8%, and -3.0% . Even at $T/L = 20$, the $L = 20, 22, 24$ gaps are 0.08833, 0.07879, and 0.06195, with residual tangent drift of 2.1–3.3% and a 40% shift between free-intercept and through-origin scaling estimates. These runs therefore demonstrate that scaling observation time with circumference does not by itself deliver the fixed-size temporal limit. Their driver also failed to pass the declared sample seed into the canonical engine's random initial state, so they are convergence diagnostics rather than replayable production evidence.

The tangent implementation itself has a strong local check: the V0 five-scale comparison between the analytic tangent and finite differences reaches a minimum relative error of 7.37×10^{-11} . This validates the differential on the tested CPU cases, but no saved production campaign yet combines that validated map with T1's replay snapshots,

TABLE X. Completed valid legacy work relevant to T1. Parentheses give the standard error in the last quoted digits where it is available. The saved spectra are useful finite-time precursors and are not pooled into the preregistered T1 scaling fit.

artifact and coverage	strongest quantitative result	T1 implication and limitation
Canonical CPU wall and uniform spectra, $N_x = 20$, $N_y = 20, 22, 24$, $S = 25$, $T = 2N_y$	Wall gaps are 0.11249, 0.10830, and 0.10142, versus 0.89365, 0.89638, and 0.89600 for uniform controls. Interface weights are 0.99365, 0.99584, and 0.99222.	Strongest provenance-complete geometry control and localization result. Only three narrow sizes, raster order, and an individual QR vector are available; no T1 temporal, rank, frame, or two-wall-subspace gate was run.
Canonical GPU wall and uniform spectra, $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$, $T = N_y$	Wall gaps fall from 0.02698 to 0.01968, while uniform gaps stay in $[0.72169, 0.76760]$. Direct wall-column weights rise from 0.9326 to 0.9435 and neighbor-inclusive weights from 0.9611 to 0.9708.	Broadest valid circumference trend and independent GPU confirmation. The observation window is short and size dependent, with no declared alignment, rank/frame repeats, or low-subspace localization.
Canonical CPU CFT-extraction histories, $S = 10$, $T/L = 2, 5, 20$	At $T/L = 5$, gaps are 0.05914, 0.04180, and 0.03651, but final-quarter drifts are 18.2%, 15.8%, and -3.0% . At $T/L = 20$, residual drift remains 2.1–3.3%.	Direct evidence that finite-time bias remains material. There is no declared alignment, localization save, or fully reproducible initial-state seed, and the long run spans only $N_y = 20, 22, 24$.
V0 analytic-tangent validation and cocycle infrastructure	The five-scale finite-difference test reaches relative error 7.37×10^{-11} ; CPU replay, resume, and serial-shard checks pass on validation sizes.	Supports the production estimator and save contract, but is method validation rather than evidence for a wall-localized thermodynamic near-marginal sector.

independent-start and alignment protocol, five accepted sizes, rank/frame repeats, two-wall low-subspace localization, and $S = 100$. Table X separates the physical precursors from this method-only validation.

g. Downgrade, figure, sharding, and legacy. If the intercept is unresolved, call the result finite-size softening rather than a thermodynamic closing. Results supply Fig. 5(b,c). Shard whole trajectories; never split one QR history across unordered shards. The valid legacy evidence summarized in Table X establishes a wall-localized finite-size slow sector with a strongly gapped uniform control, but its size-dependent observation windows do not separate finite-time drift from a thermodynamic gap closing. The principal legacy figures are under `colab_lyapunov/figs/`. T1 therefore remains planned pending the complete temporal, finite-size, rank/frame, and two-wall localization gates.

B. Campaign P2: max-mix purification and recordwise tangent bridge

a. Claim and status. Status PLANNED. The tangent singular spectrum quantitatively controls slow purification of a maximally mixed state at the wall.

b. Dependencies and run matrix. Depends on T1. Use the accepted N_x , Eq. (4), $S = 100$, wall and matched trivial controls, and a total duration of $2N_y$ cycles.

c. Initialization, duration, and observation. Set $C(0) = \mathbf{1}/2$. There is *no burn-in*: purification begins at $t = 0$. Save every early cycle and logarithmically spaced late times. Any stationary-window analysis is a separate post-purification operation.

d. Raw saves and estimator. Save global entropy, $\text{Tr}[C(\mathbf{1} - C)]$, purity gap, spatial entropy contour, Chern marker, charge cumulants, and matched tangent singular data. For each trajectory propagate the *same* max-mix record and the full start-to- t cocycle, and validate

$$\tilde{J}_t^\dagger \tilde{J}_t = 4C_t(\mathbf{1} - C_t) \quad (43)$$

record by record at the individual-mode level and again after conversion to entropy. This identity does not directly identify the stationary pure-start T1 spectrum; compare the two only after their record, initialization, and time-window differences are controlled.

Use the entropy contour and the matched wall-off run to define a wall-excess residual $S_{\text{wall}}^{\text{res}}(t, L)$. Fit

$$S_{\text{wall}}^{\text{res}}(t, L) = L^y \mathcal{F}(t/L^z), \quad (44)$$

testing the per-length hypothesis $y = 1$ and a free y . A global entropy initialized from max-mix is not collapsed without this amplitude factor or an explicitly justified normalization. Fit z independently and compare it with the appropriate pair-sum tangent prediction; do not force agreement with the one-leg T1 gap.

e. Controls, statistics, and gate. Require per-record algebraic agreement within numerical tolerance, compatible tangent and purification scaling after the one-leg/pair-sum conversion, bulk purification faster than the wall sector, and a collapse stable to time-window and amplitude-normalization changes.

f. Current legacy result. **Assessment: rapid max-mix purification and a wall-localized late residual are established, but the recordwise tangent bridge has not been run.** The strongest direct purification archive is the canonical GPU campaign in `colab_charge_fluctuations/gpu_data/purification_dynamics_maxmix/campaigns/N20_multi_geometry_nsh1_dwtrunc1_init-maxmix_S100_cycles-2Ny/`. It begins from max-mix and follows 100 perfect-correction trajectories at $N_x = 20$, $N_y = 30, 40, 50$ through $2N_y$ raster cycles. At $t = N_y$, the mean total entropies are 0.8591, 0.8571, and 0.8756, while $\text{Tr}[C(\mathbf{1} - C)]$ is 0.2552, 0.2532, and 0.2621. At $t = 2N_y$ these fall to entropies 0.2444(233), 0.2907(284), and 0.3313(271) and intrinsic charge variances 0.06647(802), 0.08464(991), and 0.09733(952). The corresponding final real-space Chern estimates are 0.9891(44), 0.9908(29), and 0.9879(49). Thus the purification clock is approximately organized by t/N_y over this range, but the nonzero and size-dependent residual at $2N_y$ is not an asymptotic collapse. This campaign saves streamed entropy, charge, and Chern observables, not covariance histories, branch records, purity gaps, entropy contours, or tangent data.

A canonical CPU mass sweep supplies the most useful control. In the same support-terminated slab geometry at $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, and $T = 2N_y$, the topological $\alpha_{\text{in}} = 1$ runs end with entropy 0.3459, 0.2076, and 0.1954 and $\text{Tr}[C(\mathbf{1} - C)] = 0.09898, 0.05444$, and 0.04793 . For the geometrically matched trivial $\alpha_{\text{in}} = 3$ controller, entropy is already only $(1.01, 1.51, 2.02) \times 10^{-8}$ at $t = N_y$ and the variance is at or below 2×10^{-14} . The trivial sector therefore purifies parametrically faster on these sizes. The comparison has only ten trajectories, three circumferences, and $2N_y$ cycles, so it is a strong wall-versus-bulk precursor rather than a scaling determination.

A matched support-construction sweep strengthens this purification inventory. Under `colab_charge_fluctuations/`, 27 untruncated-interface perfect-correction configurations keep `DW=True`, $\alpha_{\text{out}} = 30$, and `dw_truncation=False` at $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, and $T = 2N_y$. At $N_y = 32$, the untruncated final entropy is 0.10488, 0.26286, and 6.06×10^{-5} at $\alpha_{\text{in}} = 1, 1.5, 1.9$, while the matched support-terminated values are 0.18509, 0.45161, and 0.06416. The full-layer untruncated construction therefore retains a genuine spatial interface yet purifies more rapidly below the transition. Since the two arms contain different numbers of measured modes, raw totals and per-mode densities must be compared separately; neither substitutes for the recordwise tangent bridge.

The spatial entropy-contour archive under `colab_small_system_testing/gpu_data/purification_entropy_contours_maxmix/` localizes the residual directly. In the 16×32 , $n_{\text{shell}} = 1$, $S = 10$, $T = 100$ hard-wall run, the final entropy is 0.01934, of which 94.25% lies on the two wall columns and 97.46% lies within the wall columns and their nearest neighbors. The corresponding untruncated construction has final entropy 4.37×10^{-4} . An independent 30×40 , $S = 25$, $T = 100$ hard-wall run ends at entropy 0.16224, with 93.26% on the two wall columns and 96.30% including their nearest neighbors. A canonical CPU 20×20 , $S = 10$, $T = 40$ convergence study gives the same construction dependence: the final entropy is 0.3278(898) bits for the support-terminated wall and 0.01657(639) bits without support truncation. These archives demonstrate spatially concentrated slow purification, but the two constructions have different active supports and do not by themselves define the P2 wall-off subtraction.

Table XI records the resulting evidence hierarchy. No saved artifact combines the complete max-mix branch word with the start-to- t tangent cocycle, endpoint Choi frames, Gram matrix, purity gap, and the same trajectory's entropy spectrum. The available tangent-edge and flux-cocycle drivers implement useful pieces of the required record and stabilization machinery, but have no saved production outputs and do not test Eq. (43). Consequently neither the individual-mode identity nor a tangent-predicted purification exponent has legacy numerical support.

g. Downgrade, figure, sharding, and legacy. If the exact bridge passes but no universal collapse does, retain the identity and report the measured time dependence without a universal exponent. Results supply Fig. 5(d,e). Shard by complete trajectory. Table XI establishes rapid purification, a slower topological-wall sector, and a strongly wall-localized residual, but not the same-record tangent mechanism or a controlled universal exponent. P2 therefore remains planned pending the $2N_y$ wall/control family and the complete bridge save contract.

TABLE XI. Completed valid legacy work relevant to P2. Parentheses give the standard error in the last quoted digits where available. The purification archives are not pooled into a universal collapse because their sizes, durations, active supports, and saved objects differ.

artifact and coverage	strongest quantitative result	P2 implication and limitation
Canonical GPU max-mix wall dynamics, $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, $T = 2N_y$	At $t = N_y$, entropy and variance are nearly size independent at 0.857–0.876 and 0.253–0.262. At $2N_y$, entropy is 0.2444–0.3313, variance is 0.06647–0.09733, and the Chern estimate is 0.988–0.991.	Strongest ensemble purification and charge-sharpening result. It has no matched trivial arm, contour, purity gap, saved record, or tangent cocycle, and therefore lacks the complete bridge save contract.
Canonical CPU topological/trivial slab sweep, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	The topological wall retains final entropy 0.3459, 0.2076, 0.1954 and variance 0.09898, 0.05444, 0.04793; the matched trivial controller is numerically pure by $t = N_y$.	Direct evidence that the wall sector purifies more slowly than the trivial bulk control, but only at three sizes and without the bridge or a long-time collapse.
Canonical GPU entropy contours, 16×32 and 30×40	The final hard-wall residual places 94.25% and 93.26% directly on the two wall columns, rising to 97.46% and 96.30% with nearest neighbors.	Establishes spatial localization of the late entropy. Sample counts are 10 and 25; the comparison arm changes support rather than turning the wall off at fixed support.
Canonical CPU construction comparison, 20×20 , $S = 10$, $T = 40$	Final entropy is 0.3278(898) bits with support truncation versus 0.01657(639) bits without it; late-cycle means are 0.4043 and 0.09011 bits.	Independent confirmation of delayed hard-wall purification, but a one-size construction test with unequal active supports and no spatial or tangent save.
Canonical CPU overlapping-interface sweep, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	The 270-trajectory perfect-correction arm retains $\alpha_{\text{out}} = 30$ without support truncation. At $N_y = 32$, entropy falls from 0.26286 at $\alpha_{\text{in}} = 1.5$ to 6.06×10^{-5} at 1.9.	Strong multi-size evidence that the untruncated explicit interface purifies faster than the terminated slab below $\alpha = 2$; active supports differ and no recordwise tangent or purity-spectrum bundle was saved.
Recordwise tangent bridge	No saved run contains both the max-mix record and the full start-to-time tangent/Choi objects needed to test $\tilde{J}_t^\dagger \tilde{J}_t = 4C_t(\mathbf{1} - C_t)$.	The central P2 mechanism, pair-sum comparison, purity-gap gate, and universal collapse remain untested.

XII. EXACTLY THREE CHIRALITY ROUTES

The three routes use different clocks and prove different statements. Modular time probes the prepared state's entanglement generator; monitored circuit time probes physical Born-ensemble response; twist angle defines an auxiliary integer spectral-flow index. No one route substitutes for another, and none is called a quantized transport coefficient.

A. Campaign H1: modular charge drift

a. Claim and status. Status PLANNED. The wall-reduced modular generator is handed, with opposite signs on oppositely oriented walls.

b. Dependencies and run matrix. Depends on S1. Use $N_y = 32, 48, 64$, the accepted N_x , both oppositely oriented walls, the trivial control, and $S = 100$. A separately reversed-Chern controller is an escalation only if the two-wall orientation test is ambiguous; it is not a baseline trajectory family.

c. Initialization, duration, and observation. Use independent stationary snapshots from S1. Define the wall subsystem before examining the result and construct

$$h_{\xi,w} = \log \frac{1_w - C_{\xi,w}}{C_{\xi,w}}. \quad (45)$$

Launch identical localized packets and stop the modular-time fit before wraparound or loss of wall support.

d. Raw saves and estimator. Save modular spectra, clipping diagnostics, signed packet density, wall retention, width, periodic center of mass, and per-trajectory fitted velocity. Average signed densities and velocities only after per-trajectory construction.

e. Controls, statistics, and gate. Require opposite velocity signs with nonoverlapping 95% intervals, sign reversal with wall orientation, sustained wall localization, and no stable trivial-control drift. Vary the wall window, modular spectral cutoff, and fit interval.

f. Current legacy result. The exact handed target is established and trajectory-derived modular spreading is strong precursor evidence, but the H1 ensemble gate has not been run. The accepted B0 v2 campaign in `experiment_review/b0_exact_domain_wall/results/20260816_191957/` provides the cleanest deterministic calibration at $N_x = 20$, $N_y = 48$. For the coupled construction, the two endpoint-resolved modular velocities are $(-2.930, +4.536)$; for the hard-exterior construction they are $(-2.600, +4.406)$. In both cases the signs persist across all prespecified fit windows, the minimum three-column wall retention is 0.3245 and 0.3349, and the norm drift is below 4×10^{-15} . The v2 endpoint-resolved estimator supersedes the older centered-packet estimator, which mixed the two counterpropagating components and produced small same-sign slopes. This is an exact-state target with no trajectory distribution or trivial modular controller, not an H1 production result.

The strongest stochastic precursor starts from canonical GPU pure-state covariance snapshots under `colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/` and is analyzed in `colab_small_system_testing/analysis_outputs/dynamic_modular_charge_spreading/`. It covers $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, ten trajectories, and snapshots at cycles 5, 10, 20, 50. The most useful product constructs and diagonalizes h_{ξ,y_0} separately for every trajectory and translated cut, evolves each packet, and only then averages its displacement. At cycle 50 and $N_y = 40$, $n_{\text{shell}} = 2$, the four endpoint packets have

$$\Delta y(t_{\text{mod}} = 32) = (8.821(120), -8.813(107), 8.489(109), -8.520(131)), \quad (46)$$

ordered as the two longitudinal endpoints on the $x = 5$ wall followed by those on the $x = 11$ wall. Multiplying by the recorded wall orientations gives the expected opposite sign between physical walls for either endpoint convention. Charge drift is below 7×10^{-15} and the minimum retained window charge is 0.0297–0.0681 in this configuration. However, its quoted standard errors flatten ten independent trajectories and forty translated cuts into 400 observations. The cuts within one trajectory are correlated variance-reduction samples, so these errors are descriptive and are not the required trajectory-bootstrap confidence intervals. Moreover, about 77% of the modular eigenvalues are clipped at $\epsilon = 10^{-10}$ in this configuration.

A second product, `sample_y0_avg_hmod`, averages covariance-derived modular generators before packet evolution. Its cycle-50 $N_y = 40$ fits over $t_{\text{mod}} \in [0, 8]$ give the clean handed pair $(-2.769, +2.645)$ for $n_{\text{shell}} = 1$ and $(-2.138, +1.887)$ for $n_{\text{shell}} = 2$, with $R^2 = (0.846, 0.817)$ and $(0.860, 0.783)$, respectively. These are useful visualization and sign checks, but the nonlinear ordering violates H1's per-trajectory-first estimator and the remaining endpoint packets have low-quality fits. Finally, the older `notebooks/modular_charge_spreading/` archive evolves exact and eight flattened 20×40 constructions for modular time 40. It conserves charge to 4.4×10^{-7} or better, but all nine reported spectra

saturate at ± 23.718998 because of the same cutoff; it is therefore construction calibration rather than an independent stochastic result.

Table XII separates these estimator classes. None of the legacy artifacts provides the accepted-width $N_y = 32, 48, 64$, $S = 100$ trajectory family, a trivial modular control, trajectory-level velocity intervals, and simultaneous wall-window, cutoff, and fit-range sensitivity.

TABLE XII. Completed valid legacy work relevant to H1. Parentheses denote the reported standard error in the last digits. Translated cuts from one circuit trajectory are not counted as independent trajectories for the H1 acceptance gate.

artifact and coverage	strongest quantitative result	H1 implication and limitation
Accepted B0 v2 exact benchmark, $N_x = 20$, $N_y = 48$	Coupled velocities $(-2.930, +4.536)$ and hard-exterior velocities $(-2.600, +4.406)$; signs are window-stable, retention exceeds 0.32, and norm drift is below 4×10^{-15} .	Establishes the endpoint-resolved handed target and corrects the legacy estimator mismatch. It is deterministic and has neither a trajectory ensemble nor a trivial modular control.
Per-trajectory/per-cut GPU modular evolution, $N_x = 16$, $N_y = 30, 40$, $S = 10$	At $N_y = 40$, $n_{\text{shell}} = 2$, the four final displacements are $(8.821, -8.813, 8.489, -8.520)$ with nominal errors 0.107–0.131.	Strongest stochastic precursor and correct construction-before-averaging order. The 400 cut observations contain only ten independent trajectories, and spectral clipping affects about 77% of modes.
Pre-averaged modular generator, $N_x = 16$, $N_y = 40$, cycle 50	The clean endpoint pair has velocities $(-2.769, +2.645)$ for $n_{\text{shell}} = 1$ and $(-2.138, +1.887)$ for $n_{\text{shell}} = 2$ over $t_{\text{mod}} \in [0, 8]$.	Supports the sign pattern visually, but averaging before nonlinear evolution is not the H1 estimator and no trajectory confidence interval follows.
Exact and flattened legacy constructions, 20×40 , modular time 40	Charge drift is at most 4.4×10^{-7} , while all nine spectra terminate at the cutoff values ± 23.718998 .	Useful construction and conservation check; cutoff-saturated spectra and deterministic states prevent an ensemble handedness claim.
H1 production matrix and controls	No saved campaign combines accepted N_x , $N_y = 32, 48, 64$, $S = 100$, a trivial control, per-trajectory velocities, and the prescribed sensitivity grid.	The sign target is strongly motivated, but the stated H1 statistical and control gate remains open.

g. Downgrade, figure, sharding, and legacy. This proves modular handedness, not monitored-time transport. It supplies Fig. 3(a). Shard by complete circuit trajectory and retain all translated cuts within that shard. Table XII establishes exact modular handedness and compatible legacy trajectory spreading, but not the production ensemble, trivial-control null, or trajectory-bootstrap intervals. H1 therefore remains planned.

B. Campaign H2: physical Born response $\chi(k, \omega)$

a. Claim and status. Status PLANNED. The complete Born ensemble has a wall-localized causal response with a single low-energy velocity sign on each wall.

b. Dependencies and run matrix. Depends on W1 and V0 response validation. Use $N_y = 32, 48, 64$, $S = 100$, eight source positions and four stationary time origins per trajectory, and response horizon $N_y/2$. Only the production random-serial order is run.

c. Initialization, duration, and observation. Each of the 100 trajectories begins from an independent product-state draw and runs for $2N_y$ cycles with no separate physical burn-in. Apply the infinitesimal local density-potential kicks at the four preregistered late time origins whose response horizons fit inside that fixed trajectory. Sources and time origins inside one trajectory are correlated variance-reduction samples, not new trajectories.

d. Raw saves and estimator. For an observable O , evaluate the exact decomposition

$$\partial_\epsilon \mathbb{E}_{\xi, \epsilon}[O_{\xi, \epsilon}]|_0 = \mathbb{E}_\xi[\partial_\epsilon O_\xi] + \mathbb{E}_\xi[O_\xi \partial_\epsilon \log p_\xi]. \quad (47)$$

Save the conditional derivative, score term, total signed response, and cumulative score separately. Validate Eq. (47) against brute-force $\pm\epsilon$ ensembles on small systems. Compute

$$\chi_w(k, \omega) = \sum_{t, y} e^{i\omega t - iky} w_t \chi_w(y, t), \quad (48)$$

with a recorded window w_t . Zero padding is visual only; ridge fits use native resolution. Define in advance a low-energy wedge asymmetry between $\omega k > 0$ and $\omega k < 0$.

e. Controls, statistics, and gate. Bootstrap whole trajectories. Require a low-energy linear ridge, opposite slopes on the two walls, a stable sign across the declared random-serial ensemble, subdominant wrong-chirality weight, wall localization, and no corresponding trivial-control ridge. Both terms in Eq. (47) must be included in the headline panel.

f. Current legacy result. Only target calibration and conditional-response precursors exist; there is no saved full Born susceptibility. The accepted exact-domain-wall campaign under `experiment_review/b0_exact_domain_wall/results/20260816_191957/` supplies the clean Hamiltonian target. A localized density-phase impulse followed by central differencing produces paired physical velocities $(-0.8286, +0.8286)$, $(-0.8720, +0.8720)$, and $(-0.8947, +0.8947)$ at $N_x = 20$ and $N_y = 32, 48, 64$. At $N_y = 48$ the wavefront fits have $R^2 = 0.9938$, charge drift below 4×10^{-16} , and relative finite-difference error 5×10^{-7} . The archive also contains the corresponding momentum-frequency arrays. This is deterministic evolution under an exact Hamiltonian, so it calibrates orientation, Fourier conventions, and ridge extraction but contains neither adaptive records nor a likelihood-score term.

The strongest canonical-circuit precursor is the paired-reset benchmark under `topological_frustration_diagnostics/results/campaigns/20260812_233722_production/03_parallel_benchmark/response/`. It evolves eight 12×16 random-order trajectories after eight burn-in cycles and compares locally occupied and empty unit-cell resets for four response cycles. The diagonal wall velocities are $-0.252(322)$ and $-0.635(329)$, so the two walls do not have opposite signs. The paired branches share their visit schedule, but they do not share a realized outcome record. Moreover, an occupied-versus-empty reset is a finite injected charge, not the infinitesimal density-potential derivative in Eq. (47). This result is therefore a useful negative pilot for the current estimator, not evidence against Born-ensemble chirality.

The V1 smoke archive under `validation_campaigns/results/v0_v1_cpu_20260816_183244/smoke/v1/` makes the schedule problem explicit at 4×6 with only two trajectories per schedule. Random order gives two positive conditional velocities, 0.823 and 0.491; raster order gives 1.265 and -0.697 ; reverse raster gives means -0.747 and -0.560 , with intervals that include zero. The analysis classifies the result as schedule dependent, while its parent validation fails stationarity and explicitly defers the Born score to H2. A later 16×16 archive under `validation_campaigns/results/v1_cpu_N16x16_S10_20260816_225400/` contains one completed representative benchmark trajectory and a runtime projection, not the advertised 30-trajectory response campaign.

The score identity and record-Fisher increment are derived in the reference material under `tangent_edge_channel_note/`, while reusable tangent infrastructure exists under `tangent_edge_channel/`. Neither directory contains an H2 response run. In particular, no saved circuit artifact contains the conditional derivative, cumulative score, likelihood-score response, their sum, and $\chi_w(k, \omega)$ on the same trajectories. Table XIII records this limited evidence without promoting a finite-reset or fixed-record response to the physical Born susceptibility.

g. Downgrade, figure, sharding, and legacy. If only the fixed-record term is chiral, label it conditional response and do not claim physical Born-ensemble chirality. Results supply Fig. 3(b), with the two wall orientations shown inside one compound panel. Shard trajectories while keeping all sources together. Table XIII establishes the exact handed target and inventories the available conditional pilots, but no compatible legacy $\chi(k, \omega)$ result includes the score term. H2 therefore remains planned.

TABLE XIII. Completed valid legacy work relevant to H2. Parentheses give the standard error in the last quoted digits where available. Exact-Hamiltonian and conditional-response results are retained as calibration or pilot data, not as the full Born response.

artifact and coverage	strongest quantitative result	H2 implication and limitation
Accepted exact-domain-wall response, $N_x = 20$, $N_y = 32, 48, 64$	Opposite velocities evolve from ∓ 0.8286 to ∓ 0.8947 ; at $N_y = 48$, $v = \mp 0.8720$, $R^2 = 0.9938$, and charge and finite-difference checks pass.	Strong deterministic target for wall orientation and ridge analysis. It has no adaptive Born record or score contribution.
Canonical paired-reset circuit benchmark, 12×16 , $S = 8$	Random-order diagonal velocities are $-0.252(322)$ and $-0.635(329)$, so the opposite-wall test fails.	Strongest circuit response archive, but it uses a finite charge reset and separately sampled branches rather than an infinitesimal kick with full Born reweighting.
V1 conditional-response smoke, 4×6 , two trajectories per schedule	Random gives $(0.823, 0.491)$, raster gives $(1.265, -0.697)$, and reverse raster is unresolved; the saved classification is schedule dependent.	Exposes schedule and stationarity sensitivity. The score was explicitly deferred, and the sample count is diagnostic only.
Score-response implementation and H2 production data	The score increment is derived in reference notes, but no saved run contains both terms of Eq. (47) or a circuit $\chi_w(k, \omega)$.	The central physical-response estimator, trivial-control null, schedule reversal, size family, and trajectory-bootstrap gate remain untested.

C. Campaign H3: frozen-record integer twist flow

a. Claim and status. Status PLANNED. A continuous flux family constructed from a single realized record has unit wall-resolved spectral flow.

b. Dependencies and run matrix. Depends on S1 and V0 twist validation. Use $N_y = 24, 40, 64$, $S = 100$, and

$$\phi_m = 2\pi m/64, \quad m = 0, \dots, 64. \quad (49)$$

At one representative size repeat 33, 65, and 129 points and move the twist seam.

c. Initialization, duration, and observation. Sample each record physically at $\phi = 0$ from its own independent product-state start for a total of $2N_y$ cycles, with no separate physical burn-in. Freeze that entire discrete record, including the external ordered controller word and every Born outcome. Insert a one-particle $U(1)$ connection with holonomy $e^{i\phi}$ around the periodic y direction into every controller projector before support truncation, and replay the same word and outcomes at every ϕ . In a seam gauge this places the phase on controller-orbital components crossing one declared periodic seam; a uniformly distributed vector-potential gauge is equivalent and closes at 2π up to a large gauge transformation. Never draw an independent Hamiltonian, schedule, or outcome record at each twist. Appendix A gives the complete construction and explains why independently sampled records do not define a spectral-flow family.

d. Raw saves and estimator. Save bit-packed records, $p_\xi(\phi)$ in log form, purity gaps, wall entanglement eigenvalues and vectors, overlap matrices, tracking assignments, crossing tables, and 2π closure residuals. The replay covariance is used transiently to form these quantities at each twist point and is then released; it is not written to the production archive. Track

$$\varepsilon_a(\phi) = \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)} \quad (50)$$

by maximum-overlap subspace transport and count signed zero crossings separately at the two walls. The production statistic is evaluated on $S = 100$ independently sampled frozen records. A complete twist circle is an internal deterministic descendant of one record and never increases the trajectory sample count.

e. Controls, statistics, and gate. Require nonzero branch weight for the full circle, closure up to the large gauge transformation, $|\nu_{\text{flow}}| = 1$ in at least 95 of 100 records, opposite signs on the two walls, zero flow in the trivial control, and stability to grid, seam, tracking tolerance, and reference level inside the entanglement gap.

f. Downgrade, figure, sharding, and legacy. A record whose branch weight becomes exactly zero does not define this flux family and may not be replaced by another record. If too many fail, report the bulk Bott number and nonquantized spectral behavior only. Results supply Fig. 3(c). Shard by frozen record, keeping its complete ϕ circle on one shard. The directory `tangent_cocycle_flux_snapshot/` is code reference only; no compatible saved production figure is identified.

XIII. TRAJECTORY-RESOLVED UNIVERSALITY

A. Campaign R1: Born self-information and charge-resolved record spectrum

a. Claim and status. Status PLANNED. After an explicit replica/transfer normalization is validated, independent universal data $c_{\text{eff}}^{\text{rec}}$ and k_{rec} can be reconstructed from the classical measurement record without trajectory-state tomography and then compared with the state data.

b. Dependencies and run matrix. Depends on W1, T1, and an independently calibrated record velocity. Use the accepted N_x , $N_y = 16, 20, 24, 28, 32, 40, 48, 64$, $S = 100$, a matched trivial controller with identical active-mode count, and exactly $2N_y$ total cycles from each independent initial state. The available record aspect ratio is therefore $T/N_y \leq 2$; nested subblocks are correlated diagnostics and are never treated as independent trajectories.

c. Initialization, duration, and observation. Every trajectory is independently initialized and retained from $t = 0$ through $2N_y$. There is no discarded physical burn-in. Retain the trajectory as the bootstrap unit and report early-versus-late block sensitivity explicitly. If the record estimator is not stable over the fixed window, R1 fails its temporal gate rather than silently extending the run.

d. Raw saves and estimator. For each record save its self-information

$$I_\xi(t_1, t_2) = - \sum_{j \in [t_1, t_2]} \log p(m_j | \xi_{<j}). \quad (51)$$

Its Born mean is a Shannon self-information, not automatically a CFT free energy. The required bridge is the replica partition function

$$Z_n(L, T) = \sum_{\xi} p_{\xi}^n, \quad -\partial_n \log Z_n|_{n=1} = \mathbb{E}_{\xi}[I_{\xi}]. \quad (52)$$

Derive the transfer normalization and boundary conditions of Z_n for the implemented schedule before converting a coefficient to a central charge. Form a wall-minus-matched control subtraction with identical event count and define f_{def} per wall, per monitored cycle, and per wall length. First fit the agnostic form

$$f_{\text{def}}(L, T/L) = f_{\infty}(T/L) - \frac{A_{\text{rec}}(T/L)}{L^2} + \frac{b_4(T/L)}{L^4} + \dots \quad (53)$$

Only if the replica derivation fixes the chiral-cylinder convention may one identify the large-aspect-ratio amplitude as

$$A_{\text{rec}} = \frac{\pi v_{\text{rec}} c_{\text{eff}}^{\text{rec}}}{12} \quad (54)$$

per wall. For both walls before division by wall number, the coefficient doubles. The estimator is the arithmetic Born mean of $-\log p_{\xi}$, never the logarithm of an average probability.

Calibrate v_{rec} in two ways: spacetime collapse of connected local record correlations and momentum/twist dispersion of a charge-identified low transfer sector. Fit the record current level independently through

$$h_{q,n}^{\text{rec}} = \frac{q^2}{2k_{\text{rec}}} + n, \quad q \in \mathbb{Z}, \quad n \in \mathbb{N}_0, \quad (55)$$

including descendant degeneracies. At minimum resolve the $q = \pm 1$ primary and a neutral current or descendant sector. Test $k_{\text{rec}} = 1$ and $c_{\text{eff}}^{\text{rec}} = c_{\text{edge}}$ only after the independent record fits.

e. Controls, statistics, and gate. Use late-time block increments and trajectory bootstraps. Require stationarity, stable L^{-2} coefficients after deleting small sizes and adding L^{-4} , convergence in temporal aspect ratio, agreement of the two velocity calibrations, a vanishing matched-trivial defect coefficient, and at least two nontrivial charge-resolved dimensions consistent with one common k_{rec} . The normalization must also reproduce an exactly enumerated small-system Z_n and its derivative at $n = 1$.

f. Current legacy result. **Assessment: a substantive record-entropy feasibility result, but not an R1 record-CFT result.** The strongest admissible data are the canonical-GPU perfect-correction runs under `colab_charge_fluctuations/gpu_data/streaming_covariance_observables/` at $N_x = 20$, $N_y = 30, 40, 50$, with $S = 100$, respectively 60, 80, 100 cycles, and the deterministic raster- y schedule. At every site and cycle they save the four sequential premeasurement target-outcome probabilities $s_{A+}, s_{A-}, s_{B+}, s_{B-}$. Because the schedule is deterministic, these data support the chain-rule conditional-Shannon estimator

$$\hat{H}_{\text{chain}} = \sum_j h(s_j), \quad h(p) = -p \log p - (1-p) \log(1-p), \quad (56)$$

whose average over histories estimates the Born mean of the outcome self-information. The realized outcome bits were not retained, however, so Eq. (56) does not reconstruct any individual I_{ξ} , its block fluctuations, or its large-deviation statistics.

Over the final half of each run, the mean values of $\hat{H}_{\text{chain}}$ per monitored cycle are 89.661(94), 119.782(101), and 149.762(100) nats at $N_y = 30, 40, 50$, respectively, where parentheses denote trajectory standard errors in the last quoted digits. Division by N_y gives 2.9887, 2.9945, and 2.9953 nats. This near-constant ratio establishes a well-resolved extensive background; it is not the subleading wall-minus-control L^{-2} coefficient in Eq. (53). At 20×40 the first ten cycles give approximately 268.3 nats per cycle, while cycles 11–80 remain near 119.8 nats per cycle. In radius-one neighborhoods of the two controller walls, the late-time binary entropy is 0.1013 nats per measurement event, compared with 0.0101 outside those neighborhoods. Thus the available record uncertainty is strongly wall enhanced even though its proposed universal finite-size coefficient has not been isolated.

Smaller caches under `notebooks/born_probability_covariance_snapshots/` contain 16×30 and 16×40 , $S = 10$ end-of-cycle probability maps at four snapshot times. They establish spatially resolved probability histograms but are hypothetical end-of-cycle evaluations, not the chronologically realized record likelihood. The matched 12×16 wall/uniform pilot under `topological_frustration_diagnostics/results/campaigns/20260812_233722_production/04_activity_pilot_imperfect/` saves record activity and a finite-time SCGF; these are useful record-observer and control pilots, but activity is not self-information. Finally, `markov_transfer_operators/` and `colab_regularized_choi_transfer_matrix/` provide transfer-operator formalism and spectral infrastructure. Their saved spectra are not the replica-normalized, charge-resolved record spectrum required here.

TABLE XIV. Admissible legacy evidence relevant to Campaign R1. The Shannon quantities are derived from saved sequential premeasurement probabilities and use natural logarithms.

legacy source	available result	R1 interpretation
Sequential maps	probability $20 \times (30, 40, 50); S = 100; \hat{H}_{\text{chain}}/N_y = (2.9887, 2.9945, 2.9953)$	Canonical circuit data resolve a stationary extensive Shannon background, but no matched control subtraction or R1 size sequence is present.
20×40 spatial and temporal diagnostic	Wall/nonwall $h(s) = (0.1013, 0.0101)$ nats per event; transient followed by a 119.8-nat-per-cycle plateau	Record uncertainty is wall enhanced; this does not determine A_{rec} , v_{rec} , or $c_{\text{eff}}^{\text{rec}}$.
Snapshot and activity pilots	Two $N_x = 16$ probability-map sizes and one matched 12×16 activity/SCGF pair	Useful observer checks only; neither artifact supplies realized log likelihoods or the required replica estimator.
Missing products	production No outcome bits, individual I_ξ , aspect-ratio blocks, exact Z_n , record velocity, or charge-resolved tower	No legacy value may be assigned to A_{rec} , $c_{\text{eff}}^{\text{rec}}$, or k_{rec} ; R1 remains planned.

g. Downgrade, figure, sharding, and legacy. Without the replica/transfer derivation report A_{rec} as a defect self-information coefficient, not a central charge. With the derivation but without an independent velocity report $v_{\text{rec}} c_{\text{eff}}^{\text{rec}}$, not $c_{\text{eff}}^{\text{rec}}$. One low gap is not a CFT spectrum. Results supply Fig. 4. Shard by whole trajectory. No previous cumulative/no-burn central-charge output is compatible with this estimator. Table XIV instead records the available chain-rule Shannon pilot and the missing bridge to the R1 universal quantities; R1 remains planned.

XIV. MEAN, POSTSELECTION, AND NOISE CONTROLS

A. Campaign M1: trajectory ensemble versus mean channel

a. Claim and status. Status PLANNED. This campaign determines which wall signatures survive without access to the measurement record and tests the sharper possibility that a gapped unconditional map coexists with a closing conditioned tangent sector. It does not assume in advance that the mean channel is trivial or noncritical.

b. Three mean evolutions that must not be conflated. Let $P_j = |\chi_j\rangle\langle\chi_j|$, $Q_j = \mathbf{1} - P_j$, target s_j , and $\eta_j = 2s_j - 1$. For the fresh-ancilla perfect-correction primitive, averaging only the Born outcome gives the exact one-step replacement channel

$$\mathcal{E}_j(\rho) = |s_j\rangle\langle s_j|_{\chi_j} \hat{\otimes} \text{Tr}_{\chi_j} \rho, \quad \mathcal{R}_j(G) = Q_j G Q_j + \eta_j P_j. \quad (57)$$

Equation (57) uses the graded fermionic tensor factorization and is applied to the parity-even, number-conserving states of the present protocol. For a prescribed chronological schedule $\pi = (j_1, \dots, j_M)$, the exact discrete mean sweep is

$$\mathcal{R}_\pi = \mathcal{R}_{j_M} \circ \dots \circ \mathcal{R}_{j_1}. \quad (58)$$

It is the Born average of the implemented perfect-correction circuit at fixed external schedule. Since each replacement maps a Gaussian state to a Gaussian state, $\bar{\rho}_\pi$ remains Gaussian. An additional mixture over randomized schedules need not be Gaussian, although its covariance remains the corresponding average of Eq. (58). A history with a newly randomized word each cycle is a nonautonomous sequence of affine maps; it does not possess one stationary single-sweep multiplier spectrum. Spectral multipliers are defined only for a periodically repeated word, an explicitly averaged/symmetrized autonomous map, or as exponents of a separate mean-map cocycle.

Define separately the Poissonized continuous-time model

$$\mathcal{L}_{\text{pc}} = \sum_j r_j (\mathcal{E}_j - \mathbf{1}), \quad \dot{G} = \sum_j r_j (Q_j G Q_j + \eta_j P_j - G) \equiv \mathcal{A}[G] + B. \quad (59)$$

This is a valid Lindblad generator for independent Poisson applications of the local replacement channels. Use $r_j = 1$ when one unit of continuous time is to contain one expected firing of each channel. It is not the logarithm of the ordered sweep, $\mathcal{L}_{\text{pc}} \neq \log \mathcal{R}_\pi$ in general, and agreement with the discrete circuit is an empirical coarse-graining test.

The bath-density generator in `CI_Lindblad_DW.py`, with one occupation $\bar{n}_a$ controlling both gain and loss and an optional decoherence term, is a third model. It describes the earlier Markovian-bath approximation, not the exact mean of the fresh target-occupation ancillas. Its affine-vectorization, partial Fourier transform, momentum-transfer sectors, and pulse-response routines are useful infrastructure, but production must rebuild P_j from the canonical controller and implement Eqs. (57) and (59). Do not tune $\bar{n}_a$ and call the result perfect correction.

c. Dependencies and run matrix. Depends on V0, W1, S1, and T1. Validate the density-matrix channel by exact Fock-space enumeration on 2×2 and 2×3 systems and the covariance implementation on 4×6 and 6×8 . Production consumes every size already present in the primary and secondary wall/control master families; it does not fill missing protocol-size cells merely to make a rectangular M1 grid. For random serial order, use the same $S = 100$ external schedule seeds as the paired trajectory family and compute one deterministic fixed-schedule mean evolution per seed. No raster or reverse-raster production comparison is made. If the one-record-per-schedule decomposition cannot distinguish Born from schedule variance, a separate fixed-schedule $S = 32$ pilot on the two smallest primary-wall sizes may be launched, but it is not part of the minimum campaign or any manuscript gate.

For spectral work, begin with $N_x = 8, 12$ and $N_y = 16, 24, 32, 48, 64$ and proceed to the accepted production width only if matrix-free residuals and memory estimates pass. Never construct the dense $N_{\text{sp}}^2 \times N_{\text{sp}}^2$ covariance superoperator at production sizes. Apply $\mathcal{A}$ through rank-one projector actions and use matrix-free Arnoldi/Krylov methods. Translation invariance along the wall permits exact momentum-transfer sectors $q = k_y - k'_y$ only after verifying $[\mathcal{R}_\pi, \mathcal{T}_y] = 0$. This holds for the translation-invariant Poisson sum with uniform rates and for an explicitly translation-symmetrized autonomous map, but a generic random-serial or raster word breaks one-site translation. Use real space or the actual supercell momentum in that case. When the symmetry is present, the affine source lies in $q = 0$.

d. Initialization, duration, and raw saves. Use exactly the same initial covariance, controller-frame hash, domain-wall definition, schedule word, and channel ordering as the paired trajectory run. The deterministic discrete map uses the same fixed $2N_y$ trajectory window. The Poissonized model begins at the same state at $t = 0$ and is evaluated over a separately declared physical-time window; it does not inherit or manufacture a discrete burn-in.

Save fixed-schedule $\bar{C}_\pi(t)$, the schedule-averaged covariance, residuals of the affine fixed-point equation, the homogeneous discrete sweep action

$$\delta G \mapsto Q_{j_M} \cdots Q_{j_1} \delta G Q_{j_1} \cdots Q_{j_M}, \quad (60)$$

and the leading eigenpairs of both this map and $\mathcal{A}(q)$ when the corresponding map is autonomous. For randomized histories save the complete map word and accumulate mean-map-cocycle exponents rather than diagonalizing a fictitious average word. Save Hermiticity, spectral-interval, trace/charge, complete-positivity small-system, Fourier-block reconstruction, and Krylov residual diagnostics.

e. Linear-observable and record-rate identities. Compare the deterministic mean covariance with $S^{-1} \sum_{\xi=1}^S C_\xi(t)$ for all one-body densities, currents, charges, and selected matrix elements. Before channel j , the wrong-outcome probability is

$$p_j^{\text{wrong}} = \frac{1 - \eta_j \text{Tr}(P_j G)}{2}. \quad (61)$$

Use this identity to predict the mean correction-event density and signed system-charge transfer, then compare with the trajectory-resolved record ensemble globally and as a transverse wall profile. Connected temporal correlations and record large deviations are not fixed by the one-time mean covariance and remain separate observables.

f. State and topology diagnostics. At fixed schedule compare three nonlinear levels:

$$\mathbb{E}_{\xi|\pi}[S(C_{\xi,A})], \quad S(\bar{C}_{\pi,A}), \quad S_G(\mathbb{E}_\pi[\bar{C}_{\pi,A}]). \quad (62)$$

The middle quantity is the exact entropy of the fixed-schedule Gaussian mean state. The last is only the entropy of the Gaussian state sharing the schedule-averaged covariance; on small Fock-space systems compare it with the actual entropy of the schedule mixture and save Wick-factorization residuals. Likewise report $\mathbb{E}_\xi[|C_\xi(r)|^2]$ and $|\mathbb{E}_\xi C_\xi(r)|^2$ separately.

Resolve the stationary mean-covariance occupation spectrum $\nu_a(k_y)$, wall weights of states nearest $\nu = 1/2$, and bulk and wall purity gaps. A projector obtained by flattening occupations above $1/2$ may be assigned a Chern marker only where a purity gap is open. If the wall branch crosses $1/2$, quote bulk-window markers and spectral flow rather than a globally gapped mixed-state invariant. A covariance Chern marker of a non-Gaussian schedule mixture is a one-body diagnostic, not automatically a density-matrix invariant.

g. Mean relaxation and response. For the discrete map report multipliers $\mu_a(q)$ and cycle exponents $-\log |\mu_a|$; for the Poissonized map report eigenvalues $\lambda_a(q)$ and

$$\Delta_{\text{mean}} = \min_{q, a \notin \text{stationary}} [-\Re \lambda_a(q)]. \quad (63)$$

Identify every excluded stationary or symmetry mode and localize each slow eigenoperator in x . The q label is used only after the translation test above; otherwise report real-space or supercell eigenmodes. In the support-terminated geometry construct and diagonalize the relaxation map on the same active slab used by the conditioned tangent calculation. The full-space map has frozen-exterior unit multipliers, and diagonalizing it first and discarding those modes afterward is not a physical gap definition. These spectra are not the conditioned tangent Lyapunov spectrum. The central comparison is whether $\Delta_{\text{mean}} = O(1)$ while the wall-localized conditional tangent gap closes.

Use the homogeneous maps to evolve a physical small perturbation δG around the stationary mean. Prefer a local unitary kick or a validated symmetric source that keeps $0 \leq C \leq 1$ over simply adding an occupation to a saturated mode. Measure the wall-projected $\chi_{\text{mean}}(k, \omega)$, drift on both wall orientations, DW-off response, and random/raster/reverse controls. A complex linear branch or directed response would show that unconditional one-body dynamics retains chirality; it would not by itself restore the trajectory entanglement CFT or define quantized transport.

h. Controls, statistics, and gate. Linear observables and Eq. (61) must agree with the Born sample within simultaneous trajectory confidence intervals. Require exact-map agreement with Fock-space enumeration, fixed-point residual below tolerance, $0 \leq \bar{C} \leq 1$, time-step convergence for any RK implementation, and agreement between full and q -sector spectra on small systems. Compare exact discrete, Poissonized, and legacy bath models side by side without fitting their clocks after viewing the largest-size result.

This campaign is a conceptual discriminator, not a gate on the existence of the trajectory CFT. Do not describe the averaged state with an effective-temperature interpretation unless one common β fits a finite-temperature CFT correlator, occupation spectrum, and a fluctuation–dissipation or detailed-balance test. If only exponential decay or spectral broadening is observed, use finite correlation length or dephasing scale.

i. Current legacy result. **Assessment: useful mean-map infrastructure and a direct nonlinear-averaging diagnostic exist, but no saved artifact executes the M1 comparison.** The current CPU source under `src/fgtn/` contains a deterministic Markov-channel path whose `perfect_correction=true`, `decoh=true` branch implements the local occupation replacement

$$C \mapsto QCQ + sP, \quad (64)$$

for target occupation $s \in \{0, 1\}$. This is the required one-step covariance algebra, but its admissible evidence is only a 4×6 API smoke test. It has not been checked against exact Fock-space branch enumeration, paired with a saved Born ensemble, or run as a production fixed-schedule mean family.

The strongest saved comparison relevant to the ensemble distinction lies under `figs/corr_y_profiles/`. At $N = 16$ with ten trajectories, it saves both $|\overline{C_\xi}(x, r_y)|^2$ and $|\overline{C}(x, r_y)|^2$ for eight transverse positions and $1 \leq r_y \leq 8$. Averaging the pointwise ratio over the saved nonzero separations gives values between 0.185 and 0.772 across x . The missing fraction is precisely the trajectory fluctuation term in

$$|\overline{C_\xi}|^2 - |\overline{C}|^2 = \overline{|C_\xi - \overline{C}|^2} \geq 0. \quad (65)$$

This directly establishes that covariance averaging removes a spatially dependent part of the squared correlator. The artifact has only filename-level geometry and sampling provenance, however, and $\overline{C}$ was formed from sampled trajectories rather than from an independently evolved exact mean channel.

Saved deterministic histories under `cache/G_history_samples/` cover several small geometries and update-strength scans, including $N = 16$ and 21 examples. Their filenames and the better-manifested 16×24 analysis under `notebooks/markov_channel_negativity/` identify the bath-density choice $n_a = 0.5$ or `perfect_correction=false`; older caches often contain only a covariance array and no run configuration. They therefore do not instantiate Eq. (64). The reported 16×24 negativity fits also have enormous additive values and poor $R^2 \simeq 0.61$, so they are not used as physical M1 evidence.

The legacy bath generator in `CI_Lindblad.DW.py` and the synthesis under `repo_synthesis/main_pass/boss/` describe wall-localized covariance modes, k_y -resolved occupation branches, strip-like local-marker structure, and wall-centered current-like response. Their own provenance audit identifies code-version drift and treats the associated rendered summaries as secondary evidence without a primary raw-data spine. Moreover this bath-density generator is neither the ordered replacement sweep nor the Poisson sum of the canonical projectors. Finally, the extensive spectra under `choi_covariance_cpu/` and `colab_regularized_choi_transfer_matrix/` are realized-record Choi/tangent products, not unconditional mean maps.

j. Downgrade, figure, sharding, and legacy. If the mean map is gapped and noncritical, use one compact main-text panel contrasting its finite correlation length with the trajectory CFT and place its full spectrum in the Supplement. If it has a wall-localized closing or chiral response, promote the conditioned/unconditioned comparison because that changes the central interpretation. The legacy implementation is `CI_Lindblad.DW.py`; its existing occupation spectra, local-marker, current, entanglement-proxy, and pulse-response outputs guide parameter and window selection only. Its defaults use an untruncated or separately truncated spatial-mass construction and do not count as data for either

TABLE XV. Admissible legacy evidence relevant to Campaign M1.

legacy source	available result	M1 interpretation
Current CPU mean-map path	Exact local algebra $C \mapsto QCCQ + sP$; 4×6 smoke test	Required infrastructure exists, but no branch-enumeration validation or production fixed-schedule output exists.
Sampled-covariance correlator comparison	$N = 16$, $S = 10$; $ \overline{C} ^2/ \overline{C}_\xi ^2 = 0.185\text{--}0.772$ across saved x	Direct evidence that nonlinear correlations retain a substantial trajectory-fluctuation piece; not a deterministic-channel comparison.
Saved deterministic channel histories	Small-size $n_a = 0.5$ or perfect_correction=false caches; mixed provenance	Legacy bath-density evolution rather than the exact fresh-ancilla outcome average.
Legacy Lindblad and Choi spectra	Mixed-state wall phenomenology and conditioned transfer spectra	Parameter and window guidance only; the former is a different generator and the latter a different conditioned operator.
Missing M1 production products	No paired linear-observable test, exact schedule mean, Fock enumeration, mean-map gap, fixed-point residual, or canonical mean response	No conclusion about a gapped unconditional map versus a closing conditional tangent sector; M1 remains planned.

canonical wall protocol until the controller frames and parameters match exactly. Table XV separates these legacy controls from the unrun exact replacement-map campaign; M1 remains planned.

B. Campaign M2: postselection and imperfect feedback

a. Claim and status. Status PLANNED. The new wall-CFT, chirality, tangent, and record signatures occur in the physical Born ensemble rather than only in a forced favorable branch; separately, a declared interval of feedback imperfection tests their stability.

b. Dependencies and run matrix. At $N_y = 32, 48, 64$, compare perfect correction with the declared correction-error grid

$$\epsilon_{\text{corr}} = 0.01, 0.025, 0.05, 0.10, 0.20, 0.40, 0.50, \quad p_{\text{gain}} = p_{\text{loss}} = 1 - \epsilon_{\text{corr}}, \quad (66)$$

and with a forced postselected endpoint. Here the imperfect points remain physical stochastic Born trajectories; the error parameter modifies the feedback success probability rather than forcing measurement outcomes. Every stochastic point uses $S = 100$ independent trajectories. The deterministic postselected endpoint is a distinct conditioned control and is not padded with duplicate samples.

c. Initialization, saves, estimator, and controls. Use the same observable definitions as P1, S1, and T1. Store branch probabilities for postselection and never describe a zero-event record as a stochastic ensemble.

d. Current legacy result. **Assessment: several canonical GPU and CPU controls are substantial, but they span three inequivalent protocol axes and do not execute the M2 robustness grid.** The most direct interpolation is the canonical GPU campaign under `colab_partial_post-select/gpu_data/`. It uses $N_x = 20$, $N_y = 40$, $C = 40$, $S = 10$ for $p < 1$, $\alpha_{\text{in}} = 1$, $\alpha_{\text{out}} = 30$, and

$$p = 0, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95, 0.99, 1. \quad (67)$$

For $p < 1$ the engine still runs the physical perfect-correction protocol, but an independent site coin forces the favorable postselected outcome with probability p . The saved forced-site fractions agree with their declared values to within 4.9×10^{-4} . The $p = 1$ endpoint instead invokes the deterministic postselection branch and therefore contains one actual trajectory.

At the final cycle, the trajectory-mean real-space Chern markers for the eight stochastic points in increasing p order are

$$0.9808(114), 0.9582(161), 0.9596(218), 0.9252(240), 0.9276(245), 0.9475(182), 0.9194(260), 0.9283(204), \quad (68)$$

where parentheses are trajectory standard errors. The deterministic $p = 1$ endpoint gives 0.7691 without an ensemble error bar. A descriptive final-cycle fit of each saved entropy curve to

$$S(A_y) = a \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b, \quad 1 \leq A_y \leq 20, \quad (69)$$

TABLE XVI. Admissible legacy evidence relevant to Campaign M2. Forced favorable outcomes, imperfect correction, and absent feedback are distinct controls and are not placed on one unlabelled interpolation axis.

legacy source	available result	M2 interpretation
GPU partial-postselection sweep	20×40 , $S = 10$, eight stochastic $p < 1$ points; final Chern marker 0.919–0.981	Strong forced-branch interpolation with entropy, correlator, charge, probability, and frustration fields; one size and no P1/T1 data.
GPU deterministic postselection	Default-initialized $N_y = 30, 40, 50$ endpoints; one actual trajectory; final marker 0.8731, 0.7877, 0.8821	Valid favorable-branch controls without stochastic error bars; the max-mix family is kept separate.
CPU imperfect-feedback activity	12×16 , $S = 16$; wall/uniform defect rates 0.04055/0.00742 and interface/interior rates 0.08017/0.00885	Genuine imperfect correction and matched spatial control, but only one small-geometry activity point.
GPU no-feedback transfer sweep	$N_y = 20, 30, 40$, $S = 10$; no-feedback and perfect-correction particle–Choi gaps	Endpoint sensitivity of a conditioned transfer construction, not a declared imperfection interval or the physical covariance tangent.
Missing M2 production products	No matched $N_y = 32, 48, 64$, $S = 100$ grid with P1, S1, and T1 observables	No robustness threshold, common finite-size interval, or equivalence between Born and postselected ensembles can be claimed; M2 remains planned.

gives $a = 0.4398(61)$ at $p = 0$, $0.4126(38)$ at $p = 0.5$, and $0.3860(95)$ at $p = 0.99$; the single $p = 1$ curve gives $a = 0.3769$. This mild, nonmonotonic drift is a finite-geometry diagnostic, not evidence for a postselection transition or a precision CFT coefficient. The campaign also saves squared correlators, charge fields, branch probability maps, and measurement-frustration fields, but no covariance histories from which the P1 chirality and T1 tangent observables can be reconstructed.

Larger canonical GPU postselection controls under `colab_charge_fluctuations/gpu_data/` cover $N_y = 30, 40, 50$ for $C = 2N_y$. The default-initialized deterministic branches give final real-space Chern markers 0.8731, 0.7877, and 0.8821, respectively. Although their metadata request $S = 100$, each correctly records one actual trajectory. The corresponding max-mix postselected runs approach a Chern marker of one and total charge variance near 0.5; they are purification controls with a different initialization and are not combined with the default-initialized values.

The one genuine imperfect-feedback legacy point is the matched CPU activity pilot under `topological_frustration_diagnostics/results/campaigns/`. At 12×16 , $S = 16$, $C = 64$, and burn-in 32, the mean defect rate is 0.04055 in the wall geometry versus 0.00742 in the uniform control. Within the wall run it is 0.08017 in the interface region and 0.00885 in the interior. This establishes a strongly wall-localized cost of imperfect correction, but supplies neither an imperfection-strength interval nor the state, chirality, and tangent observables required by M2.

Finally, the no-feedback/perfect-correction GPU sweep already summarized in Table IX supplies an endpoint comparison for a conditioned particle–Choi transfer gap. For example, at $\alpha_{\text{in}} = 1.9$ its no-feedback mean gap decreases from 0.01215(229) to 0.00249(90) between $N_y = 20$ and 40, while the perfect-correction values remain near 0.10. This is evidence that feedback changes the conditioned transfer sector, not a measurement of stability across a continuously tunable feedback error.

e. Acceptance and downgrade. The physical Born-ensemble wall claim requires the perfect-correction point only. Robustness is claimed solely over the tested imperfect interval. Put this comparison in the supplement unless it is indispensable to a mechanism. Table XVI records the available controls without conflating their protocol axes; none supplies the declared imperfect-feedback size grid, so M2 remains planned.

C. Campaign M3: noise and the BPJ bulk transition

a. Claim and status. Status PLANNED. This campaign tests, but does not assume, bulk–boundary locking under the noise perturbation used for the BPJ domain-wall-free bulk transition.

b. Dependencies and run matrix. First run the domain-wall-free geometry at $N = 16, 20, 24, 28, 32$, $S = 100$, on the declared grid

$$\sigma = 0, 0.025, 0.050, \dots, 0.700. \quad (70)$$

Locate and bootstrap the bulk transition using the topological marker, purity gap, and bulk tangent diagnostic. The wall campaign does *not* repeat this complete dense grid. After the bulk gate, select a preregistered bracket consisting

TABLE XVII. Legacy evidence relevant to Campaign M3. Only Eq. (71) defines the BPJ coherent-noise axis used by this campaign.

legacy source	available result	M3 interpretation
Published BPJ bulk study	Uniform $\alpha = 1$ circuit; trajectory and averaged transitions near $\sigma \simeq 0.2$ and 0.55	Defines the perturbation, observables, and scan scale; external baseline rather than a saved repository campaign.
Historical coherent-term notebook	10×10 , $S = 1$, $C = 20$ domain-wall smoke comparison using deterministic $e^{-iH_{\text{DW}}}$	Protocol mismatch with no σ grid, uniform control, finite-size analysis, or current canonical entry point.
Lindblad and click-record controls	Deterministic dephasing and zero-external-noise gain/loss activity	Different generator and record axes; neither realizes onsite random phases between cycles.
Mass and feedback sweeps	Existing topological-to-trivial and perfect/no-feedback comparisons	Useful clean-system controls only; changing α or feedback is not coherent noise.
Missing M3 production products	No uniform $N = 16, 20, 24, 28, 32$ σ grid, purity gap, bulk tangent, or matched noisy wall families	No local estimate of either bulk threshold or bulk-boundary locking; M3 remains planned.

of the nearest points below, at, and above the inferred bulk crossing (expanded by one neighboring point on each side when the uncertainty interval requires it), and run only that bracket over the primary N_y sequence. This staged rule prevents spending the wall GPU budget before the bulk transition has been located.

c. Initialization, saves, estimator, and controls. Use independent initial states and a total of $2N_y$ cycles at every noise value. Save all P1, S1, and T1 observables and the exact noise convention. Estimate bulk and boundary pseudocritical points independently in each bootstrap resample.

d. Current legacy result. **Assessment: the external BPJ baseline fixes the perturbation and expected bulk scales, but the present repository contains no admissible run of that noise protocol.** BPJ inserts, between complete adaptive cycles, the onsite charge-preserving unitary

$$U_\sigma(\theta) = \prod_{\mathbf{r}, \mu} \exp \left[4\pi i \theta_{\mathbf{r}, \mu} \left(c_{\mathbf{r}, \mu}^\dagger c_{\mathbf{r}, \mu} - \frac{1}{2} \right) \right], \quad \theta_{\mathbf{r}, \mu} \stackrel{\text{iid}}{\sim} \text{Uniform}[0, \sigma), \quad (71)$$

with a fresh independent spatial field at each application. In the domain-wall-free $\alpha = 1$ system, BPJ reports loss of the trajectory-resolved topological phase near $\sigma_{c,1} \simeq 0.2$ and a distinct closing of the trajectory-averaged occupation spectral gap, accompanied by a change of its regularized Chern marker, near $\sigma_{c,2} \simeq 0.55$ [1]. Those values calibrate the M3 scan; they are not imported as boundary critical points.

The closest local notebook is under `notebooks/test_coherent_term_markov_circuit.ipynb`. Its historical coherent branch used 10×10 , one trajectory, 20 cycles, a domain wall, and the deterministic full domain-wall Hamiltonian step $G \mapsto e^{-iH_{\text{DW}}} G e^{iH_{\text{DW}}}$ after every cycle. This is neither the independent onsite phase field in Eq. (71) nor a noise-strength scan. The current notebook explicitly states that the canonical CPU driver no longer accepts `coherent_term`; it retains only a baseline smoke run and stale rendered cells that refer to a no-longer-defined coherent output. It therefore supplies implementation history, not M3 data.

Other superficially nearby assets vary different physical axes. The dephasing terms in `CI_Lindblad_DW.py` act in a deterministic trajectory-averaged generator rather than through recordwise coherent phases. The click sequences under `topological_frustration_diagnostics/results/local_gain_loss_clicks/` resolve the intrinsic gain/loss feedback record at zero external noise. The mass sweeps and no-feedback controls change the target Hamiltonian or correction law, respectively. None can be placed on the BPJ σ axis by a clock rescaling or parameter relabeling.

e. Acceptance gate and downgrade. Bulk-boundary locking is accepted only if extrapolated critical noise values have compatible joint confidence intervals. A preemptive boundary crossover or transition is explicitly allowed and must not be forced to coincide with the bulk point. This campaign belongs in the supplement unless it reveals a central new phenomenon. Table XVII records the completed path audit: no saved repository artifact matches both the BPJ observable and Eq. (71), so M3 remains planned.

XV. FIGURE ARCHITECTURE

A. Main figures

a. Figure 1: inherited BPJ platform and validated wall implementation. (a) Vector schematic of the four overcomplete-Wannier occupation measurements and the outcome-conditioned ancillary fSWAP; (b) explicit controller-interface and support-terminated walls, two wall orientations, periodic direction, and flux seam; (c) representative prepared Chern profile; (d) topological and trivial finite-size indices; (e) transverse-converged comparison of wall implementations. The figure validates the inherited topological platform and fixes the two microscopic wall implementations on which the new universal claims are built.

b. Figure 2: universal state data. (a) entropy versus log chord with a $q = 1, 2, 3$ coefficient inset; (b) quantum interval variance and k , with $\delta_{SF}(N_y)$ and its zero target in the inset; (c) the unnormalized trajectory-averaged entanglement contour; (d) max-mix wall-on/wall-off purification; (e) transverse profile of the residual entropy or purity defect. The squared correlator, small- k structure factor details, and topological sweep move to the supplement. Exact-Hamiltonian calibration appears only as a visually distinct gray reference.

c. Figure 3: chirality in three clocks. (a) opposite modular drifts, with the modular clock marked in the panel; (b) total Born $\chi(k, \omega)$ on both walls, including a score/conditional decomposition inset; (c) frozen-record spectral flow with an integer histogram inset. These are exactly three primary panels. A compact three-clock key may sit above them, but is not a fourth result panel. The caption states what each route proves and explicitly denies a quantized transport claim.

d. Figure 4: record-only defect CFT. (a) record spacetime raster and the replica bridge from p_ξ to $\mathcal{Z}_n$; (b) defect self-information versus L^{-2} before and after the fixed subtraction; (c) record spacetime collapse and v_{rec} ; (d) charge-resolved tower; (e) independent state-versus-record universal-number summary. Panels (c)–(e) are shown as CFT data only after the small-system replica normalization has passed.

e. Figure 5: tangent edge channel and purification. (a) tangent-cocycle construction with independent starts and tangent-frame alignment; (b) wall/trivial gap scaling; (c) globally normalized low-mode localization; (d) max-mix purification collapse; (e) recordwise tangent bridge; optional inset comparing trajectory and mean covariance.

B. Supplement and schematic requirements

Supplement figures contain the exact benchmark contact sheet, transverse gate, fixed-window transient diagnostics, fit-window sensitivity, update-order and n_{shell} controls, conditional and score response pieces, twist closure, tangent rank/null diagnostics, postselection, mean-channel comparisons, and noise transition. Every caption states S , the independent sampling unit, initialization, total trajectory window, estimator order, uncertainty type, and fit window.

All conceptual schematics should be vector/TikZ or native illustration assets rather than screenshots. The key sketches are: the actual measurement-and-feedback tile; the two-wall cylinder with entanglement interval, response source, and twist seam; a taxonomy from one record to C_ξ , p_ξ , and $D\Phi_\xi$ versus $\overline{C}$; and the three-clock chirality diagram. A generic brick-wall circuit is not an adequate substitute for the implemented overlapping-orbital schedule.

XVI. PRODUCTION ORDER AND A100 RESOURCE STAGING

Execute in the following stop/go order:

1. V0 small-system validation.
2. B0 exact benchmark and the optional B1 controller-frame pilot.
3. P1 bulk topology.
4. W1 transverse gate and selection of the actual production N_x .
5. S1 state CFT, including the matched entropy–variance spectral test, and S2 topological sweep.
6. T1 and P2, with tangent-frame rank pilots before full sizes.
7. B2 mechanism analysis, reusing the late-time activity records and the completed T1 and P2 outputs.
8. H1, H2, and H3 as three separate datasets.

9. R1 fixed- $2N_y$ record campaign.

10. M1–M3 controls.

Each stage begins with one parameter point and four samples, then a complete $S = 100$ pilot at the smallest accepted size, then the full family. Batch size is autotuned against the 40 GB A100 budget with at least 20% headroom, but a trajectory is never divided in a way that changes its ordered updates or random stream. Colab notebooks import the canonical GPU source as a separate file and end by disconnecting the runtime.

A. Colab bundle and Drive-staging contract

Production artifacts are organized by shared trajectory family rather than by manuscript claim. The intended Colab packages are:

1. **validation**: V0 engine, branch normalization, frozen replay, observer, device-parity, tangent-QR, and compact-archive checks;
2. **bulk_width_gate**: P1 and W1, executed before accepting a production width;
3. **pure_wall_master**: the common S1, T1, and B2 observer run, followed shard-locally by H1 modular observables and H2 response continuations while the required parent states remain live; it emits the reconstructible parent IDs, pre-run RNG states, and ordered records consumed by H3 and R1; the same package also carries the $N_y = 16, 20, 28$ primary wall/control R1 record families and its compact SCGF merger;
4. **chirality_replay**: H3 frozen-record twist circles attached to parent sample IDs, together with exact replay recovery when a completed compact parent archive is resumed in a new runtime;
5. **maxmix_master**: P2, the max-mix S1 observables, and the recordwise tangent bridge; and
6. **scans_and_controls**: S2 and M1–M3, launched only after the corresponding core gates remain viable.

Two or three claims are bundled only when they consume the same trajectory family or a deterministic descendant of it. Bundling never licenses pooling different initial ensembles, controllers, schedule laws, or noise values. The codemaxmix_master and codescans_and_controls runners also require a checksummed schema-versioned launch-decision file. P2 cannot launch until T1 is marked viable; each S2 or M1–M3 case checks only the P1, S1, or T1 decisions on which that case actually depends. A template remains codepending by default, so copying a later bundle to Drive cannot silently bypass the campaign order.

The speed-optimized execution unit is one live five-trajectory shard. S1, T1, and B2 are streamed during its single physical evolution; H1 consumes the declared late covariances before they are released, and H2 forks its finite-difference continuations from the four live origin states instead of replaying the common prefix for every source. Each stage writes an atomic compact receipt, so a later-stage failure does not invalidate completed parent products. H3 is intentionally not fused into this all-or-nothing invocation: its 65–129 full twist replays dominate runtime and use independent restart checkpoints based on the compact RNG state and frozen record.

Google Drive is a staging area rather than the permanent raw-data archive. At most one geometry–size–control arm is live at a time. The $S = 100$ sample list is assigned before execution to deterministic five-trajectory shards. A shard runs first in Colab local storage, computes all streaming observables, writes an atomic manifest and checksums, and only then copies one compact archive to Drive. The permanent shard contains ensemble- resolved observables, replay-capable ordered controller/outcome records, the pre-run RNG state, provenance, and replay hashes. Per-event float64 probabilities are used live where H2 requires them but are reduced to per-cycle sums before Drive archival. A dense parent covariance may exist only as local in-memory or scratch state while observables or immediate descendants are evaluated. It is never an ensemble-level archival product: descendants that cannot run in the same process regenerate the parent exactly from the pre-run RNG state and frozen record and must pass replay-residual and checksum tests. Dense covariance histories are never saved merely for convenience.

The 15 GB Drive allocation may be filled deliberately: after one experiment arm is complete, its checksummed archives are downloaded to the cluster and the staging area is cleared before the next arm. Compact replay records are therefore retained whenever they make an experiment independently reproducible; Drive scarcity is not used to weaken a record. The routine ceiling for any one staged arm or live auxiliary bank is 12 GiB, with 12–14 GiB allowed only as an explicitly reported edge case. A preflight above 14 GiB must split the arm or immediate descendant stages. These storage limits are separate from the 40 GB A100 device-memory contract and do not license sample covariance archives.

The canonical engine has $N_{\text{layer}} = 2N_x N_y$. At $N_x = 20, N_y = 80$, one dense $N_{\text{layer}} \times N_{\text{layer}} = 3200 \times 3200$ complex128 one-particle covariance therefore occupies 163.84 MB (decimal units), not 41 MB. Three such checkpoints for 100 trajectories require 49.152 GB for one arm before records, tangents, or controls; even exact Hermitian-triangle storage requires approximately 24.584 GB. Consequently wall and matched-control arms are processed separately, and a five-trajectory shard is reduced to compact observables before the next shard is admitted. Parent covariances are not copied to Drive, including as a nominally temporary production artifact. H3 keeps every record’s complete twist circle on one shard; T1 keeps every ordered QR history on one trajectory shard.

The implementation exposes these contracts directly in the canonical GPU engine. `record_observer` receives the ordered controller word, Born outcomes, branch probabilities, feedback targets, and cycle/update/sample identifiers without changing the dynamics. `frozen_schedule` together with `frozen_outcomes` replays that exact conditional branch. `set_controller_twist(phi, gauge, seam_y)` applies an absolute rather than cumulative controller twist, with seam and uniform gauges used as mutual controls. `onsite_phase_noise_sigma` inserts the M3 onsite phase kick once after each complete adaptive cycle, while `noise_observer` records per-cycle moments and a SHA-256 digest of the realized phase field. Exact regeneration uses the pre-run RNG state, engine hash, and configuration; the full field is not copied to Drive. Production archives save these compact records and API conventions, not samplewise covariance matrices.

The fixed-schedule M1 descendant is also routed through the canonical `run_markov_circuit` entry point with `mean_replacement=True` and a declared `frozen_schedule`. For each ordered controller channel it applies the exact affine replacement $G \mapsto Q_\chi G Q_\chi + \eta P_\chi$ and streams mean-state observables and homogeneous QR products before discarding the mean covariance. It is paired to the same initial RNG state and controller word as its Born parent; it is not mislabeled as a new stochastic ensemble. The same bundle carries two deterministic M1 jobs that do not consume trajectory shards. The first enumerates the two Kraus branches of every replacement channel in exact Fock space on 2×2 and 2×3 , then compares complete 4×6 and 6×8 mean sweeps with the canonical covariance implementation. The second constructs $\mathcal{A}[X] = \sum_j (Q_j X Q_j - X)$ from the canonical controller frame and uses matrix-free complex128 Lanczos in declared momentum-transfer sectors on the $N_x = 8, 12, N_y = 16, 24, 32, 48, 64$ grid. It saves only leading decay rates, Ritz residuals, translation-sector residuals, and provenance—never a dense covariance superoperator.

B. Audited legacy implementation lineage

Tables XVIII and XIX identify the actual runners and observer packages behind every substantial legacy result used above. An output directory by itself is not treated as an implementation specification. The production bundle inherits a legacy component only when the runner, canonical engine call, observable definition, and output schema have all been identified. *Rebuild* below means that the legacy artifact supplies a diagnostic or formal target but no conforming production runner.

At this audit revision, the bundled `classA_U1FGTN_gpu.py` files in `colab_lyapunov/`, `colab_large_entanglement_scaling_N20/`, and `colab_charge_fluctuations/` are byte-identical to the canonical source. The copies in `colab_small_system_testing/`, `colab_no_feedback_alpha_sweep_transfer/`, `colab_regularized_choi_transfer_matrix/`, and `colab_partial_post-select/` match as well. The canonical file is `src/fgtn/classA_U1FGTN_gpu.py`, and the common SHA-256 is

$$h_{\text{GPU}} = 19\text{b}200\text{dce}7\text{b}5\text{fd}004\text{a}627770492\text{d}32\text{c}573 \\ 8\text{ef}19\text{ff}8\text{f}94066\text{eebad}5\text{ad}66\text{d}9\text{f}23\text{d}. \quad (72)$$

This hash is an audit fact, not a permanent pin: every new bundle copies the then-current canonical source, records its hash, and refuses to run if its observer expects an incompatible engine signature.

TABLE XVIII. Core trajectory and state-data implementation lineage. Paths are repository-relative.

campaign	substantial legacy artifact	audited runner or observer	production use
B0; exact H1/H2	experiment_review/b0_exact_domain_wall/results/20260816_191957/	experiment_review/b0_exact_domain_wall/run_b0_campaign.py with experiment_review/b0_exact_domain_wall/b0lib.py and locked experiment_review/b0_exact_domain_wall/campaign_config.v2.json	Keep as the deterministic CPU calibration and V0 reference. Its entropy, modular, response, and twist routines provide exact tests, not a stochastic Colab runner.
B1	experiment_review/b1_controller_frame/	experiment_review/b1_controller_frame/run_b1_campaign.py calling classA_U1FGTN.run_markov_circuit	Keep as an optional CPU mechanism pilot. Reuse its controller-frame and matched-control constructors only after GPU parity tests.
P1, W1, T1	colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/	colab_lyapunov/notebooks/data_generation/run_lyapunov_spectra_N20_Ny30-50_S25_cyclesNy.ipynb with colab_lyapunov/src/lyapunov_observables_gpu.py	Reuse the canonical A100 launch, streamed Chern estimator, charge observer, and tangent QR plumbing in bulk_width_gate and pure_wall_master. Replace the single selected QR vector by the declared aligned low-subspace products.
B2 and H2	topological_frustration_diagnostics/results/campaigns/20260813_121340_Nx20_perfect_Ny20_22_24/ and topological_frustration_diagnostics/results/campaigns/20260812_233722_production/03_parallel_benchmark/response/	topological_frustration_diagnostics/launch_campaign.py dispatching topological_frustration_diagnostics/run_cpu.py modes activity, spectral, and response	Port the eventwise X_j, Y_j , branch-probability, activity-profile, and paired-response observers. Rebuild H2's likelihood-score term on GPU; the legacy response runner contains only the conditional/reset precursor.
S1	colab_large_entanglement_scaling_N20/ and colab_charge_fluctuations/gpu_data/streaming_covariance_observables/	colab_large_entanglement_scaling_N20/notebooks/run_slope_vs_system_size.ipynb with colab_large_entanglement_scaling_N20/src/strip_entropy_streaming_gpu.py; and colab_charge_fluctuations/notebooks/data_generation/run_streaming_covariance_observables_multi_geometry.ipynb with colab_charge_fluctuations/src/streaming_covariance_observables_gpu.py	This is the primary streaming-observer skeleton for pure_wall_master. Preserve the sample axis and add matched interval-charge spectra and the three declared late checkpoints. sample_average_testing/ contributes only a TMI estimator cross-check, not dynamics.
P2	colab_charge_fluctuations/gpu_data/purification_dynamics_maxmix/campaigns/N20_multi_geometry_nsh1_dwtrunc1_init-maxmix_S100_cycles-2Ny/	colab_charge_fluctuations/notebooks/data_generation/run_purification_dynamics_maxmix_multi_geometry.ipynb with colab_charge_fluctuations/src/purification_dynamics_observables_gpu.py	Use as the maxmix_master base. Add the matched trivial arm, spatial contour, purity-gap output, realized record, and record-wise tangent bridge.
H1	colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/ and colab_small_system_testing/analysis_outputs/dynamic_modular_charge_spreading/	colab_small_system_testing/notebooks/data_generation/run_pure_state_markov_snapshots.ipynb followed by colab_small_system_testing/src/dynamic_modular_charge_spreading.py	Reuse the snapshot validation and modular-Hamiltonian construction in chirality_replay. Perform packet evolution and velocity fitting per trajectory; do not reuse the legacy product that averages modular generators before evolution.

TABLE XIX. Derived, record, scan, and control implementation lineage.

campaign	substantial legacy artifact	audited runner or observer	production use
H3	<code>tangent_cocycle_flux_snapshot/</code> and the exact twist validation in B0	<code>tangent_cocycle_flux_snapshot/run_flux_cocycle_snapshot.py</code> calling the canonical CPU engine; exact reference routines in <code>experiment_review/b0_exact_domain_wall/b0lib.py</code>	Port record freezing, replay checksums, twist checkpointing, and large-gauge closure to <code>chirality_replay</code> . Rebuild the controller-projector twist before support truncation and add wall entanglement-spectrum subspace tracking. The existing tangent flux snapshot is not the H3 estimator.
R1	<code>colab_charge_fluctuations/gpu_data/streaming_covariance_observables/</code> , <code>notebooks/born_probability_covariance_snapshots/</code> , and <code>markov_transfer_operators/</code>	The charge-fluctuation streaming notebook above supplies the canonical GPU trajectory; the other two directories contain analysis caches and transfer formalism rather than a production runner	Extend <code>pure_wall_master</code> with chronological conditional log probabilities, bit-packed outcomes, and charge transfers. Build the replica normalization and charge-resolved transfer analysis as new downstream code.
S2	<code>colab_no_feedback_alpha_sweep_transfer/</code>	<code>colab_no_feedback_alpha_sweep_transfer/notebooks/data_generation/run_no_feedback_alpha_sweep_dwtrunc1_N20_Ny20-30-40_S10_cycles2Ny_gpu.ipynb</code> with <code>colab_no_feedback_alpha_sweep_transfer/src/choi_transfer_observables_gpu.py</code>	Reuse the A100 configuration-grid, manifest, resume, and matrix-free near-gap machinery in <code>scans_and_controls</code> . Replace the legacy no-feedback emphasis by the declared $S = 100$ perfect-correction S2 observables.
T1/P2 method checks	<code>tangent_edge_channel/</code> and <code>colab_regularized_choi_transfer_matrix/</code>	<code>tangent_edge_channel/run_tangent_edge_channel.py</code> ; and <code>colab_regularized_choi_transfer_matrix/notebooks/data_generation/run_particle_choi_transfer_gap_N20_Ny30-50_S25_cycles2Ny_gpu.ipynb</code> with <code>colab_regularized_choi_transfer_matrix/src/choi_transfer_observables_gpu.py</code>	Reuse positive-diagonal QR gauges, scale-separated products, rank diagnostics, and reduced Choi Gram checks. These are validation components, not independent manuscript observables or trajectory families.
M1	<code>src/fgtn/classA_U1FGTN.py</code> , <code>CI_Lindblad_DW.py</code> , <code>figs/corr_y_profiles/</code> , and <code>cache/G_history_samples/</code>	<code>classA_U1FGTN.run_markov_channel</code> is the exact local-replacement CPU path; <code>CI_Lindblad_DW.py</code> is the inequivalent legacy bath generator	Pair the exact deterministic replacement sweep with the same initial state and schedule metadata as the trajectory shard. Treat the bath generator as a labelled control and write new matrix-free production analysis.
M2	<code>colab_partial_post-select/gpu_data/</code>	<code>colab_partial_post-select/notebooks/run_partial_postselection_sweep_N20x40_gpu.ipynb</code> with <code>colab_partial_post-select/src/streaming_covariance_observables_gpu.py</code>	Reuse the complete Colab sweep, streamed observables, and partial-postselection book-keeping. Replace its pilot grid by the declared M2 configurations and $S = 100$ independent trajectories at every stochastic point.
M3	<code>notebooks/test_coherent_term_markov_circuit.ipynb</code>	No conforming runner: rendered cells refer to a removed <code>coherent_term</code> argument, and <code>CI_Lindblad_DW.py</code> implements a different noise model	Rebuild. Add the BPJ onsite random-phase step to the canonical GPU class, validate it in V0, and only then create the M3 portion of <code>scans_and_controls</code> .

TABLE XX. Mandatory interpretation when a gate fails.

observation	licensed wording
c fit works but charge/correlator fails	entropy is consistent with a $c = 1$ wall; no full CFT identification
c and k fit separately but the matched $\pi^2/3$ relation fails	report separate entropy and current coefficients; no minimal $U(1)_1$ identification
wall tangent gap decreases but intercept unresolved	finite-size softening and wall-localized slow mode
modular drift only	handed modular flow, not physical chiral propagation
conditional response only	fixed-record response, not Born-ensemble response
χ chiral but twist not robust	chiral response without an extracted integer wall index
bulk Bott is quantized but $n_{\text{shell}} = 1$ marker does not converge	topological index with nonquantized local-marker finite-range corrections
replica mapping fails	report a defect self-information coefficient, not a record central charge
replica mapping passes but v_{rec} is unresolved	report $v_{\text{rec}} c_{\text{eff}}^{\text{rec}}$ only
one record gap only	slow record sector, not an operator spectrum
explicit-interface and support-terminated walls disagree	two distinct programmable interfaces
bulk and boundary noise points differ	preemptive boundary crossover/transition, not bulk–boundary locking

XVII. CLAIM-DOWNGRADE MATRIX

XVIII. DEFERRED WORK

The core campaign excludes a Chern number assigned to an ad hoc quenched-log tangent Hamiltonian, polar-unitary alignment of independently sampled states, a full scattering network, regularized-Choi endpoint spectroscopy, an absorbing-transition phase diagram, high-rank many-body reconstruction, a broad non-Hermitian survey, and a full Lindblad phase diagram. It also excludes promoting the natural-orbital coarse graining in `notebooks/flattened_hamiltonian_analysis/gaussian_entanglement_rg/` to a MERA result. That legacy analysis usefully tests whether locally mixed modes retained under quasi-one-dimensional Gaussian coarse graining move toward the walls, but its isometries are scale dependent, its repeated-tensor fits have order-one mismatch, and purified variants explicitly project the coarse covariance back to a Slater projector. It becomes main-text relevant only after a repeated finite-depth tensor has controlled discarded-mode error, stable state fidelity or correlators, and a nontrivial prediction beyond re-extracting $c \simeq 1$. These topics are follow-up projects unless a core gate specifically requires one.

Appendix A: Frozen-record $U(1)$ twist family and wall spectral flow

This appendix defines the branchwise twist used in Campaign H3. It is an auxiliary spectral-flow construction for a single realized Born record. It is not a physical monitored-time pump, and its integer is not called a quantized transport coefficient.

1. Boundary holonomy and twisted controller projectors

Let $y \in \mathbb{Z}_{N_y}$ denote the periodic direction and let ϕ be the flux through the cylinder. A one-particle $U(1)$ connection $A_{ij}^{(\phi)}$ is chosen so that the oriented product of Peierls phases around one complete y cycle is

$$\prod_{(ij) \in S_y^1} e^{iA_{ij}^{(\phi)}} = e^{i\phi}. \quad (\text{A1})$$

In a seam gauge all links are untwisted except those crossing a declared bond $y = N_y - 1 \rightarrow 0$, which carry $e^{+i\phi}$ in the positive orientation and $e^{-i\phi}$ in the reverse orientation. A uniform gauge distributes the same holonomy as a phase

$e^{i\phi/N_y}$ per positive- y step. The two descriptions are related by a one-particle gauge transformation U_ϕ and must give identical spectra.

For a normalized controller orbital

$$|\chi_j(0)\rangle = \sum_i w_{ji}|i\rangle, \quad P_j(0) = |\chi_j(0)\rangle\langle\chi_j(0)|, \quad (\text{A2})$$

parallel transport each component from the declared controller center to site i using $A^{(\phi)}$. This defines

$$|\chi_j(\phi)\rangle = \sum_i e^{i\vartheta_{ji}(\phi)} w_{ji}|i\rangle, \quad P_j(\phi) = |\chi_j(\phi)\rangle\langle\chi_j(\phi)|. \quad (\text{A3})$$

The phase difference $\vartheta_{ji} - \vartheta_{jk}$ equals the line integral of the connection between the two components. This prescription matters for a controller whose finite support straddles the periodic seam: multiplying the whole orbital by one common phase would cancel from P_j and would not insert a twist. Construct the full twisted controller frame first and only then apply the support mask. Twisting after truncation can omit seam-crossing components and defines a different protocol.

At $\phi = 2\pi$ the seam-gauge projectors close directly. In a uniform gauge they obey

$$P_j(2\pi) = U_{2\pi} P_j(0) U_{2\pi}^\dagger, \quad (\text{A4})$$

where $U_{2\pi}$ is the large gauge transformation. Every covariance and overlap closure test must therefore be performed in one common gauge, applying $U_{2\pi}$ where required. Moving the seam is a gauge-relocation control rather than a new physical flux family.

2. One record, one continuous conditional branch

At $\phi = 0$, independently prepare and evolve trajectory s and save its complete record

$$\xi_s = [(j_1, m_1), (j_2, m_2), \dots, (j_M, m_M)], \quad (\text{A5})$$

including the external controller/channel word j_e , every Born outcome m_e , the pre-run RNG state (or equivalent counter-based initial-state seed map), and the replay hash. Feedback targets and correction rules remain those of the physical protocol. For the twisted projectors define the unnormalized branch operator and branch weight

$$K_{\xi_s}(\phi) = K_{m_M|\xi_s, < M}(\phi) \cdots K_{m_1}(\phi), \quad (\text{A6})$$

$$p_{\xi_s}(\phi) = \text{Tr}[K_{\xi_s}(\phi) \rho_0 K_{\xi_s}^\dagger(\phi)]. \quad (\text{A7})$$

Whenever $p_{\xi_s}(\phi) > 0$, normalize this same branch and obtain $C_{\xi_s}(\phi)$. The physical sampling occurs only once, at $\phi = 0$; all other twist points are deterministic replays of that record. The resulting family asks how one conditional branch changes under boundary holonomy. It does not assert that $p_{\xi_s}(\phi)$ is uniform or that the frozen record is typical of the Born ensemble at nonzero ϕ .

Drawing a new outcome record at each ϕ would instead produce unrelated random covariances $C_{\xi(\phi)}(\phi)$. Sorting their spectra can manufacture apparent jumps or crossings, but there is no continuous eigenbundle and hence no branchwise spectral-flow integer. Likewise, the external random-serial word is frozen along with the outcomes; a new word at every ϕ changes the microscopic conditional map. If $p_{\xi_s}(\phi) = 0$ anywhere on the circle, the normalized branch is undefined there and record s does not define the required flux family. Such a failure is saved and counted; the record may not be replaced after its twist behavior is known.

3. Entanglement flow, wall resolution, and crossing count

For the preregistered wall subsystem A_w , diagonalize the restricted covariance at every twist point,

$$C_{\xi_s, A_w}(\phi) u_{sa}(\phi) = \nu_{sa}(\phi) u_{sa}(\phi), \quad \varepsilon_{sa}(\phi) = \log \frac{1 - \nu_{sa}(\phi)}{\nu_{sa}(\phi)}. \quad (\text{A8})$$

The reference level $\varepsilon = 0$ is equivalent to $\nu = 1/2$. Between adjacent twist points, form the overlap matrix

$$\mathcal{O}_{ab}^{(m)} = \langle u_{sa}(\phi_m) | u_{sb}(\phi_{m+1}) \rangle. \quad (\text{A9})$$

Track isolated levels by maximum absolute overlap. For a near-degenerate cluster, track the entire subspace by its singular values and polar transport before assigning individual branches; an arbitrary basis rotation inside the cluster is not a physical crossing. Save every overlap matrix, assignment, minimum tracking singular value, and ambiguity flag.

Assign a tracked branch to wall w using its preregistered wall-window weight or a projected transverse-position operator. A crossing between ϕ_m and ϕ_{m+1} contributes the sign of the tracked spectral slope. The wall-resolved flow is

$$\nu_{\text{flow},w}(\xi_s) = \sum_{a,c \in w} \text{Sgn}[\varepsilon_{sa}(\phi_{m_c+1}) - \varepsilon_{sa}(\phi_{m_c})], \quad (\text{A10})$$

where c runs over unambiguous zero crossings of tracked branch a assigned to wall w . The global sign convention is fixed once using the B0 orientation labels. Opposite walls of the periodic domain-wall geometry should carry opposite flow, so the target is $(-1, +1)$ up to the declared orientation convention and the net flow is zero.

Repeat the crossing count after changing the twist grid, moving the seam, varying the tracking tolerance, and moving the reference level within the entanglement gap. A valid integer cannot depend on any of these choices. Verify

$$C_{\xi_s}(2\pi) \simeq U_{2\pi} C_{\xi_s}(0) U_{2\pi}^\dagger \quad (\text{A11})$$

and the corresponding subspace closure. Failure of closure, an unresolved overlap assignment, or a zero branch weight is not silently repaired by smoothing the spectrum.

Production uses $S = 100$ independently Born-sampled records at $\phi = 0$. Each record and its complete twist circle form one statistical unit and one shard. Report the signed-flow histogram, the fraction of records with a well-defined full circle, and the fraction with $|\nu_{\text{flow},w}| = 1$ separately. The 65 twist points, the two walls, and multiple reference levels are correlated deterministic descendants and do not increase S .

Appendix B: Quenched finite-time tangent spectra and localization modes

This appendix fixes the tangent-product convention used in Campaigns T1 and P2. It also separates three objects that are easily conflated: the quenched Lyapunov spectrum, the eigenmodes of a trajectory-averaged finite-time log-strain operator, and an annealed second-moment spectrum.

1. Cycle Jacobian, product, and time normalization

Along realized trajectory s , let $\Phi_{s,t}$ denote one complete normalized circuit cycle and let

$$\delta C_{t+1} = J_{s,t}[\delta C_t], \quad J_{s,t} = D\Phi_{s,t}|_{C_{s,t}}, \quad (\text{B1})$$

after restriction to the declared physical tangent space. The T -cycle product is

$$M_s(T) = J_{s,T-1} \cdots J_{s,1} J_{s,0}. \quad (\text{B2})$$

Here T is the number of complete circuit cycles. The finite-time singular exponents are

$$\lambda_{sa}(T) = \frac{1}{T} \log \sigma_a[M_s(T)]. \quad (\text{B3})$$

Equivalently, on the nonsingular physical subspace define the right log-strain operator

$$L_{s,T}^R = \frac{1}{2T} \log[M_s^\dagger(T) M_s(T)]. \quad (\text{B4})$$

Its eigenvalues are $\lambda_{sa}(T)$ and its eigenvectors are the right finite-time singular vectors, namely input perturbations at the left endpoint of the window. The analogous left operator

$$L_{s,T}^L = \frac{1}{2T} \log[M_s(T) M_s^\dagger(T)] \quad (\text{B5})$$

has the same finite eigenvalues and output singular vectors. The normalization is multiplication by $1/(2T)$, not division by $1/(2T)$. Exact null directions have $\log 0 = -\infty$ and are recorded in a null mask rather than replaced by an arbitrary finite floor.

The reported unit is inverse circuit cycles. The scaled variable T/N_y may be used to display convergence, and the size dependence may subsequently be fitted to $\Delta(N_y) \sim N_y^{-z}$, but N_y is not inserted into Eq. (B3). Normalizing by elementary update count or hardware layer depth defines a different clock and must be reported as a separate conversion.

2. Stabilized QR accumulation and finite-time singular vectors

The dense product in Eq. (B2) is not formed at production sizes. Strong contraction, exact projector nulls, and expanding directions make it ill-conditioned. Starting from an orthonormal rank- r physical frame $Q_{s,0}$, iterate

$$Y_{s,t} = J_{s,t} Q_{s,t}, \quad Y_{s,t} = Q_{s,t+1} R_{s,t}, \quad (\text{B6})$$

with a fixed diagonal-sign convention for $R_{s,t}$. After the declared alignment interval, the QR estimator is

$$\ell_{sa}(T) = \frac{1}{T} \sum_{t=0}^{T-1} \log |(R_{s,t})_{aa}|. \quad (\text{B7})$$

The asymptotic limit gives the Oseledec exponents represented by the retained frame. QR vectors themselves are backward Lyapunov vectors and need not equal covariant Lyapunov vectors or finite-window singular vectors, particularly inside a near-degenerate band.

To reconstruct finite-window singular modes, save block- R factors together with their diagonal log scales and the input/output frames. A stabilized SVD of the coefficient-space product

$$\mathcal{R}_s(T) = R_{s,T-1} \cdots R_{s,0} = \tilde{U}_s \Sigma_s \tilde{V}_s^\dagger \quad (\text{B8})$$

gives retained input and output singular vectors

$$v_{sa}^{\text{in}} = Q_{s,0} \tilde{V}_{sa}, \quad v_{sa}^{\text{out}} = Q_{s,T} \tilde{U}_{sa}. \quad (\text{B9})$$

The rank- r result is exact only within the propagated subspace. Rank doubling and independent initial frames test whether the near-zero band has been captured.

3. Quenched spectrum, mean log strain, and annealed spectrum

The primary quenched spectrum takes the logarithm trajectory by trajectory and then averages continuously tracked indices,

$$\bar{\lambda}_a^q(T) = \frac{1}{S} \sum_{s=1}^S \lambda_{sa}(T) = \frac{1}{ST} \sum_{s=1}^S \log \sigma_a[M_s(T)]. \quad (\text{B10})$$

The gap is formed only after this indexwise average. Means and medians of the per-trajectory minimum gap are retained as distinct sensitivity statistics. Averaging $M_s(T)$ before taking singular values is neither this quenched estimator nor the tangent spectrum of a typical realized record.

If all right log-strain operators have been embedded or parallel transported into one declared common physical tangent space, one may also define

$$\bar{L}_T^R = \frac{1}{S} \sum_{s=1}^S L_{s,T}^R. \quad (\text{B11})$$

This is a useful mean finite-time log-strain operator. Its eigenvectors answer which common directions are slow or expanding under the averaged logarithmic strain. In general, however,

$$\text{eig}(\bar{L}_T^R) \neq \frac{1}{S} \sum_s \text{eig}(L_{s,T}^R), \quad (\text{B12})$$

because the sample log-strain operators do not commute. Consequently there is no unique single vector whose eigenvalue is the indexwise quenched average in Eq. (B10). The eigenmodes of $\bar{L}_T^R$ may be plotted, but they are labelled “mean quenched log-strain modes,” not “the quenched Lyapunov modes.” Since physical tangent spaces at different covariances need not coincide, the embedding or transport convention and its residual are part of this definition.

For comparison, the annealed second-moment operator is

$$L_T^{\text{ann}} = \frac{1}{2T} \log \left[\frac{1}{S} \sum_s M_s^\dagger(T) M_s(T) \right]. \quad (\text{B13})$$

It places enhanced weight on rare records with large stretch and is not substituted for the quenched spectrum. Equations (B10), (B11), and (B13) are saved and named separately whenever more than one is reported.

4. Localization of a nonunique slow band

For an isolated nondegenerate exponent one may plot its input or output singular-vector density, provided the endpoint convention is stated. Near degeneracy, individual vectors rotate under numerical noise, QR gauge, and sample-to-sample wall exchange. The stable trajectory-level object is the projector onto the declared slow band $\mathcal{L}_s$,

$$P_{\mathcal{L},s} = \sum_{a \in \mathcal{L}_s} |v_{sa}\rangle \langle v_{sa}|, \quad \rho_s(x, y) = \frac{1}{r_s} \sum_{a \in \mathcal{L}_s, \mu} |v_{sa, \mu}(x, y)|^2. \quad (\text{B14})$$

Average $P_{\mathcal{L},s}$ only after expressing it in the same ambient real-space basis; otherwise average the gauge-invariant density ρ_s and wall-window weights. Resolve the two walls by diagonalizing the projected transverse-position operator inside $\mathcal{L}_s$ or by fixed left/right windows. Do not average signed eigenvectors, select the most localized QR column, or count a basis rotation inside a degenerate subspace as motion of a physical mode.

When the tangent action factorizes as a one-leg congruence,

$$\delta C \longmapsto K_s(T) \delta C K_s^\dagger(T), \quad (\text{B15})$$

and $K_s(T)$ has one-leg exponents ℓ_{sa} , the full covariance-tangent exponents are pair sums

$$\lambda_{s,ab}^{(C)} = \ell_{sa} + \ell_{sb}, \quad (\text{B16})$$

subject to the physical Hermiticity/projector tangent constraints. A purification rate must therefore be compared with the appropriate covariance pair-sum sector, not automatically with the smallest one-leg gap.

For every trajectory save the QR log increments, block- R factors, frame ranks, null and gauge masks, input/output endpoint convention, slow-band membership, projector density, wall weights, orthogonality residuals, and replay hash. The independent sample count is the number of Born trajectories S ; tangent-frame repeats, QR checkpoints, and multiple vectors within one slow band are correlated numerical diagnostics and do not increase S .

Appendix C: Charge full counting statistics and the level-one current algebra

For a Gaussian trajectory with subsystem correlation matrix $C_{\xi,A}$, the quantum FCS of Q_A is

$$\mathcal{Z}_{\xi,A}(\lambda) = \det[\mathbf{1}_A - C_{\xi,A} + e^{i\lambda} C_{\xi,A}], \quad \mathcal{K}_{\xi,A}(\lambda) = \log \mathcal{Z}_{\xi,A}(\lambda). \quad (\text{C1})$$

With $\kappa_{n,\xi}^q = (-i\partial_\lambda)^n \mathcal{K}|_{\lambda=0}$,

$$\kappa_{1,\xi}^q = \text{Tr } C_{\xi,A}, \quad (\text{C2})$$

$$\kappa_{2,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})], \quad (\text{C3})$$

$$\kappa_{3,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})(\mathbf{1}_A - 2C_{\xi,A})], \quad (\text{C4})$$

$$\kappa_{4,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})(\mathbf{1}_A - 6C_{\xi,A} + 6C_{\xi,A}^2)]. \quad (\text{C5})$$

The state-CFT observable is $\mathbb{E}[\kappa_{n,\xi}^q]$. It is distinct from the FCS of the Born mixture,

$$\mathcal{Z}_A^{\text{mix}}(\lambda) = \mathbb{E}_\xi[\mathcal{Z}_{\xi,A}(\lambda)], \quad (\text{C6})$$

whose logarithm includes classical trajectory fluctuations. In particular,

$$\text{Var}_{\text{mix}}(Q_A) = \mathbb{E}_\xi[\kappa_{2,\xi}^q] + \text{Var}_\xi(\kappa_{1,\xi}^q). \quad (\text{C7})$$

Both terms in Eq. (C7) are saved and plotted separately.

For a chiral $U(1)$ current, the normalization convention is fixed by

$$J(z)J(0) \sim \frac{k}{z^2}, \quad [J_m, J_n] = k m \delta_{m+n,0}. \quad (\text{C8})$$

A free chiral complex fermion has $k = 1$. A single wall contributes

$$X_L(\ell) = \log \left[\frac{N_y}{\pi a} \sin \left(\frac{\pi \ell}{N_y} \right) \right], \quad \kappa_2^{(1w)}(\ell) = \frac{k}{2\pi^2} X_L(\ell) + b + \dots, \quad (\text{C9})$$

whereas two well-separated, oppositely oriented walls contribute

$$\kappa_2^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b + \dots. \quad (\text{C10})$$

The common correlation-spectrum derivation of the matching entropy coefficients and the $\pi^2/3$ relation is given in Appendix D. The charged primaries obey $h_q = q^2/(2k)$, giving Eq. (55) at $k = 1$. The analysis must declare whether a subsystem contains one or two walls before converting a fit coefficient to k or c_{edge} . The coefficient k identifies the current algebra; by itself it does not determine chirality, which is tested only by Campaigns H1–H3.

Signed feedback-transfer FCS is a different classical observable. If $Y_e \in \{-1, 0, 1\}$ is the system-layer charge transfer at event e , save

$$Q_{\text{rec}}(R, T) = \sum_{e \in R \times T} Y_e, \quad \theta_R(s, T) = \frac{1}{T} \log \mathbb{E}_\xi [e^{-s Q_{\text{rec}}}], \quad (\text{C11})$$

The empirical finite-time θ_R is not called a tilted-channel eigenvalue. Use log-sum-exp stabilization, trajectory bootstrap intervals, and a reweighting effective sample-size flag. Record FCS and quantum subsystem FCS may be compared only after their distinct definitions are displayed.

Appendix D: Common correlation-spectrum origin of entropy and charge fluctuations

This appendix fixes the exact Gaussian relation between interval entanglement and quantum charge fluctuations and then states the stronger CFT test used in Campaign S1. It applies to a number-conserving Gaussian reduced state whether or not the reduced density matrix is pure. It does not identify the within-trajectory quantum variance with classical trajectory-to-trajectory charge wandering.

1. Spectral calculus of the restricted covariance

Let

$$C_{\xi, A} u_{\xi a} = \nu_{\xi a} u_{\xi a}, \quad 0 \leq \nu_{\xi a} \leq 1, \quad (\text{D1})$$

and define the positive charge-susceptibility operator

$$\mathcal{V}_{\xi, A} = C_{\xi, A} (\mathbf{1}_A - C_{\xi, A}), \quad q_{\xi a} = \nu_{\xi a} (1 - \nu_{\xi a}) \in [0, 1/4]. \quad (\text{D2})$$

The interval quantum variance and von Neumann entropy are

$$F_{\xi, A}^q = \text{Tr } \mathcal{V}_{\xi, A} = \sum_a q_{\xi a}, \quad (\text{D3})$$

$$\begin{aligned} S_{\xi, A} &= -\text{Tr} [C_{\xi, A} \log C_{\xi, A} + (\mathbf{1}_A - C_{\xi, A}) \log (\mathbf{1}_A - C_{\xi, A})] \\ &= \sum_a h(\nu_{\xi a}) = \text{Tr } g(\mathcal{V}_{\xi, A}), \end{aligned} \quad (\text{D4})$$

where

$$h(\nu) = -\nu \log \nu - (1 - \nu) \log (1 - \nu), \quad g(q) = h\left(\frac{1 + \sqrt{1 - 4q}}{2}\right). \quad (\text{D5})$$

The branch of the square root is immaterial because $h(\nu) = h(1 - \nu)$. More generally,

$$S_{\xi, A}^{(n)} = \text{Tr } g_n(\mathcal{V}_{\xi, A}), \quad g_n(q) = \frac{1}{1 - n} \log [\nu(q)^n + (1 - \nu(q))^n], \quad (\text{D6})$$

with $\nu(q) = (1 + \sqrt{1 - 4q})/2$. Thus entropy and variance are different spectral functionals of the same operator. The scalar $F_{\xi,A}^q$ alone does not determine $S_{\xi,A}$: distinct multisets $\{q_{\xi a}\}$ can have the same first moment. The full spectrum of $\mathcal{V}_{\xi,A}$ does determine every Gaussian Rényi entropy.

The corresponding single-particle modular Hamiltonian is

$$h_{\xi,A}^{\text{mod}} = \log \left[(\mathbf{1}_A - C_{\xi,A}) C_{\xi,A}^{-1} \right], \quad \nu_{\xi a} = \frac{1}{1 + e^{\epsilon_{\xi a}}}, \quad (\text{D7})$$

with zero and unit occupations understood by continuity. In modular-energy variables,

$$q(\epsilon) = \frac{1}{4 \cosh^2(\epsilon/2)}, \quad (\text{D8})$$

$$s(\epsilon) = \log \left[2 \cosh \left(\frac{\epsilon}{2} \right) \right] - \frac{\epsilon}{2} \tanh \left(\frac{\epsilon}{2} \right). \quad (\text{D9})$$

If $\rho_{\xi,A}(\epsilon) = \sum_a \delta(\epsilon - \epsilon_{\xi a})$, then

$$F_{\xi,A}^q = \int d\epsilon \rho_{\xi,A}(\epsilon) q(\epsilon), \quad S_{\xi,A} = \int d\epsilon \rho_{\xi,A}(\epsilon) s(\epsilon). \quad (\text{D10})$$

Both kernels select entanglement levels with $|\epsilon| = O(1)$, and

$$\int_{-\infty}^{\infty} d\epsilon q(\epsilon) = 1, \quad \int_{-\infty}^{\infty} d\epsilon s(\epsilon) = \frac{\pi^2}{3}. \quad (\text{D11})$$

Consequently a common slowly varying, logarithmically growing low-modular-energy density produces $S = (\pi^2/3) F^q + O(1)$. Equation (D11) explains the spectral origin of the coefficient but is not an exact finite-size equality for an arbitrary covariance spectrum.

2. CFT coefficient and level-one test

For the interval convention of Appendix C, one chiral wall contributes

$$S_n^{(1w)}(\ell) = \frac{c}{12} \left(1 + \frac{1}{n} \right) X_L(\ell) + b_{S_n} + \dots, \quad (\text{D12})$$

$$F_q^{(1w)}(\ell) = \frac{k}{2\pi^2} X_L(\ell) + b_F + \dots, \quad (\text{D13})$$

whereas two well-separated oppositely oriented walls contribute twice these logarithmic coefficients,

$$S_n^{(2w)}(\ell) = \frac{c}{6} \left(1 + \frac{1}{n} \right) X_L(\ell) + b'_{S_n} + \dots, \quad (\text{D14})$$

$$F_q^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b'_F + \dots. \quad (\text{D15})$$

The geometry factor cancels in the ratio. Denoting the corresponding log-chord slopes by a_{S_n} and a_F gives

$$\frac{a_{S_n}}{a_F} = \frac{\pi^2 c}{6k} \left(1 + \frac{1}{n} \right), \quad \frac{a_S}{a_F} = \frac{\pi^2 c}{3k}. \quad (\text{D16})$$

For a charge-one chiral complex fermion, $c = k = 1$, and therefore

$$a_S - \frac{\pi^2}{3} a_F = 0, \quad a_{S_n} - \frac{\pi^2}{6} \left(1 + \frac{1}{n} \right) a_F = 0. \quad (\text{D17})$$

Equivalently,

$$S(\ell) - \frac{\pi^2}{3} F^q(\ell) = \text{constant} + \text{finite-size corrections} \quad (\text{D18})$$

has no logarithmic term. This is a leading-scaling statement, not a claim that the two spectral kernels agree mode by mode.

Under a unitary chiral-CFT interpretation and a fixed microscopic normalization of charge, the $U(1)$ current generates a $c = 1$ Sugawara sector. The combination $c \simeq 1$, $k \simeq 1$, Eq. (D17), and an independent H1–H3 chirality test supports a minimal $U(1)_1$ charged wall without an additional neutral gapless sector. Entropy and variance alone do not determine the sign of the chiral central charge or the direction of propagation.

3. Cross-boundary coherence and intrinsic mixedness

For a pure Slater trajectory the full correlation matrix is a projector. With the spatial block decomposition

$$C_\xi = \begin{pmatrix} C_{\xi,A} & C_{\xi,AB} \\ C_{\xi,BA} & C_{\xi,B} \end{pmatrix}, \quad C_\xi^2 = C_\xi, \quad (\text{D19})$$

the AA block of the projector identity gives

$$\mathcal{V}_{\xi,A} = C_{\xi,A}(\mathbf{1}_A - C_{\xi,A}) = C_{\xi,AB}C_{\xi,BA}. \quad (\text{D20})$$

Hence the $q_{\xi a}$ are the squared singular values of the cross-boundary correlation block,

$$F_{\xi,A}^q = \|C_{\xi,AB}\|_F^2, \quad S_{\xi,A} = \text{Tr } g(C_{\xi,AB}C_{\xi,BA}). \quad (\text{D21})$$

The variance is the total squared cross-boundary coherence, whereas the entropy retains the complete singular-value distribution.

For a globally mixed Gaussian state, $0 \leq C_\xi \leq \mathbf{1}$ but $C_\xi^2 \neq C_\xi$, and the exact decomposition becomes

$$\mathcal{V}_{\xi,A} = C_{\xi,AB}C_{\xi,BA} + (C_\xi - C_\xi^2)_{AA}. \quad (\text{D22})$$

Both terms are positive. The first measures cross-boundary coherence and the second is intrinsic mixedness already present in the global state. Their traces add in the charge variance, but the entropy is not generally the sum of separate functions of the two terms because they need not commute. Equation (D22) is the appropriate diagnostic during max-mix purification; Eq. (D20) is used only after the pure-trajectory gate passes.

The shared natural orbitals also define matched real-space contours,

$$f_{\xi,A}(i) = \sum_a |u_{\xi a}(i)|^2 q_{\xi a}, \quad s_{\xi,A}(i) = \sum_a |u_{\xi a}(i)|^2 h(\nu_{\xi a}), \quad (\text{D23})$$

with the exact sum rules

$$\sum_{i \in A} f_{\xi,A}(i) = F_{\xi,A}^q, \quad \sum_{i \in A} s_{\xi,A}(i) = S_{\xi,A}. \quad (\text{D24})$$

Comparing these unnormalized contours tests whether the same wall-localized natural orbitals carry both logarithms. Each contour is averaged over trajectories only after its per-trajectory construction.

4. Full charge statistics and ensemble decomposition

Writing $z = e^{i\lambda}$, the interval FCS polynomial factorizes as

$$\chi_{\xi,A}(z) = \det[\mathbf{1}_A - C_{\xi,A} + zC_{\xi,A}] = \prod_a (1 - \nu_{\xi a} + z\nu_{\xi a}). \quad (\text{D25})$$

Its zeros obey

$$z_{\xi a} = -\frac{1 - \nu_{\xi a}}{\nu_{\xi a}} = -e^{\epsilon_{\xi a}}. \quad (\text{D26})$$

Thus the exact full interval-charge distribution, whose probabilities are the coefficients of $\chi_{\xi,A}(z)$, determines the single-particle entanglement spectrum up to permutation and hence every Gaussian entropy. Numerical inversion of high-degree noisy counting polynomials is ill conditioned and is not a production gate; direct covariance eigenvalues remain the primary simulated estimator.

Finally, if one first samples a trajectory and then measures its interval charge, the law of total variance is

$$\text{Var}_{\text{total}}(N_A) = \mathbb{E}_\xi[\text{Tr } \mathcal{V}_{\xi,A}] + \text{Var}_\xi[\text{Tr } C_{\xi,A}]. \quad (\text{D27})$$

Only the first term is the quantum interval variance governed by the spectrum in Eq. (D2) and used to infer k . Boundary localization of the second term is meaningful dynamical information about the trajectory ensemble, but it is not substituted for the current-algebra coefficient.

5. Production estimator and acceptance rule

Campaign S1 computes $S_{\xi,A}$, $F_{\xi,A}^q$, their contours, and the $q_{\xi a}$ spectrum from the same restricted covariance for every declared interval. Origin and stationary-snapshot averages are internal to a trajectory and are performed identically for both observables. The common linear fit then produces $m_{1,\xi}$, f_ξ , and the paired difference $\delta_{SF,\xi}$ in Eq. (32). Whole trajectories are resampled jointly, preserving the covariance between the two slopes and among all interval lengths.

The primary test is the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ and the equivalent residual curve in Eq. (33). The ratio a_S/a_F and the Rényi relations in Eq. (34) are secondary. Repeat the complete paired analysis after each preregistered fit-window deletion and on the exact-Hamiltonian, matched trivial, and one-versus-two-wall controls. A nonzero residual logarithmic coefficient forbids the minimal $U(1)_1$ identification even if entropy and variance separately admit logarithmic fits. It does not erase the separate estimates of c and k , and it says nothing by itself about chirality.

a. Legacy comparison. The canonical pure-state covariance archive under `colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/` does contain the information needed for matched full-width interval entropy and within-trajectory quantum charge variance. A direct cycle-50 reduction at $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, and $S = 10$ gives $c_{\text{eff}} = 1.032\text{--}1.086$ and $k_{\text{eff}} = 1.033\text{--}1.085$, with the four matched values tracking one another. This is a genuine legacy point in the leading $\pi^2/3$ relation, but it is not the production test: only two sizes and ten trajectories are present, the Rényi slopes and one-wall charge decomposition are absent, and no thermodynamic residual extrapolation can be performed. The max-mix campaign's total charge variance during purification remains a different observable.

Appendix E: Circuit depth, Lu bound, wall hybridization, and fixed duration

For a product input and a depth- D local circuit, the rank-entropy growth across region A obeys the Lu–Lessa–Kim–Hsieh bound [2]

$$S_0(A, T) - S_0(A, 0) \lesssim D |\partial A|. \quad (\text{E1})$$

For the full-width strip $A = [0, N_x) \times [0, \ell)$, $|\partial A| \sim 2N_x$. A two-wall CFT contribution therefore requires parametrically

$$\frac{1}{3} \log N_y \lesssim \kappa D N_x. \quad (\text{E2})$$

Here D counts layers of mutually nonoverlapping local maps, not circuit sweeps. The production random-serial schedule visits overlapping orbitals sequentially, so one cycle has a size-dependent implementation depth. For $n_{\text{shell}} = 1$, an update centered at $u = (x, y)$ has the periodic 3×3 cell support

$$\mathcal{S}_u = \{u + (\delta_x, \delta_y) : \delta_x, \delta_y \in \{-1, 0, 1\}\}. \quad (\text{E3})$$

The conflict graph joins u and v when $\mathcal{S}_u \cap \mathcal{S}_v \neq \emptyset$, equivalently when their cyclic separations obey

$$d_{N_x}(x, x') \leq 2, \quad d_{N_y}(y, y') \leq 2, \quad d_L(a, b) = \min_{m \in \mathbb{Z}} |a - b + mL|. \quad (\text{E4})$$

It has maximum degree at most 24 and hence at most 25 colors by a greedy bound. If $q_x \mid N_x$, $q_y \mid N_y$, and $q_x, q_y \geq 3$, the explicit residue coloring

$$c(x, y) = (x \bmod q_x, y \bmod q_y) \quad (\text{E5})$$

uses $q_x q_y$ colors and is proper: distinct equal-color centers have separation at least three in one coordinate. A commensurate 3×3 coloring therefore uses nine colors. For every size in Eq. (4), the conservative 4×4 coloring wraps cleanly and gives $\chi_{\text{conf}} \leq 16$.

Let $d_{\text{inst}} = O(1)$ denote the physical layer depth of one occupation-measurement–feedforward–fresh-ancilla–fSWAP instrument block. Keeping the four controller channels as separate blocks gives

$$d_{\text{cyc}} \leq 4\chi_{\text{conf}} d_{\text{inst}} \leq 64 d_{\text{inst}}, \quad D = d_{\text{cyc}} T_{\text{cyc}}. \quad (\text{E6})$$

This is an $O(1)$ bound on the physical depth of one colored sweep, not on the total preparation depth. The BPJ bulk can converge in $O(1)$ sweeps while the additional critical wall requires a cycle count that grows with N_y ; the two statements concern different target entanglement structures. Equation (E2) gives only

$$T_{\text{cyc}} \gtrsim \frac{\log N_y}{3\kappa d_{\text{cyc}} N_x}. \quad (\text{E7})$$

The unknown coefficient makes this too weak to choose 10 versus 50 versus $2N_y$ cycles. The production duration is therefore a declared numerical choice rather than a consequence of this bound.

This coloring gives an optional quantum-simulation compilation. Within a color, disjoint local instruments commute, so simultaneous hardware readout and sequential numerical evaluation define the same joint Born instrument even though its outcomes need not be statistically independent. Reordering different color blocks is not a parallel reinterpretation of a saved random-serial trajectory because overlapping conditional maps generally do not commute. The production campaign therefore makes no constant-depth-per-sweep claim for its random-serial word; it reports Eq. (E6) only as a realizable resource bound for the corresponding color-block hardware schedule.

The circuit-specific operator-Schmidt-rank bounds are

$$r(a_\chi), r(a_\chi^\dagger) \leq 2, \quad r(n_\chi), r(\mathbf{1} - n_\chi) \leq 4. \quad (\text{E8})$$

Along record ξ , a sharper bookkeeping bound is

$$\Delta S_0^\xi \leq 2 \log 2 N_{=}^\partial(\xi) + \log 2 N_{\neq}^\partial(\xi), \quad (\text{E9})$$

with the two counts distinguished according to the boundary-crossing branch type. These bounds assume an initially product state. A random Slater determinant is already entangled and cannot be used to validate a product-input depth bound. The maximally mixed state is treated separately as a purification problem beginning at $t = 0$.

Finite transverse width produces an independent infrared scale. If the opposing-wall hybridization behaves as

$$m(w) \sim m_0 e^{-w/\xi_\perp}, \quad (\text{E10})$$

then one requires

$$m(w) \ll \frac{2\pi v}{N_y}, \quad w \gtrsim \xi_\perp \log N_y \quad (\text{E11})$$

up to nonuniversal constants. B0 determines $\xi_\perp$ and W1 verifies the same condition in the stochastic circuit.

Every production trajectory begins from its own independently sampled declared initial state and runs for exactly $2N_y$ complete cycles. There is no separate physical burn-in, no shared equilibrated parent state, and no cloning of one evolved covariance into 100 continuations. Time-resolved bulk markers, entropy coefficients, charge statistics, wall profiles, purity diagnostics, and successive-covariance changes are retained as transient checks inside this fixed window.

The three pure-state checkpoints used by S1 are $t = N_y, 3N_y/2, 2N_y$. Agreement among them is a sensitivity diagnostic, not a device for converting cycle snapshots into independent samples. If the late checkpoints are incompatible beyond the preregistered tolerance, the relevant stationarity or temporal gate fails. Production is not silently extended beyond $2N_y$; a longer campaign requires a versioned amendment and a new output directory.

Max-mix purification likewise starts at $C = \mathbf{1}/2$ at $t = 0$ and retains its complete $2N_y$ clock. T1 initializes its physical trajectory and tangent frame at $t = 0$, uses the first N_y cycles for tangent-frame alignment only, resets only the QR log accumulator, and accumulates exponents over the final N_y cycles. The physical trajectory is never reset and the alignment interval is not called burn-in. This procedure is tailored to the production serial implementation and must be re-evaluated if the actual overlap-conflict schedule changes.

Appendix F: Campaign-card checklist

Before a campaign is marked complete, its final note must answer all of the following:

1. Was the canonical GPU entry point recorded?
2. Were all 100 trajectories independently initialized and run for the declared total duration without cloning an evolved state?
3. Were max-mix and tangent clocks treated according to Appendix E?
4. Can every plotted scalar be reconstructed from per-trajectory raw data?
5. Is the order of nonlinear evaluation, time/source averaging, and trajectory averaging explicit?
6. Are fit windows, excluded points, and correction models preregistered and saved?
7. Do paired controls share declared random streams without being called the same record?

8. Did every shard pass manifest, seed, replay/restart, and numerical-residual checks?
9. Does the caption state sample count, independent unit, uncertainty, and geometry?
10. If the gate failed, was the prescribed downgrade used without changing the estimator?

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- [1] A. Bhuiyan, H. Pan, and C.-M. Jian, [Physical Review Research](#) **8**, 023147 (2026), [arXiv:2507.13437 \[quant-ph\]](#).
[2] T.-C. Lu, L. A. Lessa, I. H. Kim, and T. H. Hsieh, [PRX Quantum](#) **3**, 040337 (2022).